

HR / POWER

Report di identificazione

Run 9c9076e0e2b94c218e7ce7e5e45e895b

Dataset: G1&G2_UtD

Creazione: 2026-09-28T18:35:31.018624+00:00 | Stato: completed

Struttura: p1d_full | $\tau dx/dt + x = K (P(t-L)-P_0)$; $HR_{hat} = B+x$

Inizializzazione: estimated_equilibrium | $P_0 = \text{mean}(P_{pre})$; $HR_0 = HR(0)$; $x(0) = HR_0 - B$; B estimated; $P(t < 0) = P_0$

Dataset: 123 segmenti; train richiesti: 92; risultati: 92; validi: 92; falliti: 0.

Ambito: fit in-sample sui soli segmenti train. Val/test non utilizzati. Aggregati non pesati per durata, calcolati solo sui record fitted; n indica i valori numerici disponibili. n.d. = non disponibile o non applicabile. Nessuna conclusione fisiologica automatica.

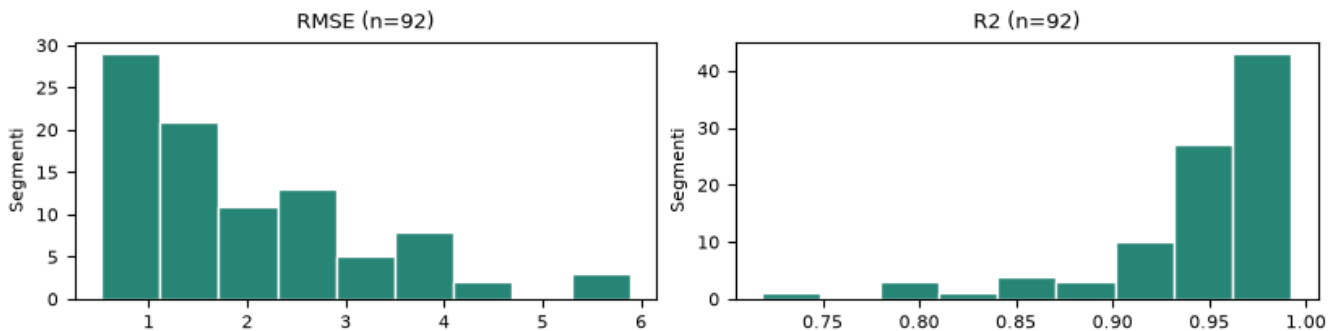
Pre-window richiesta: si; durata: 10 s. P0: pre_window_mean; HR0: first_segment_sample

Pre-window disponibili: 92 / 92 segmenti train inventariati.

Qualita della traiettoria

RMSE, MAE e bias in bpm; R2 e amplitude_ratio adimensionali. Bias positivo: HR osservata maggiore della prevista.

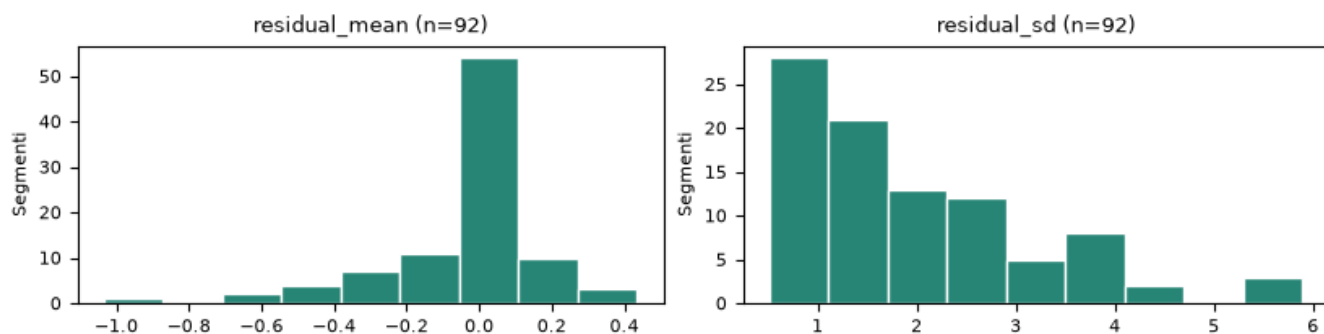
Metrica	n	Media	Mediana	SD	Min	Max
RMSE	92	1.966	1.652	1.2305	0.51228	5.8886
MAE	92	1.5628	1.3218	0.94876	0.42722	4.4097
bias	92	-0.032401	0.0038962	0.21449	-1.0337	0.43523
R2	92	0.94347	0.9606	0.050215	0.71871	0.99245
amplitude_ratio	92	1.0015	1.0081	0.09448	0.54369	1.149



Residui

e = HR osservata - HR prevista. SD campionaria (ddof=1); ACF centrata, lag in campioni. Le segnalazioni sono conservate e non impongono nuove esclusioni.

Metrica	n	Media	Mediana	SD	Min	Max
residual_mean	92	-0.032401	0.0038962	0.21449	-1.0337	0.43523
residual_sd	92	1.9648	1.6458	1.2287	0.51529	5.8958
residual_autocorrelation_lag1	92	0.85387	0.89926	0.12672	0.45661	0.98507
residual_autocorrelation_lag5	92	0.35858	0.32795	0.34652	-0.36138	0.89787



Segnalazione	Segmenti
at_bounds	32
autocorrelated_residuals	91
equivalent_predictions_different_parameters	23
irregular_sampling	3
multistart_disagreement	92
near_bounds	6
recording_gaps	3

Multistart

Ricerca: profile_multistart. I tentativi comprendono gli eventuali raffinamenti da griglia; n_multistart indica solo gli start originali.

Start registrati: 1472; convergenti: 1465. Accordo di tutti gli start: 0 / 92 fit validi.

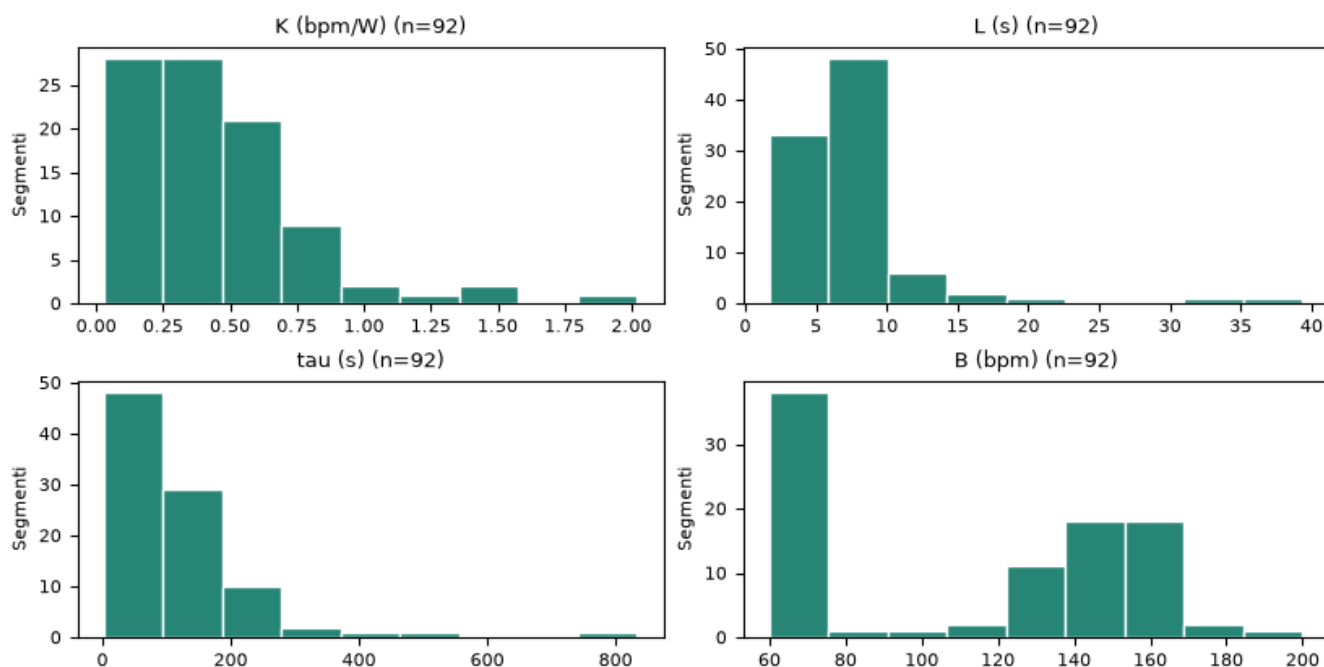
Diagnostica	n	Media	Mediana	SD	Min	Max
SSE	1472	8749.3	439.55	38380	11.809	3.172e+05
nfev	1472	15.892	12	34.622	3	500

Distribuzione e precisione dei parametri

SE e CI95 locali di Wald, condizionati ai riferimenti iniziali. RSE in percentuale. Bounds, autocorrelazione e rango ridotto possono limitare la precisione. Un parametro fissato non ha SE/CI stimabili.

Parametro	n	Media	Mediana	SD	Min	Max
K (bpm/W)	92	0.45885	0.38572	0.32396	0.02886	2.0211
L (s)	92	7.6676	6.446	5.2543	1.7117	39.367
tau (s)	92	120.79	85.415	115.07	2.4107	833.49
B (bpm)	92	110.62	130.34	44.985	60	200

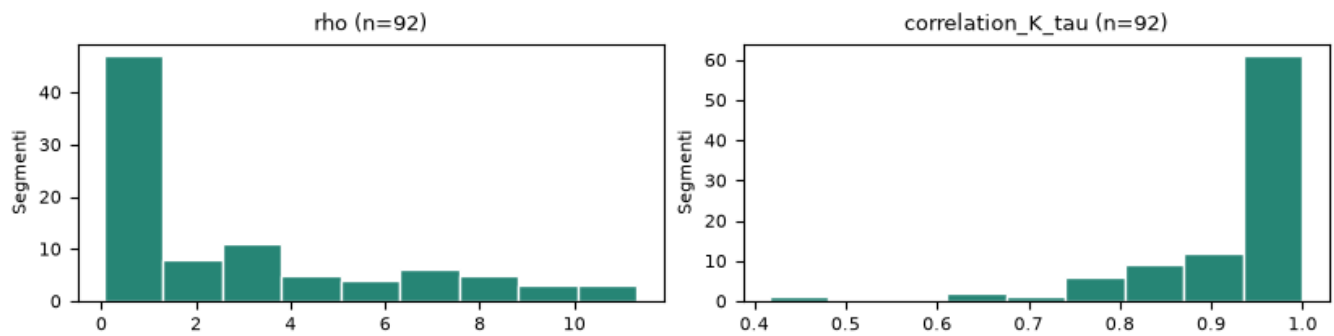
Parametro	SE mediana	RSE mediana %	AI bound	Vicino al bound
K	0.037435	15.507	0	0
L	0.37294	5.8955	0	0
tau	11.999	21.646	0	0
B	3.1343	2.1313	32	38



Identificabilit 

- $\rho = (T-L)/\tau$: tempo disponibile dopo il ritardo in unit  di τ . Valori piccoli descrivono una frazione breve del transitorio; nessuna soglia automatica.
- Correlazioni parametriche vicine a -1 o +1 segnalano compensazione locale tra stime; non misurano la qualit  della traiettoria.
- Rango Jacobiano: confrontare con p , il numero di parametri liberi. Rango $< p$ indica direzioni non identificabili localmente.
- Multistart: SSE simili con parametri differenti indicano ambiguit . CI e correlazioni restano approssimazioni locali.

Diagnostica	n	Media	Mediana	SD	Min	Max
ρ	92	2.8046	1.2053	3.1529	0.053456	11.326
correlation_K_tau	92	0.93194	0.98687	0.10305	0.41665	0.99995
Jacobian rank	92	4	4	0	4	4
Jacobian condition	92	28464	5421.9	36573	242.17	1.6643e+05



Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_022-023.csv

ID: 00412a3a52332d7ce2dc0d21 | Stato: fitted

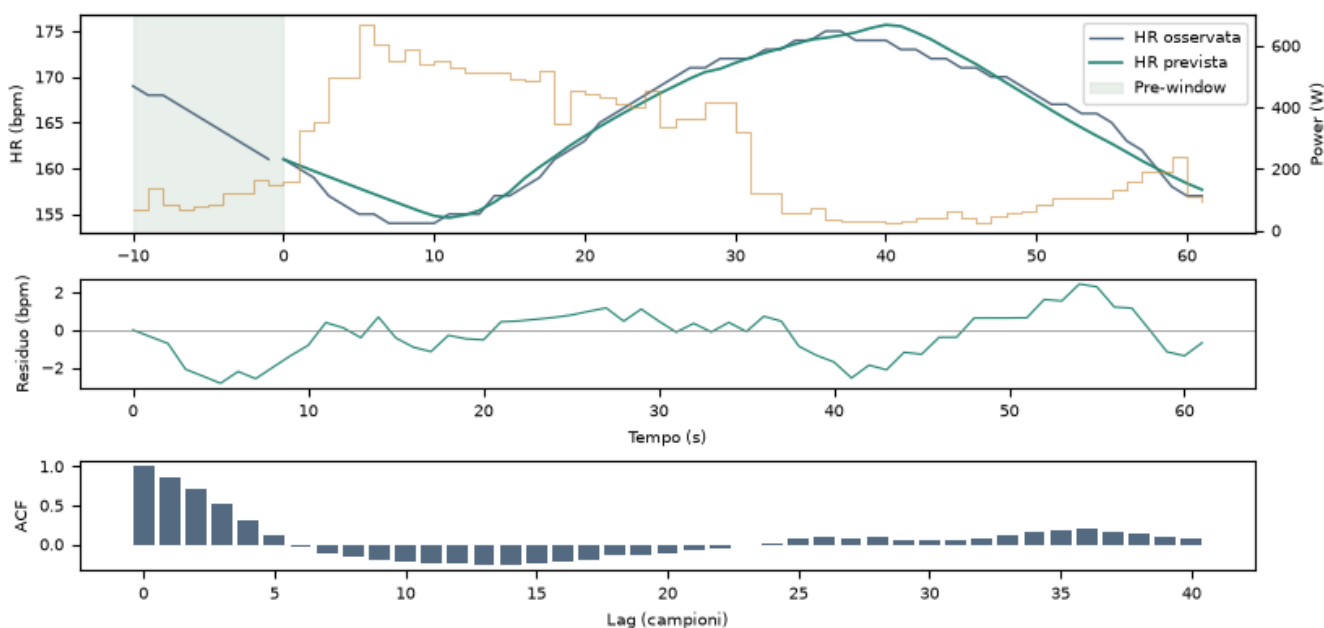
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=108.5 W; HR0=161 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=101 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.2374	0.99663	-0.24998	0.96891	1.0044

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.24998	1.2217	0.85707	0.11291



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.61421	0.66374	-0.68673 / 1.9152	108.06	no	no
L	9.6925	0.58413	8.5476 / 10.837	6.0266	no	no
tau	156.51	169.64	-175.98 / 489	108.39	no	no
B	60	113.78	-163.01 / 283.01	189.64	si	si

rho=0.32782; Jacobian rank=4/4; cond=52673; accordo multistart=no.

Correlazioni: {"K_L": -0.9506973654540947, "K_tau": 0.9996775044374357, "K_B": -0.9999585781692797, "L_tau": -0.9521475953364921, "L_B": 0.9508650896544035, "tau_B": -0.9995058279478921}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 94.925 / 2237

Griglia: 42441 coppie L/tau; SSE multistart originale: 94.925; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_089-090.csv

ID: 00416dc80bc919c10976bef2 | Stato: fitted

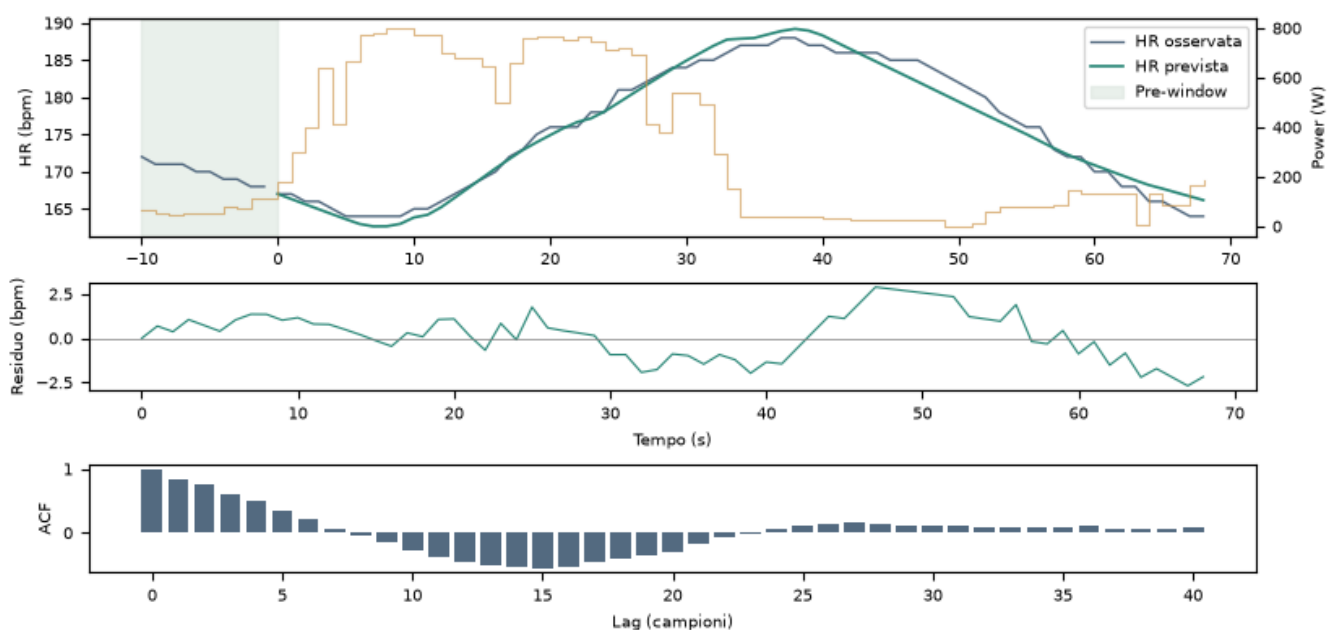
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=69.1 W; HR0=167 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=107 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.3479	1.1115	0.16841	0.97445	1.107

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.16841	1.3471	0.83058	0.34077



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.43288	0.43383	-0.41742 / 1.2832	100.22	no	no
L	5.8998	0.6221	4.6805 / 7.1192	10.544	no	no
tau	155.04	154.79	-148.36 / 458.43	99.843	no	no
B	60	116.73	-168.78 / 288.78	194.54	no	si

rho=0.40055; Jacobian rank=4/4; cond=88512; accordo multistart=no.

Correlazioni: {"K_L": -0.9595361387626504, "K_tau": 0.9997735878527462, "K_B": -0.9999630620893007, "L_tau": -0.9613482231391955, "L_B": 0.9593907155953654, "tau_B": -0.9996240682752542}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 125.36 / 3572

Griglia: 42441 coppie L/tau; SSE multistart originale: 125.94; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250625105513__laps_018-020.csv

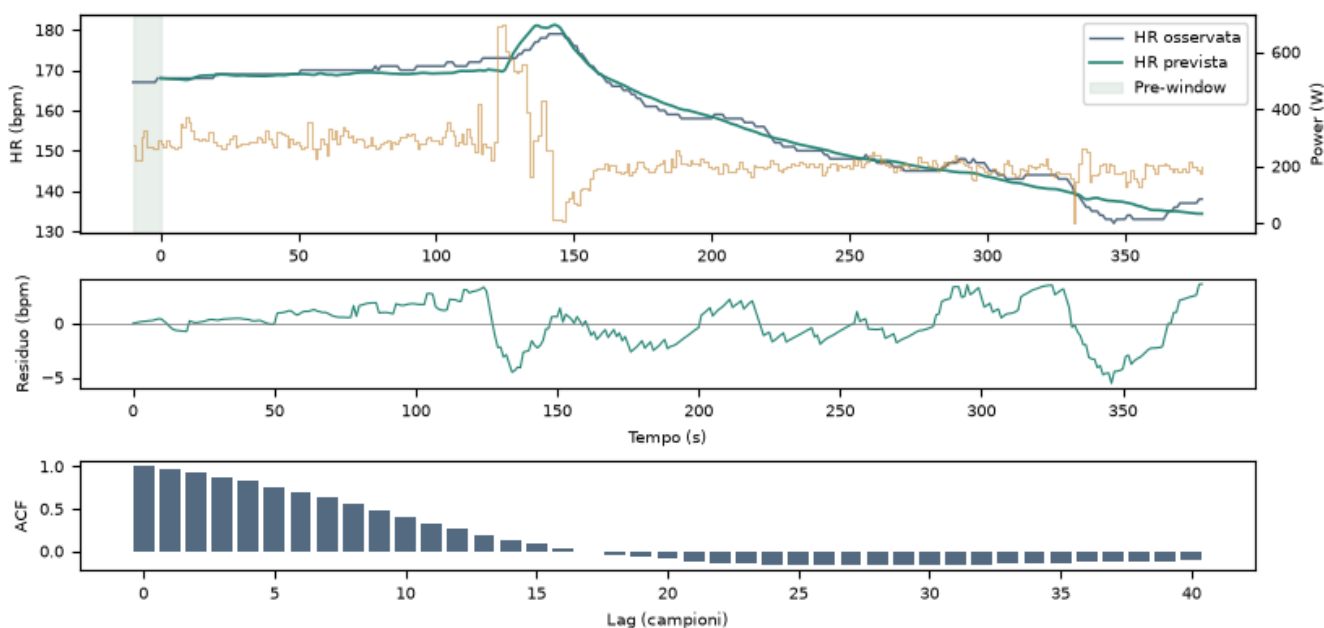
ID: 02b3e2fd2237ca893114ea2a | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=269.8 W; HR0=168 bpm.

HR(0) osservata fissata; B=160.79 bpm stimato; $x(0)=7.2141$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.8789	1.521	0.089793	0.98097	0.99756
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.089793	1.8792	0.964	0.76435	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.43705	0.0098547	0.41773 / 0.45636	2.2548	no	no
L	2.7263	0.36423	2.0124 / 3.4402	13.36	no	no
tau	123.43	4.8802	113.86 / 132.99	3.9539	no	no
B	160.79	0.26154	160.27 / 161.3	0.16267	no	no

rho=3.0404; Jacobian rank=4/4; cond=1531; accordo multistart=no.

Correlazioni: {"K_L": -0.21534832950661253, "K_tau": 0.9152168583787871, "K_B": -0.5530066321789514, "L_tau": -0.2602322851390675, "L_B": 0.05063049644704169, "tau_B": -0.7637654496267599}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1337.9 / 25713

Griglia: 42441 coppie L/tau; SSE multistart originale: 1337.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_010-011.csv

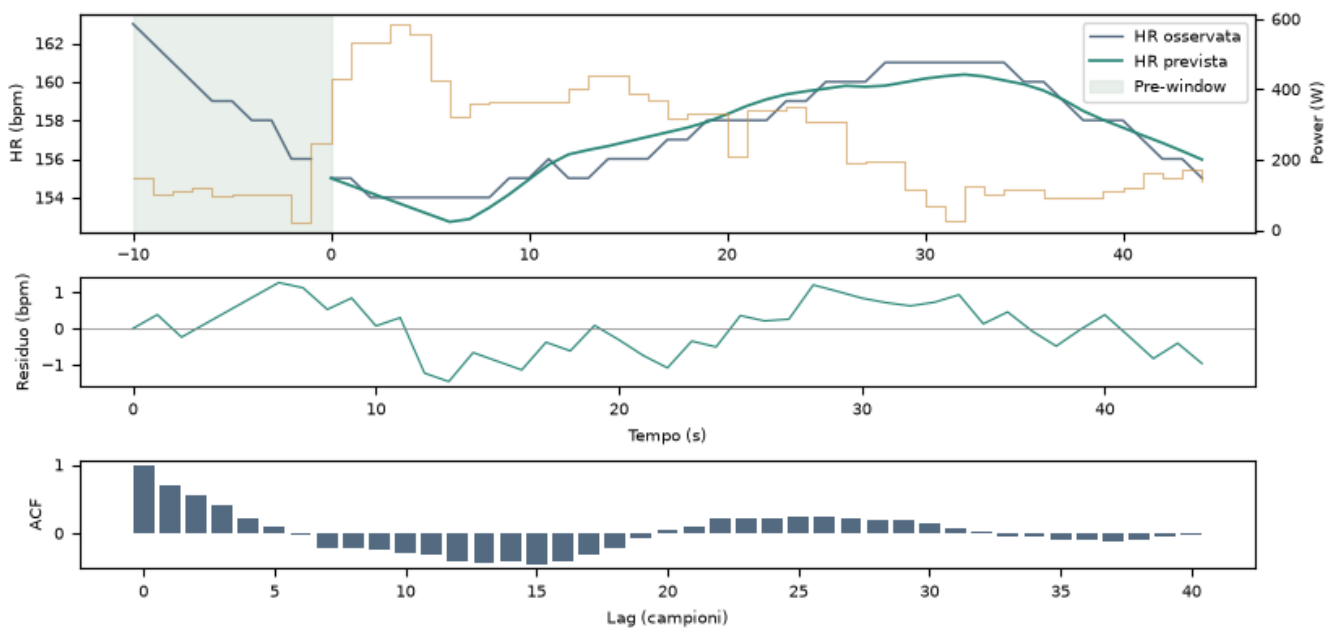
ID: 07f68be91359a07d09c5b596 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=114.9 W; HR0=155 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=95 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.70469	0.59152	0.02952	0.91662	1.0917
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.02952	0.71202	0.70615		0.098244



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.63306	1.8358	-2.9651 / 4.2312	289.99	no	no
L	6.3593	0.53772	5.3054 / 7.4133	8.4557	no	no
tau	248.58	719.01	-1160.7 / 1657.8	289.25	no	no
B	60	282.22	-493.16 / 613.16	470.37	si	si

rho=0.15143; Jacobian rank=4/4; cond=1.3403e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.8979163209302764, "K_tau": 0.9997438175602127, "K_B": -0.9999717702338244, "L_tau": -0.8996572422354177, "L_B": 0.8982567800303918, "tau_B": -0.9995801590088702}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 22.346 / 143.65

Griglia: 42441 coppie L/tau; SSE multistart originale: 22.346; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_074-075.csv

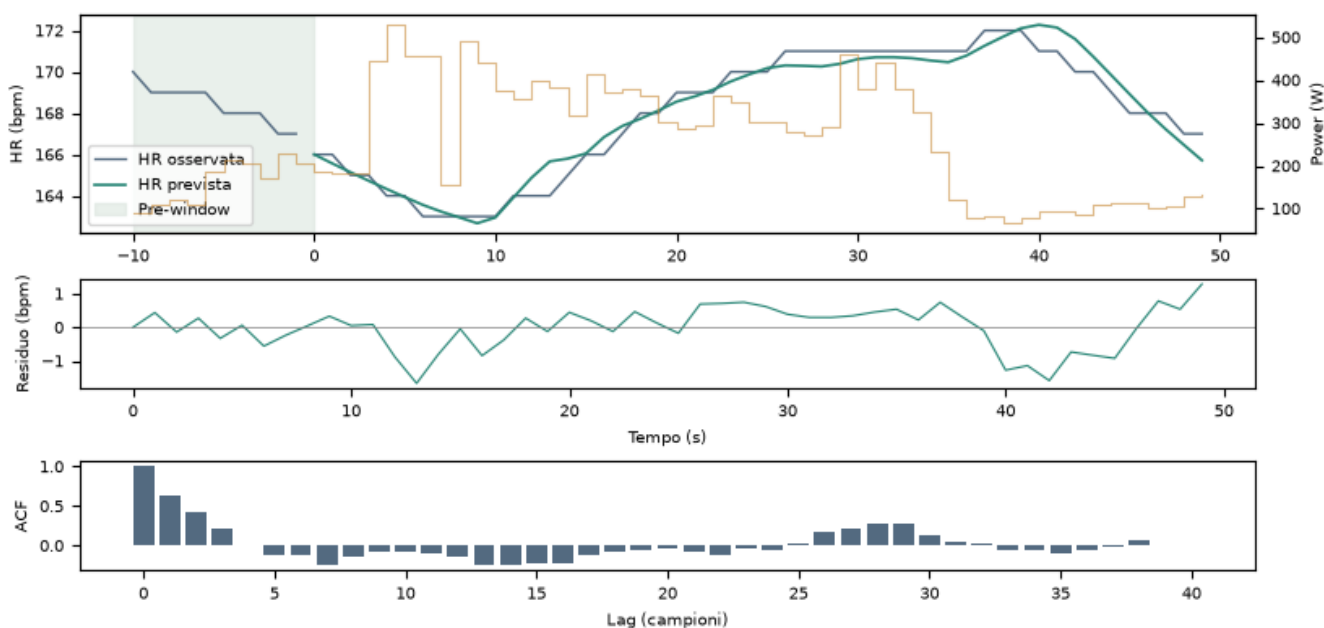
ID: 0898bb8abd4b4f4dd42982da | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=164.4 W; HR0=166 bpm.

HR(0) osservata fissata; B=149.79 bpm stimato; x(0)=16.209 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.63422	0.49236	-0.024287	0.95429	1.067
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.024287	0.64019	0.62814	-0.12912	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.1523	0.033006	0.087611 / 0.217	21.671	no	no
L	6.4882	0.27568	5.9478 / 7.0285	4.249	no	no
tau	36.766	8.4646	20.176 / 53.357	23.023	no	no
B	149.79	3.9021	142.14 / 157.44	2.605	no	no

rho=1.1563; Jacobian rank=4/4; cond=7180.7; accordo multistart=no.

Correlazioni: {"K_L": -0.8020490902787116, "K_tau": 0.9796378773900383, "K_B": -0.9977764790557746, "L_tau": -0.8135038749821687, "L_B": 0.80010056760771, "tau_B": -0.9685058092190437}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 20.112 / 246.78

Griglia: 42441 coppie L/tau; SSE multistart originale: 20.112; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_035-036.csv

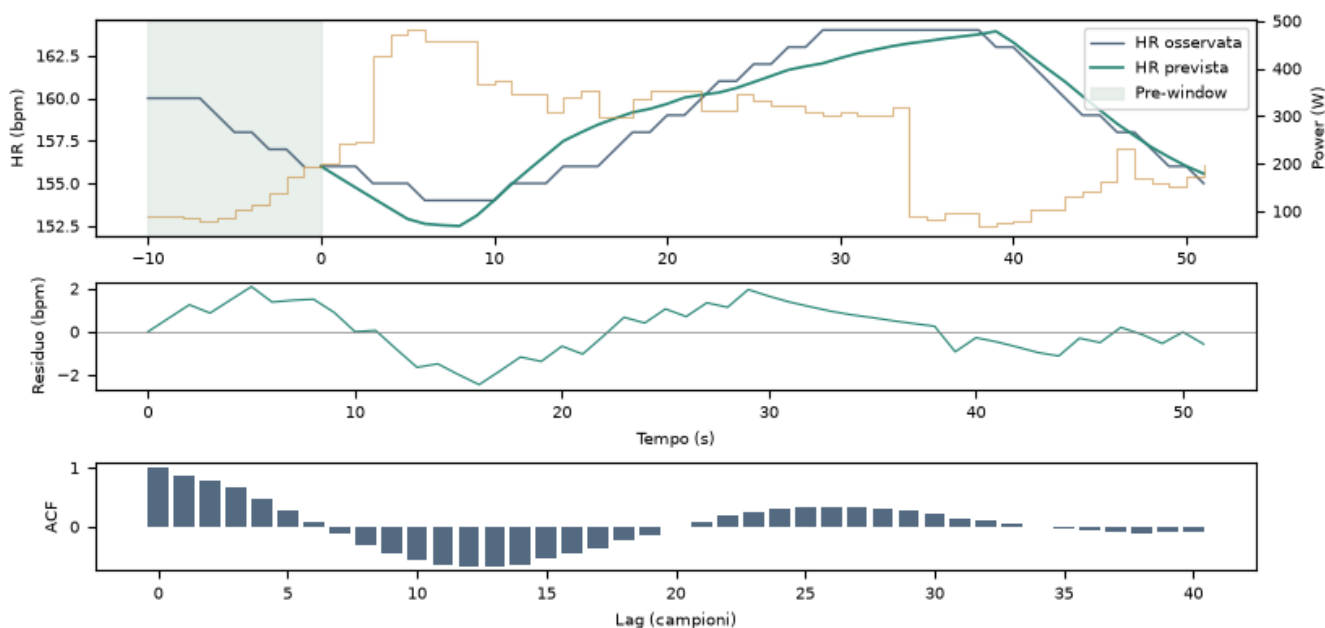
ID: 0981db84c676c17b48528d4a | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=114.9 W; HR0=156 bpm.

HR(0) osservata fissata; B=66.128 bpm stimato; x(0)=89.872 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.0928	0.92014	0.10832	0.90812	1.1429
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.10832	1.098	0.879	0.29159	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.614	0.98102	-1.3088 / 2.5368	159.78	no	no
L	5.1458	0.6014	3.967 / 6.3245	11.687	no	no
tau	143.09	228.82	-305.4 / 591.58	159.92	no	no
B	66.128	148.41	-224.75 / 357.01	224.43	no	no

rho=0.32047; Jacobian rank=4/4; cond=56505; accordo multistart=no.

Correlazioni: {"K_L": -0.9157784219785985, "K_tau": 0.9993679831313669, "K_B": -0.9999612066127225, "L_tau": -0.9185549354070075, "L_B": 0.9163256529938115, "tau_B": -0.9991196051219252}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 62.101 / 445.74

Griglia: 42441 coppie L/tau; SSE multistart originale: 62.101; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_014-015.csv

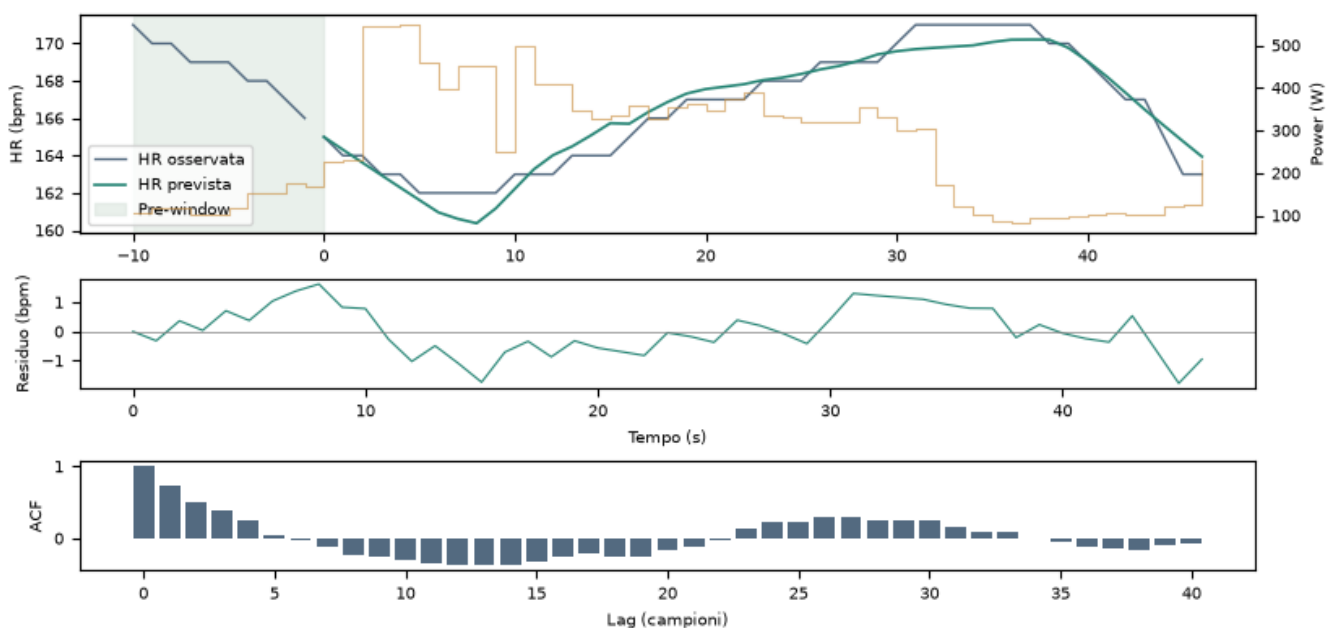
ID: 0b4df52239328d0be144febd | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=130.7 W; HR0=165 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=105 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.79699	0.65339	0.037836	0.93302	1.0925
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.037836	0.8047	0.7229	0.040406	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.63238	0.91395	-1.159 / 2.4237	144.53	no	no
L	6.2007	0.38447	5.4471 / 6.9543	6.2005	no	no
tau	152.93	222.44	-283.05 / 588.91	145.45	no	no
B	60	153.68	-241.22 / 361.22	256.14	no	si

rho=0.26025; Jacobian rank=4/4; cond=66285; accordo multistart=no.

Correlazioni: {"K_L": -0.8894136855603829, "K_tau": 0.9993848496087439, "K_B": -0.9999726808385296, "L_tau": -0.8892468944966927, "L_B": 0.8906874236682611, "tau_B": -0.9991993763703687}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 29.854 / 302.48

Griglia: 42441 coppie L/tau; SSE multistart originale: 29.854; selezione: multistart.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_016-017.csv

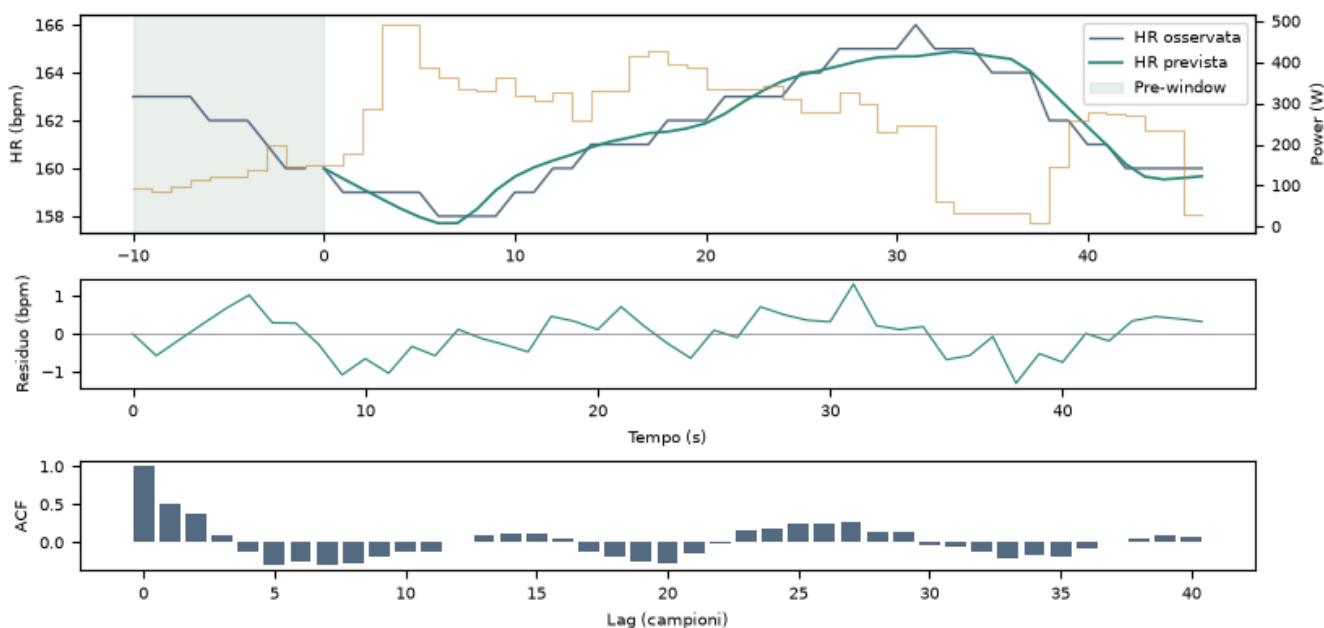
ID: 0f3e732fa081e464158d73f0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=126.2 W; HR0=160 bpm.

HR(0) osservata fissata; B=132.13 bpm stimato; x(0)=27.872 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.54215	0.43691	-0.013059	0.94676	0.8971
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.013059	0.54785	0.50436	-0.30568	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21422	0.11816	-0.01738 / 0.44582	55.16	no	no
L	4.3583	0.34519	3.6817 / 5.0348	7.9204	no	no
tau	64.489	35.507	-5.1052 / 134.08	55.059	no	no
B	132.13	16.353	100.08 / 164.18	12.377	no	no

rho=0.64572; Jacobian rank=4/4; cond=24572; accordo multistart=no.

Correlazioni: {"K_L": -0.8653971301192538, "K_tau": 0.9969078058511611, "K_B": -0.9997904275161842, "L_tau": -0.8702439823610376, "L_B": 0.8670916230485707, "tau_B": -0.9957863126493385}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 13.814 / 180.84

Griglia: 42441 coppie L/tau; SSE multistart originale: 13.814; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_028-029.csv

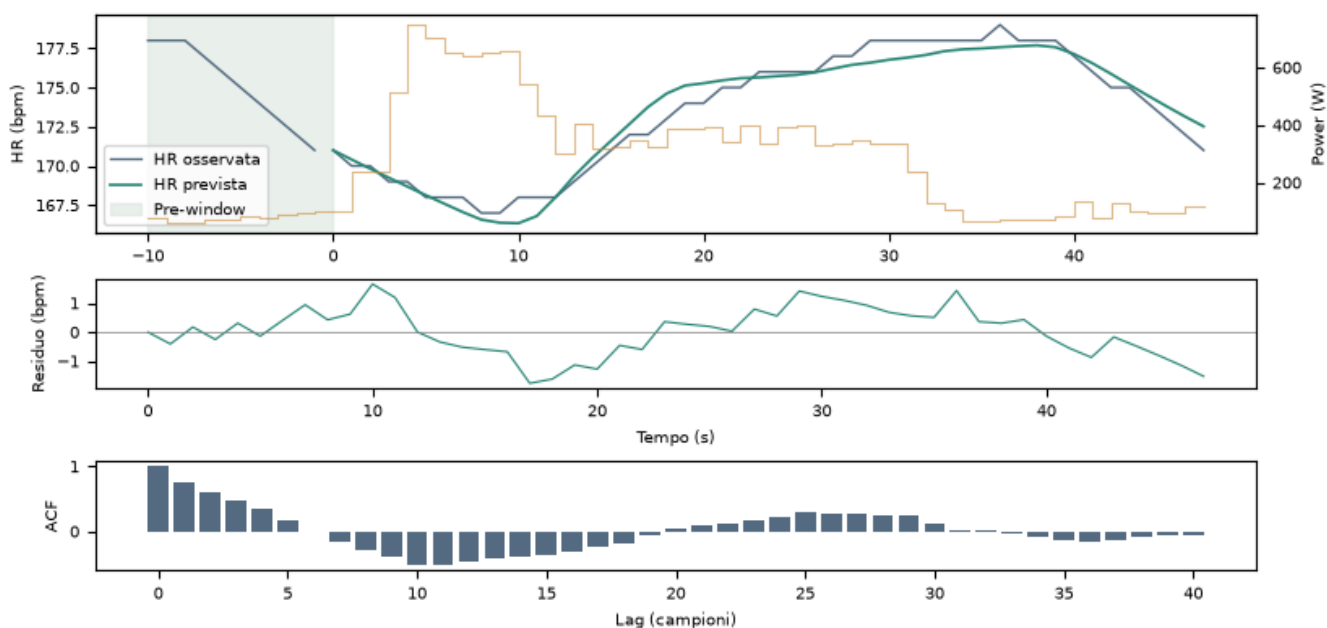
ID: 101b42598bd2beff0cbfd3e6 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=78.7 W; HR0=171 bpm.

HR(0) osservata fissata; B=135.99 bpm stimato; x(0)=35.008 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.82073	0.67276	0.03001	0.9538	0.94558
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.03001	0.82886	0.7594	0.18258	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.17405	0.075484	0.0261 / 0.322	43.369	no	no
L	7.3798	0.38504	6.6251 / 8.1345	5.2175	no	no
tau	58.602	25.824	7.9862 / 109.22	44.067	no	no
B	135.99	16.187	104.27 / 167.72	11.903	no	no

rho=0.67609; Jacobian rank=4/4; cond=22667; accordo multistart=no.

Correlazioni: {"K_L": -0.8790353172606559, "K_tau": 0.9944860520202572, "K_B": -0.9994396675617272, "L_tau": -0.8756507947899206, "L_B": 0.8829132102205652, "tau_B": -0.9914715559950233}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 32.333 / 394.56

Griglia: 42441 coppie L/tau; SSE multistart originale: 32.333; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_019-020.csv

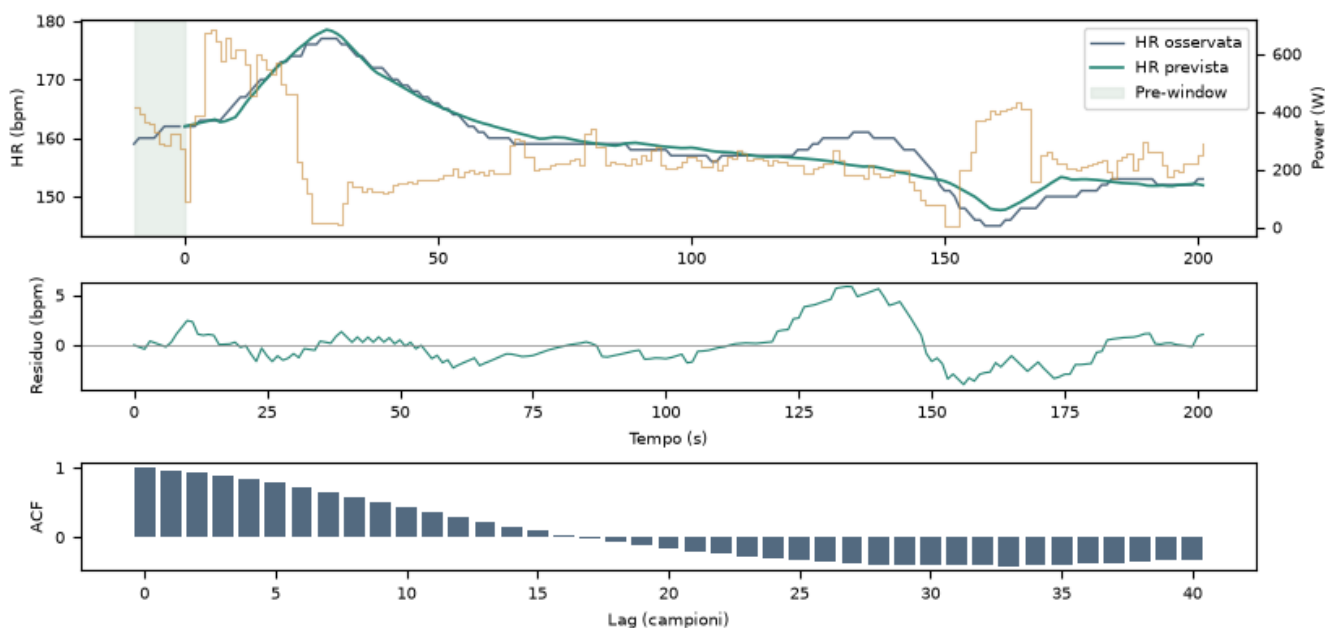
ID: 10d53032bbe7b0aee7fafa34 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=336.5 W; HR0=162 bpm.

HR(0) osservata fissata; B=182.58 bpm stimato; $x(0)=-20.584$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0218	1.4517	-0.013668	0.93212	0.96323
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.013668	2.0267	0.96743	0.78852	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.31107	0.021542	0.26885 / 0.3533	6.9249	no	no
L	6.1254	0.41325	5.3154 / 6.9354	6.7465	no	no
tau	101.18	8.5907	84.34 / 118.02	8.4907	no	no
B	182.58	1.5033	179.64 / 185.53	0.82334	no	no

$\rho=1.9261$; Jacobian rank=4/4; cond=3635.5; accordo multistart=no.

Correlazioni: {"K_L": -0.4392110644409544, "K_tau": 0.940101717602568, "K_B": 0.9768343613896237, "L_tau": -0.4034509055947034, "L_B": -0.4826571148328546, "tau_B": 0.871877417911439}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 825.68 / 7396.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 825.68; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_066-067.csv

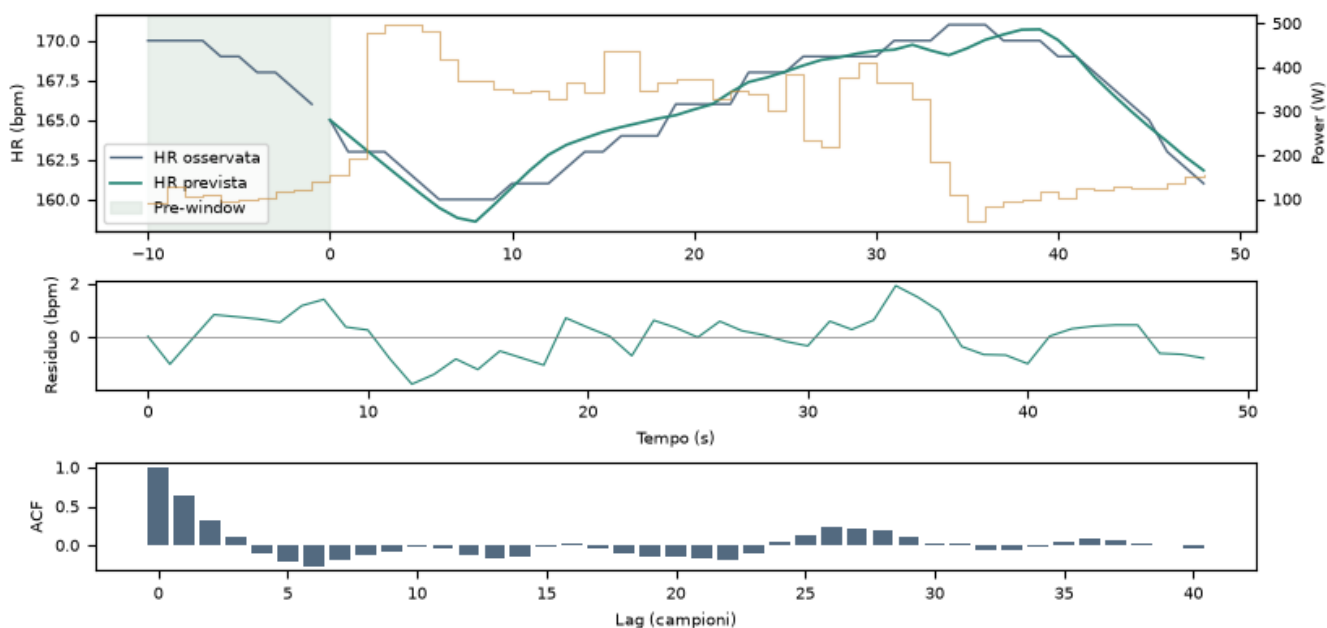
ID: 12c32d6f14eb454003daf51e | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=111 W; HR0=165 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=105 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.79686	0.65629	-0.00095915	0.95056	1.1007
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.00095915	0.80511	0.63597	-0.219	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.57702	0.49304	-0.38934 / 1.5434	85.446	no	no
L	5.8476	0.28958	5.28 / 6.4152	4.952	no	no
tau	110.2	94.321	-74.668 / 295.07	85.59	no	no
B	60	90.338	-117.06 / 237.06	150.56	si	si

rho=0.3825; Jacobian rank=4/4; cond=46627; accordo multistart=no.

Correlazioni: {"K_L": -0.8825685584881103, "K_tau": 0.9989883712092557, "K_B": -0.9999693979535124, "L_tau": -0.8793224194490339, "L_B": 0.8843049284361488, "tau_B": -0.9987978174913155}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 31.114 / 519.53

Griglia: 42441 coppie L/tau; SSE multistart originale: 31.114; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_012-013.csv

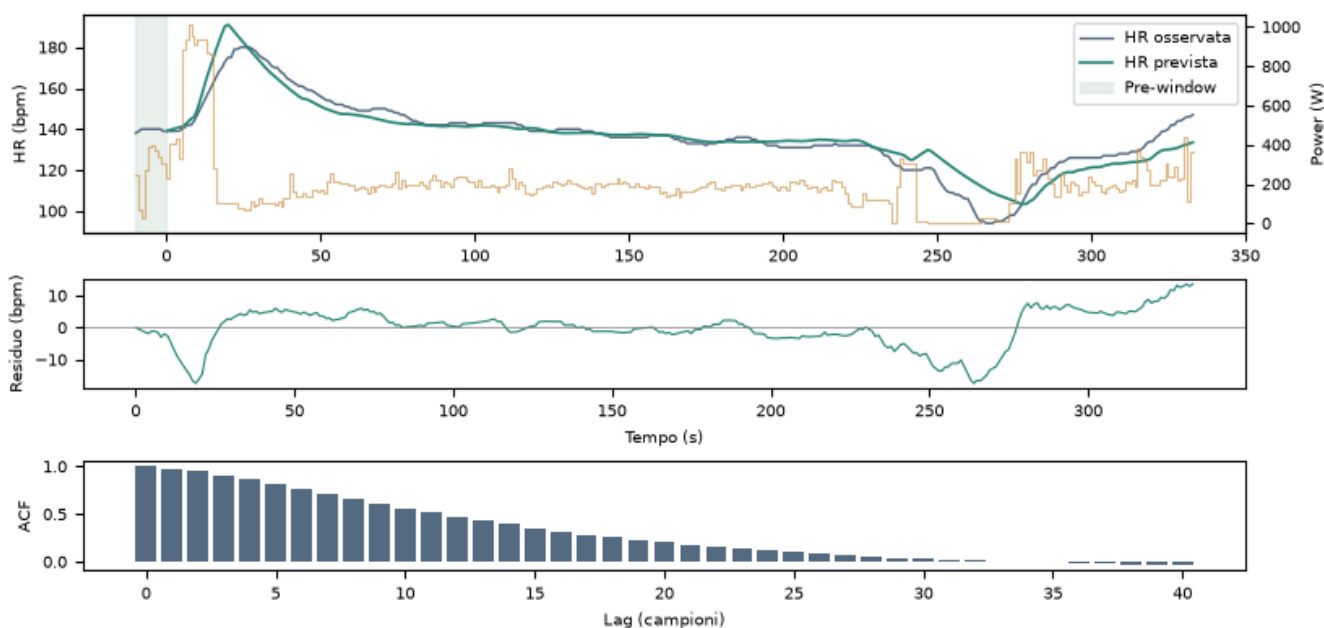
ID: 16099d4d95a84b6f5b36c5fe | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=273.4 W; HR0=139 bpm.

HR(0) osservata fissata; B=160.25 bpm stimato; x(0)=-21.246 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
5.8886	4.306	-0.13957	0.88168	1.0178
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.13957	5.8958	0.98029	0.81486	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.3176	0.010188	0.29763 / 0.33757	3.2079	no	no
L	4.305	0.35308	3.613 / 4.997	8.2016	no	no
tau	43.51	2.1676	39.262 / 47.759	4.9817	no	no
B	160.25	0.82286	158.63 / 161.86	0.5135	no	no

rho=7.5544; Jacobian rank=4/4; cond=617.86; accordo multistart=no.

Correlazioni: {"K_L": -0.34361637040371856, "K_tau": 0.755108236177321, "K_B": 0.8854275286513119, "L_tau": -0.38365443847324054, "L_B": -0.3216980769207415, "tau_B": 0.5748097074422325}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 11582 / 76073

Griglia: 42441 coppie L/tau; SSE multistart originale: 11582; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_051-052.csv

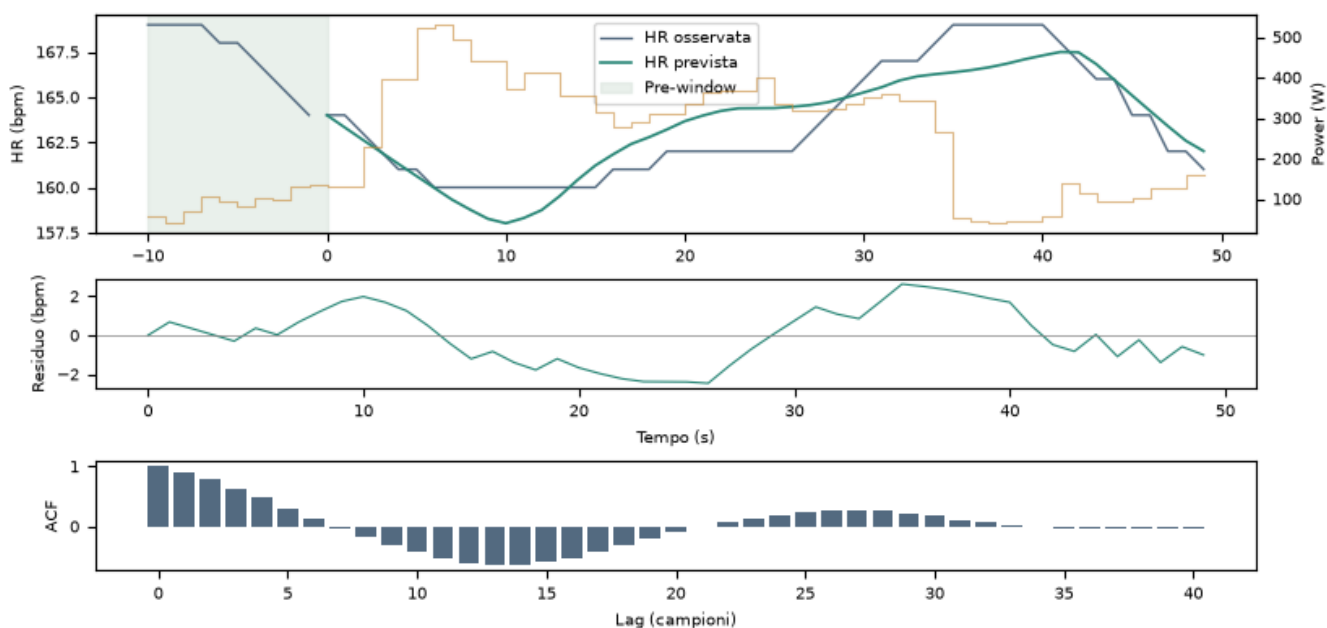
ID: 17ad642582dd9c31d929048c | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=92.1 W; HR0=164 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=104 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.444	1.2149	-0.0049834	0.78213	1.0538
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.0049834	1.4586	0.90619	0.29639	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.54209	1.546	-2.488 / 3.5722	285.19	no	no
L	7.251	0.77928	5.7236 / 8.7784	10.747	no	no
tau	151.98	436.86	-704.28 / 1008.2	287.45	no	no
B	60	294.87	-517.95 / 637.95	491.45	si	si

rho=0.2747; Jacobian rank=4/4; cond=82169; accordo multistart=no.

Correlazioni: {"K_L": -0.8817228619292905, "K_tau": 0.9995807449611412, "K_B": -0.9999834811521081, "L_tau": -0.8777700452858281, "L_B": 0.8833978951740366, "tau_B": -0.9995047612798825}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 104.25 / 476.51

Griglia: 42441 coppie L/tau; SSE multistart originale: 104.25; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_026-027.csv

ID: 1822b3e6cfec7ec77aa201 | Stato: fitted

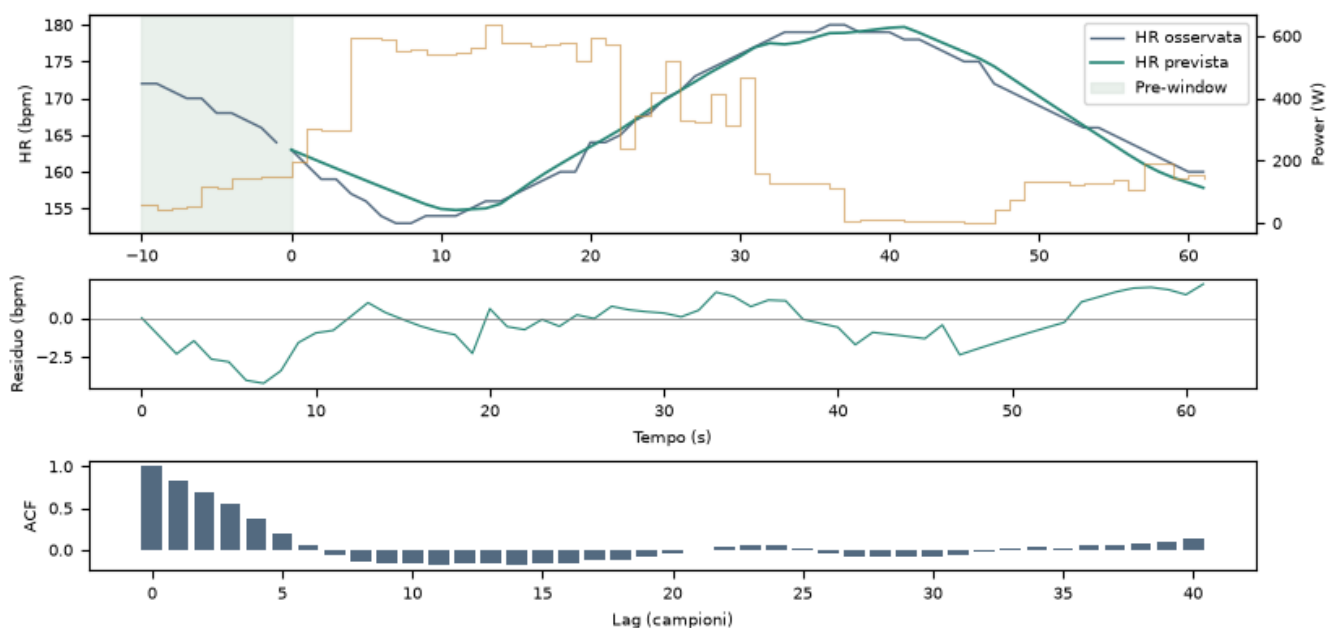
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=102.7 W; HR0=163 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=103 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.4945	1.1753	-0.39637	0.97048	0.92276

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.39637	1.4527	0.82141	0.19879



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.5504	0.46539	-0.36177 / 1.4626	84.555	no	no
L	9.6086	0.57886	8.474 / 10.743	6.0244	no	no
tau	120.33	102.42	-80.415 / 321.08	85.116	no	no
B	60	90.744	-117.86 / 237.86	151.24	si	si

rho=0.42707; Jacobian rank=4/4; cond=40057; accordo multistart=no.

Correlazioni: {"K_L": -0.9539487223581111, "K_tau": 0.9995212166852065, "K_B": -0.9999457427333427, "L_tau": -0.9553071072245846, "L_B": 0.9542843592394364, "tau_B": -0.9993120166287358}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 138.47 / 3629.6

Griglia: 42441 coppie L/tau; SSE multistart originale: 138.47; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_024-025.csv

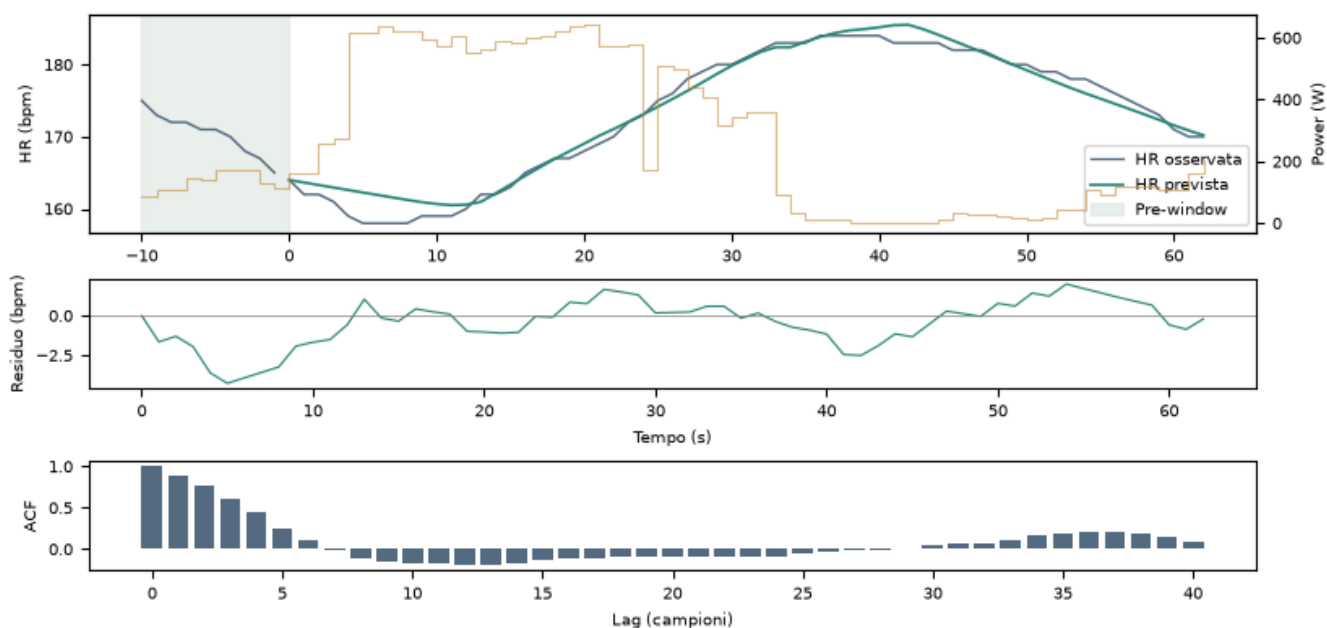
ID: 1ab958790772ba8ceb12c0ab | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=135 W; HR0=164 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=104 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.4991	1.1285	-0.42444	0.97223	0.96165
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.42444	1.4493	0.8873	0.25589	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.9381	1.4356	-1.8757 / 3.7519	153.04	no	no
L	8.7273	0.63879	7.4752 / 9.9793	7.3195	no	no
tau	297.41	453.56	-591.57 / 1186.4	152.5	no	no
B	60	172.53	-278.15 / 398.15	287.54	si	si

rho=0.17912; Jacobian rank=4/4; cond=62361; accordo multistart=no.

Correlazioni: {"K_L": -0.9325027452084493, "K_tau": 0.9997662589109708, "K_B": -0.9998028563462217, "L_tau": -0.9342886478275598, "L_B": 0.931440561967018, "tau_B": -0.9992022514500025}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 141.58 / 2289.8

Griglia: 42441 coppie L/tau; SSE multistart originale: 141.58; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250708171459__laps_002-003.csv

ID: 1b4ade4ad920633a01b0d825 | Stato: fitted

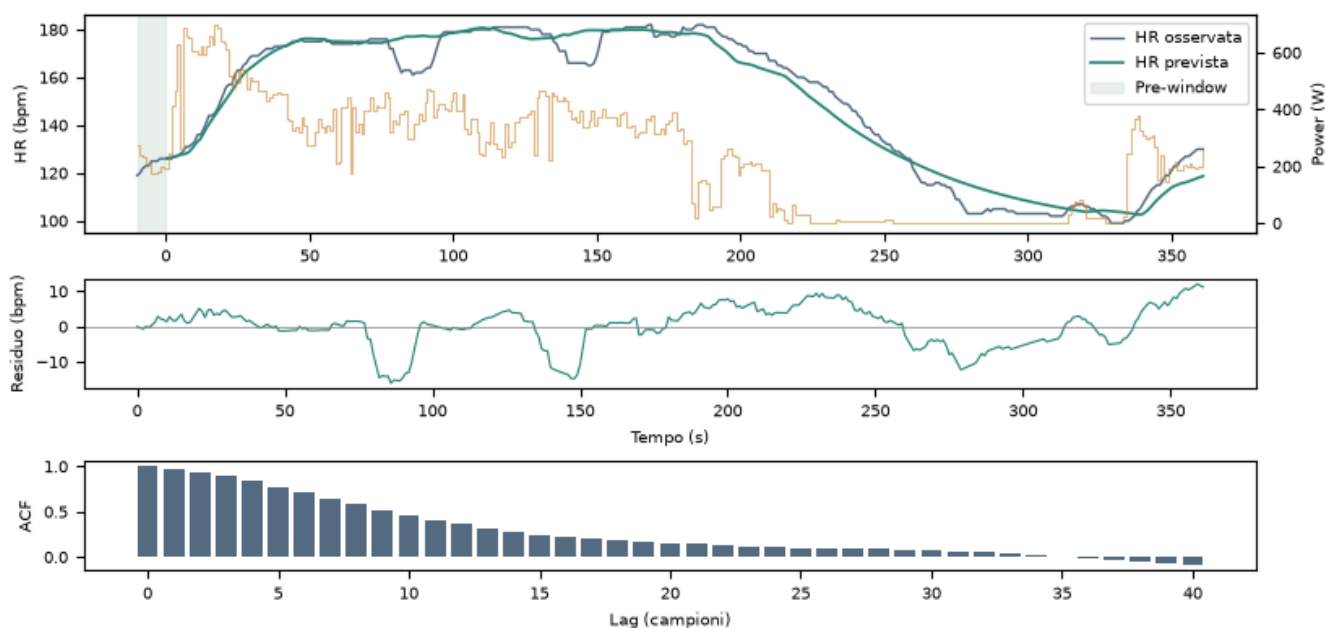
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=209.7 W; HR0=126 bpm.

HR(0) osservata fissata; B=143.16 bpm stimato; x(0)=-17.163 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
5.7862	4.4097	0.054184	0.96133	0.94225

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.054184	5.794	0.9755	0.77562



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23444	0.0049444	0.22475 / 0.24413	2.109	no	no
L	5.6908	0.73831	4.2438 / 7.1379	12.974	no	no
tau	52.83	2.1326	48.65 / 57.01	4.0368	no	no
B	143.16	0.44579	142.29 / 144.04	0.31139	no	no

rho=6.7256; Jacobian rank=4/4; cond=968.13; accordo multistart=no.

Correlazioni: {"K_L": -0.5836233117021838, "K_tau": 0.8796160615069097, "K_B": -0.5880862450606206, "L_tau": -0.7016235289966136, "L_B": 0.23868554719073898, "tau_B": -0.5643438618655493}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 12120 / 3.0877e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 12120; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250708171459__laps_006-007.csv

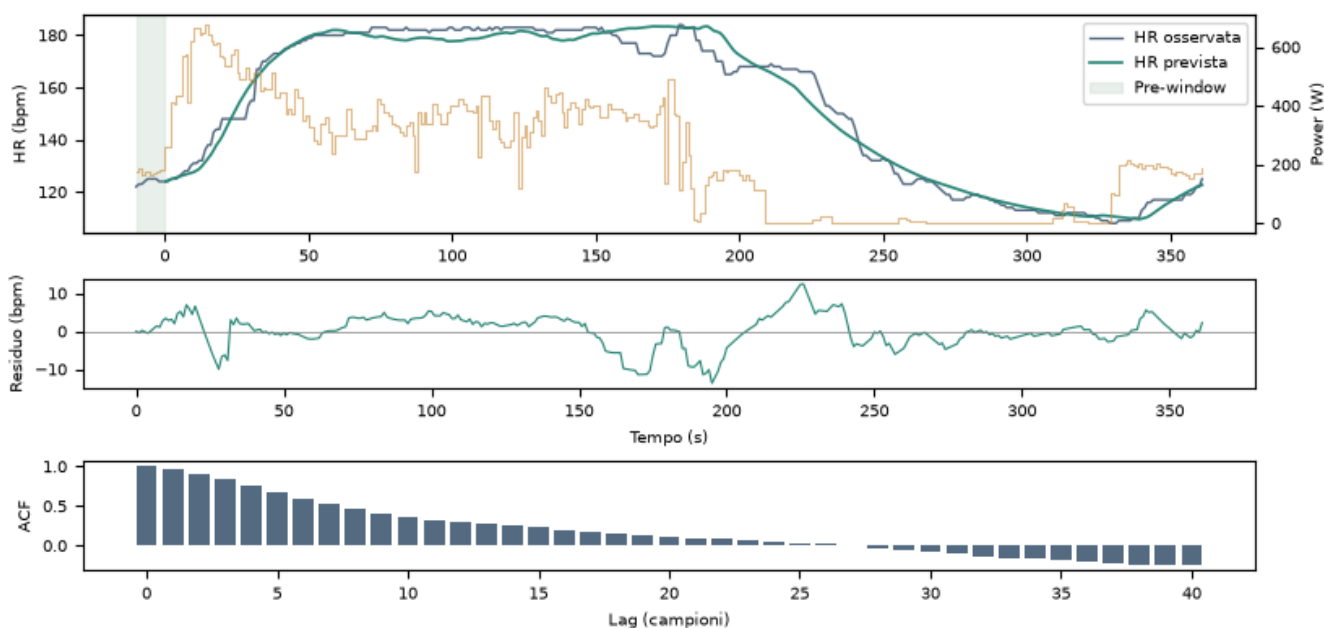
ID: 1f7a8c47b00f00f482c8b29c | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=174.4 W; HR0=124 bpm.

HR(0) osservata fissata; B=141.97 bpm stimato; x(0)=-17.967 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
4.2496	3.1519	0.10716	0.97763	0.97206
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.10716	4.2542	0.95769	0.67746	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21631	0.0031347	0.21017 / 0.22245	1.4492	no	no
L	10.36	0.553	9.276 / 11.444	5.3379	no	no
tau	45.288	1.3286	42.684 / 47.892	2.9336	no	no
B	141.97	0.33806	141.3 / 142.63	0.23813	no	no

rho=7.7424; Jacobian rank=4/4; cond=915.34; accordo multistart=no.

Correlazioni: {"K_L": -0.5723958019340911, "K_tau": 0.8480603516583144, "K_B": -0.6543800877796445, "L_tau": -0.7252699414032607, "L_B": 0.2844205542907332, "tau_B": -0.5777650258520491}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 6537.5 / 2.8353e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 6543.6; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250708171459__laps_008-009.csv

ID: 201c1b0432475dd79dca4e7d | Stato: fitted

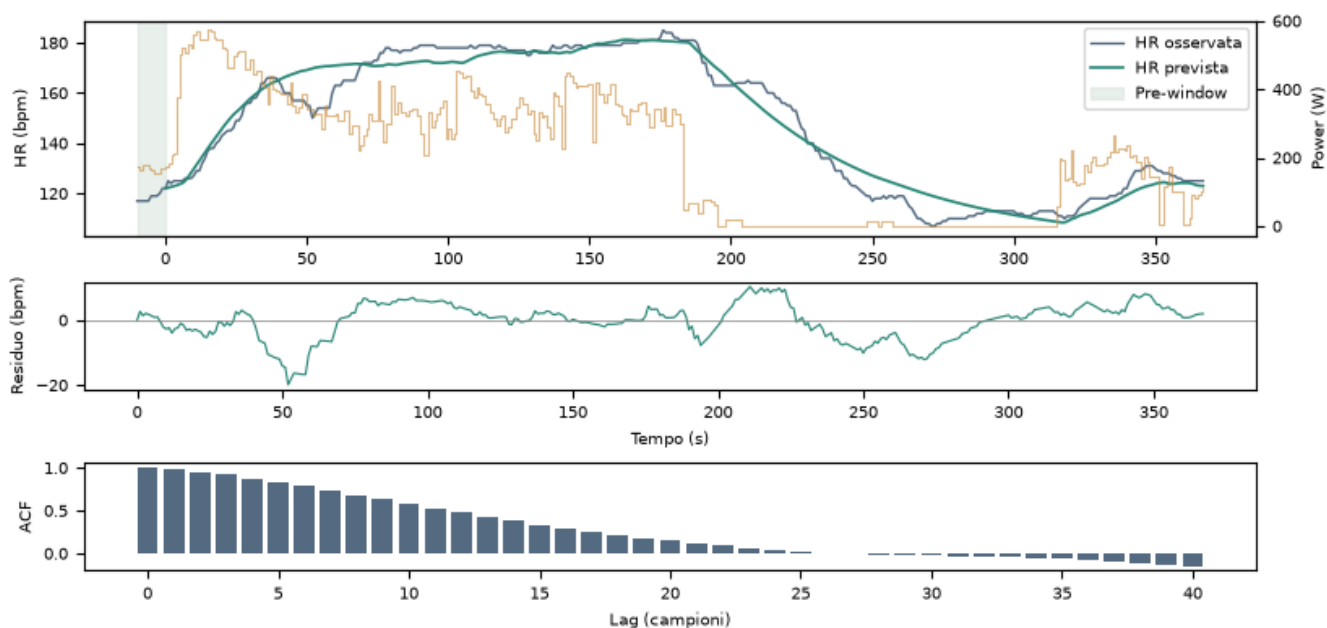
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=166.8 W; HR0=122 bpm.

HR(0) osservata fissata; B=139.38 bpm stimato; x(0)=-17.379 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
5.5477	4.2429	-0.26966	0.95693	0.93301

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.26966	5.5487	0.97838	0.83458



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23726	0.0052478	0.22697 / 0.24754	2.2119	no	no
L	2.3506	0.80406	0.77468 / 3.9266	34.206	no	no
tau	57.377	2.2256	53.015 / 61.74	3.8789	no	no
B	139.38	0.4394	138.52 / 140.24	0.31526	no	no

rho=6.3553; Jacobian rank=4/4; cond=988.18; accordo multistart=no.

Correlazioni: {"K_L": -0.598151454680679, "K_tau": 0.8806650651921514, "K_B": -0.6271598931504827, "L_tau": -0.7197539080343114, "L_B": 0.2594208308174258, "tau_B": -0.5650782647630898}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 11326 / 2.5642e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 11326; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260919065417__laps_002-003.csv

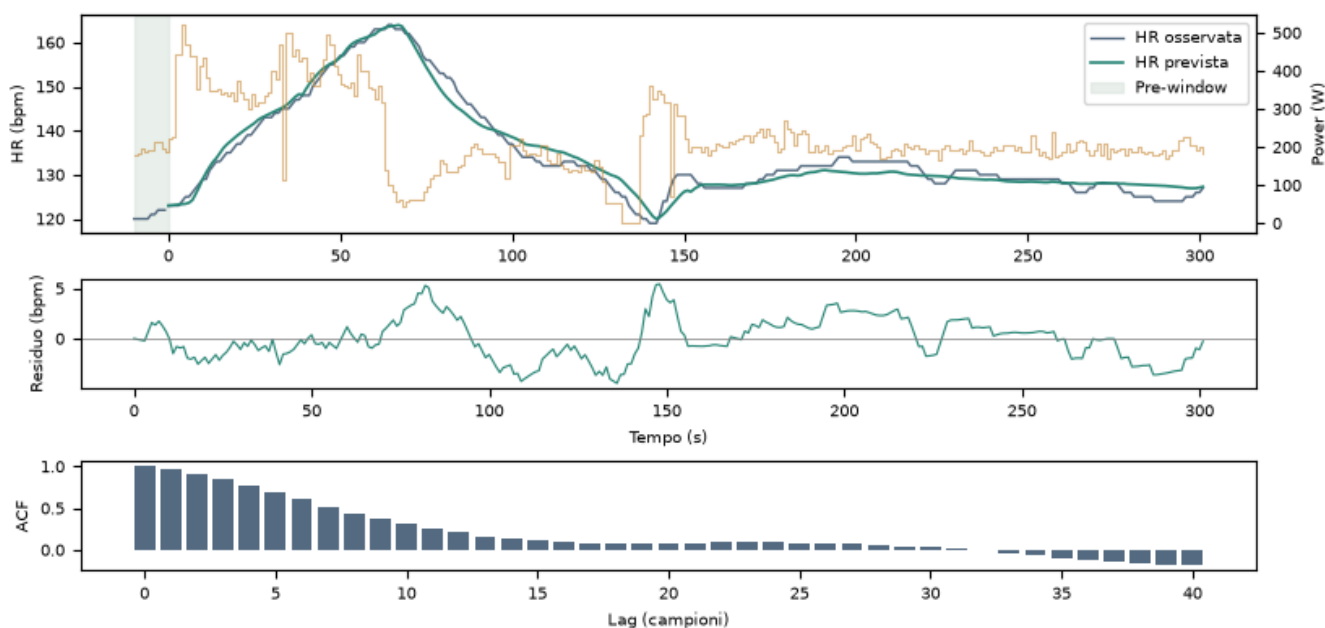
ID: 20ddf0cf07abe30440443e67 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=193.8 W; HR0=123 bpm.

HR(0) osservata fissata; B=126.91 bpm stimato; $x(0)=-3.9054$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.065	1.6786	-0.060244	0.96475	0.9778
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.060244	2.0676	0.96021	0.69328	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.28979	0.0063492	0.27734 / 0.30223	2.191	no	no
L	4.7012	0.27602	4.1602 / 5.2422	5.8713	no	no
tau	50.872	1.5042	47.924 / 53.821	2.9568	no	no
B	126.91	0.23201	126.45 / 127.36	0.18282	no	no

rho=5.8243; Jacobian rank=4/4; cond=616.22; accordo multistart=no.

Correlazioni: {"K_L": -0.47148756310733514, "K_tau": 0.8942477490896319, "K_B": -0.7994964199129296, "L_tau": -0.5500424677217625, "L_B": 0.3266339097947674, "tau_B": -0.7228235409133268}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1287.8 / 35584

Griglia: 42441 coppie L/tau; SSE multistart originale: 1287.8; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250625105513__laps_009-012.csv

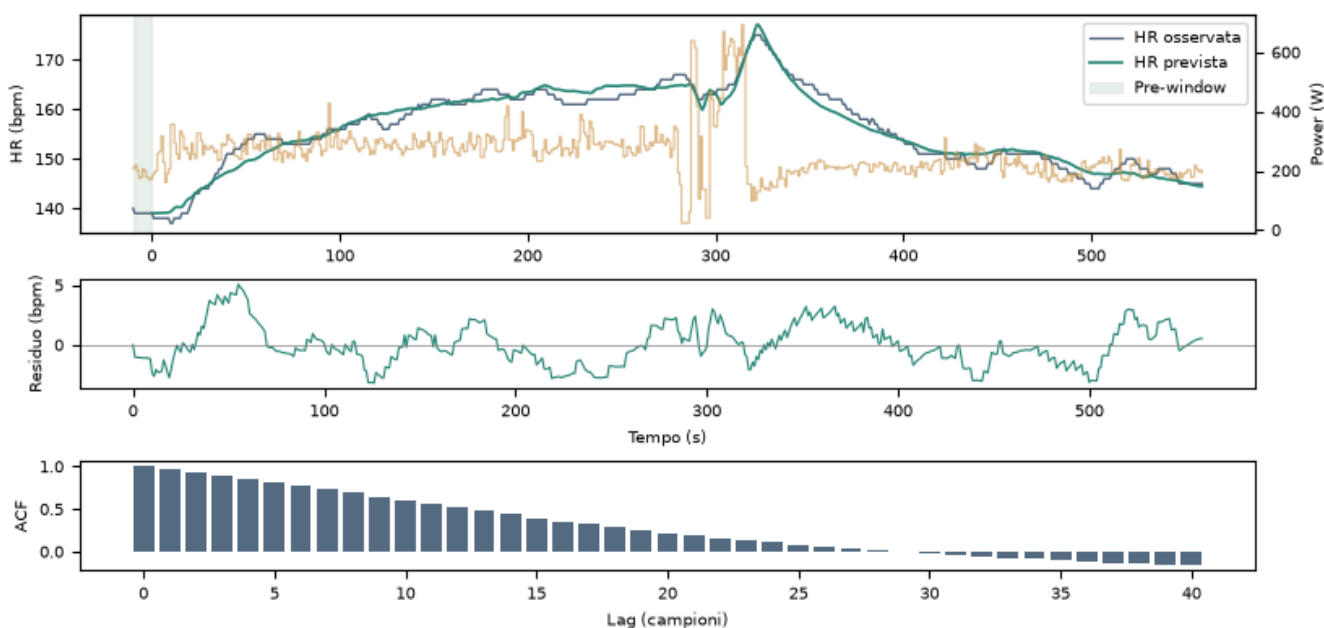
ID: 212207f51ebc4eb88edf1bc0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=190.9 W; HR0=139 bpm.

HR(0) osservata fissata; B=140.22 bpm stimato; x(0)=-1.2247 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.6992	1.3648	0.020954	0.9558	1.0021
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.020954	1.7006	0.96314	0.81461	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.24905	0.0037335	0.24173 / 0.25637	1.4991	no	no
L	6.3128	0.25838	5.8064 / 6.8192	4.0929	no	no
tau	66.865	1.2643	64.387 / 69.343	1.8908	no	no
B	140.22	0.2743	139.69 / 140.76	0.19562	no	no

rho=8.2657; Jacobian rank=4/4; cond=1272.8; accordo multistart=no.

Correlazioni: {"K_L": -0.176229126239317, "K_tau": 0.7164696564630139, "K_B": -0.9562487871135082, "L_tau": -0.3372765260132711, "L_B": 0.1479381712290638, "tau_B": -0.6597700325911963}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1616.9 / 26964

Griglia: 42441 coppie L/tau; SSE multistart originale: 1616.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_021-023.csv

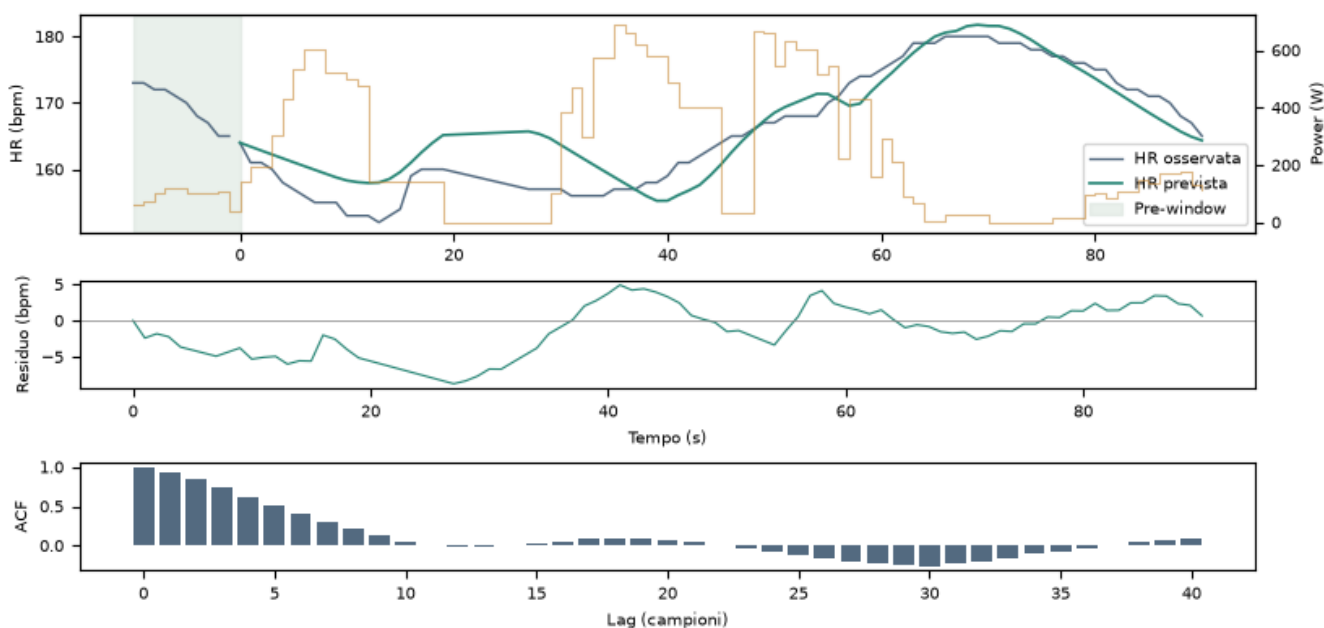
ID: 239cf3ea29845ae82a992d37 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=92.9 W; HR0=164 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=104 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.438	2.8053	-1.0337	0.85207	0.94684
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-1.0337	3.2986	0.94051	0.51886	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.73084	0.62914	-0.50227 / 1.9639	86.084	no	no
L	9.5541	0.51859	8.5377 / 10.571	5.428	no	no
tau	174.66	153.9	-126.99 / 476.3	88.116	no	no
B	60	90.819	-118.01 / 238.01	151.37	si	si

rho=0.46059; Jacobian rank=4/4; cond=20682; accordo multistart=no.

Correlazioni: {"K_L": -0.7078549755199889, "K_tau": 0.9976971185530081, "K_B": -0.99976571407595, "L_tau": -0.7044599470383692, "L_B": 0.7117174058545168, "tau_B": -0.9965874614437544}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 992.87 / 4687.8

Griglia: 42441 coppie L/tau; SSE multistart originale: 992.87; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, recording_gaps, irregular_sampling, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_068-069.csv

ID: 24774320a7b915b740dde850 | Stato: fitted

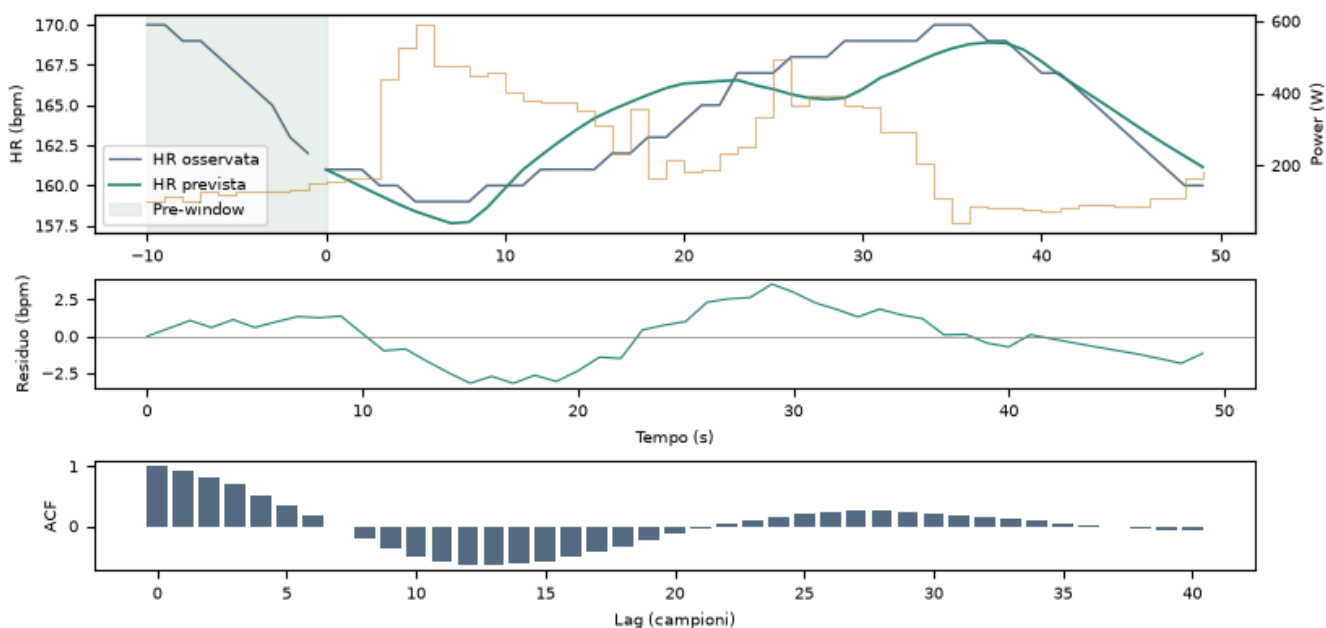
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=123.2 W; HR0=161 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=101 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.7002	1.417	0.011809	0.79256	1.0225

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.011809	1.7174	0.92554	0.3594



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.75406	2.1279	-3.4166 / 4.9248	282.19	no	no
L	4.6006	0.6747	3.2782 / 5.923	14.665	no	no
tau	187.24	525.84	-843.41 / 1217.9	280.84	no	no
B	60	293.98	-516.19 / 636.19	489.96	si	si

rho=0.23713; Jacobian rank=4/4; cond=57052; accordo multistart=no.

Correlazioni: {"K_L": -0.8449403404665354, "K_tau": 0.9993824862285795, "K_B": -0.9999473269998435, "L_tau": -0.8485178853208796, "L_B": 0.8452775656012955, "tau_B": -0.9990555237780401}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 144.53 / 447.55

Griglia: 42441 coppie L/tau; SSE multistart originale: 144.53; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_019-020.csv

ID: 24e7951d1c4d446ab88cdd4e | Stato: fitted

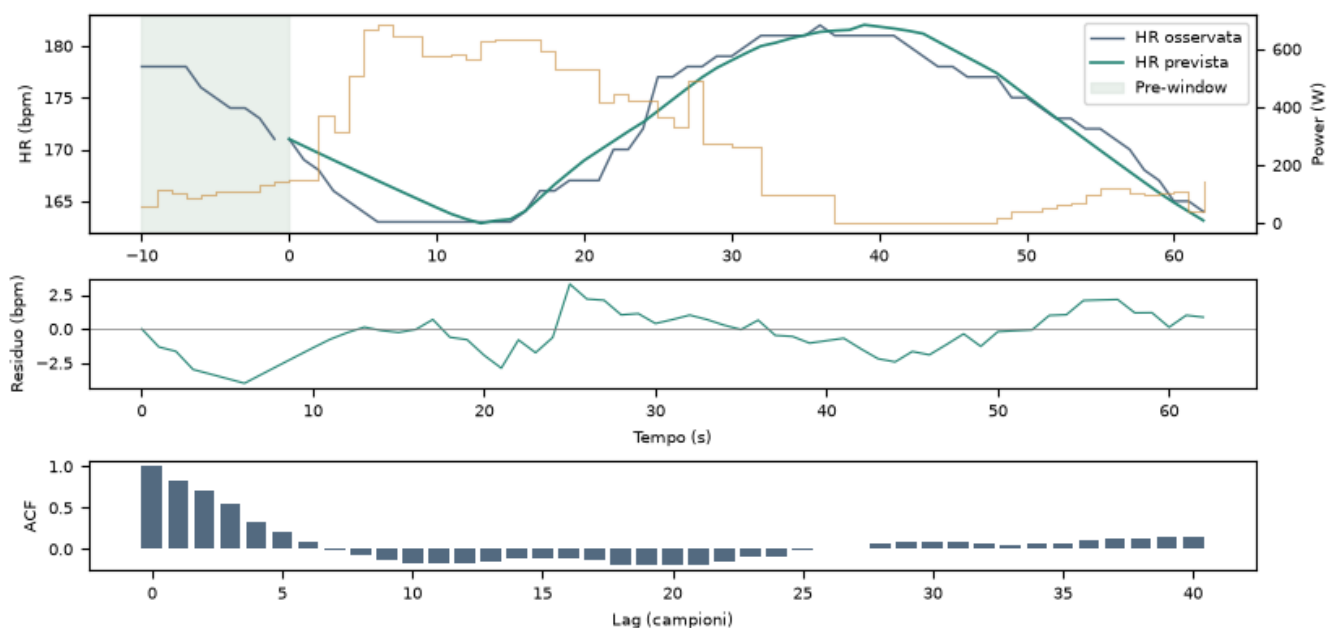
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=107.2 W; HR0=171 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=111 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.623	1.2788	-0.42131	0.94092	1.009

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.42131	1.5799	0.83803	0.20361



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.55462	1.0652	-1.5331 / 2.6423	192.05	no	no
L	10.897	0.89967	9.1341 / 12.661	8.2558	no	no
tau	162.75	314.74	-454.13 / 779.64	193.39	no	no
B	60	216.1	-363.57 / 483.57	360.17	no	si

rho=0.31399; Jacobian rank=4/4; cond=79500; accordo multistart=no.

Correlazioni: {"K_L": -0.965148697659342, "K_tau": 0.9998372541097109, "K_B": -0.9999851203637004, "L_tau": -0.9656909265803799, "L_B": 0.9653799531698898, "tau_B": -0.9997940292580222}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 165.94 / 2520.6

Griglia: 42441 coppie L/tau; SSE multistart originale: 165.94; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_009-010.csv

ID: 26a8c904c169cd2064ccba9e | Stato: fitted

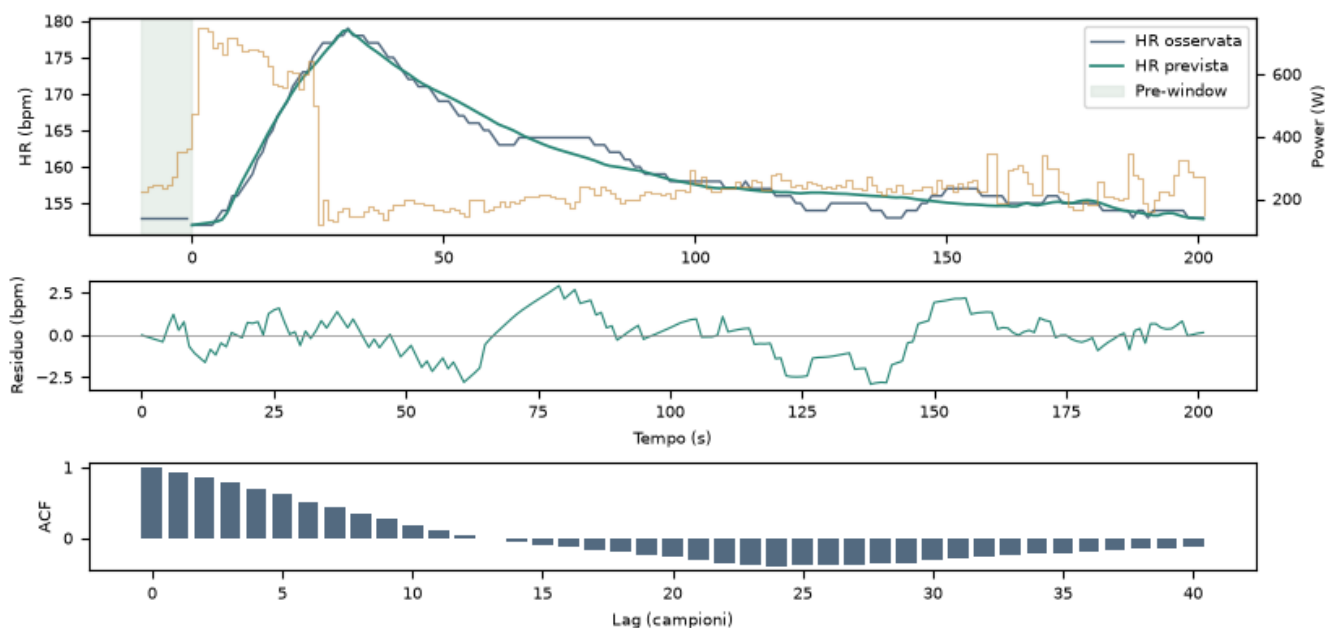
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=277.1 W; HR0=152 bpm.

HR(0) osservata fissata; B=162.3 bpm stimato; x(0)=-10.3 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.2476	0.96339	-0.0054132	0.97086	0.99135

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.0054132	1.2507	0.91783	0.61508



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.30691	0.016605	0.27436 / 0.33945	5.4104	no	no
L	5.6906	0.24241	5.2155 / 6.1657	4.2599	no	no
tau	101.55	6.2014	89.395 / 113.7	6.1068	no	no
B	162.3	0.15601	161.99 / 162.61	0.096127	no	no

rho=1.9233; Jacobian rank=4/4; cond=2124.1; accordo multistart=no.

Correlazioni: {"K_L": -0.484623942060502, "K_tau": 0.9843680537171852, "K_B": 0.30697283360760963, "L_tau": -0.49526412742423137, "L_B": -0.3418015048025866, "tau_B": 0.30409209953489336}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 314.41 / 10402

Griglia: 42441 coppie L/tau; SSE multistart originale: 314.41; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_020-021.csv

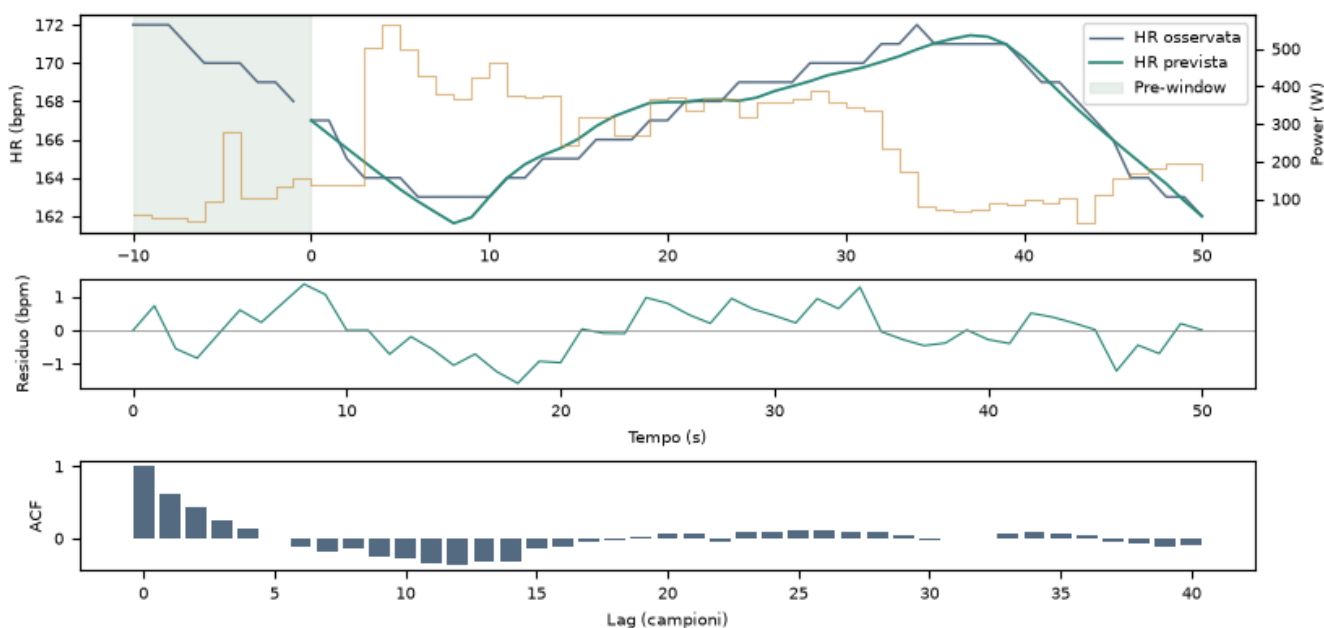
ID: 2c5eda7172b63991a6053062 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=106.4 W; HR0=167 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=107 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.67689	0.5379	0.0043722	0.94565	0.98267
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.0043722	0.68362	0.62797	-0.004208	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.59551	0.75363	-0.8816 / 2.0726	126.55	no	no
L	5.4088	0.30075	4.8194 / 5.9983	5.5603	no	no
tau	145.88	185.93	-218.53 / 510.3	127.45	no	no
B	60	135.61	-205.79 / 325.79	226.01	si	si

rho=0.30566; Jacobian rank=4/4; cond=76271; accordo multistart=no.

Correlazioni: {"K_L": -0.8852607385462687, "K_tau": 0.9996038259754939, "K_B": -0.9999890800074985, "L_tau": -0.8857747869866325, "L_B": 0.886166515840378, "tau_B": -0.9995726059485074}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.367 / 415.41

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.367; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_010-011.csv

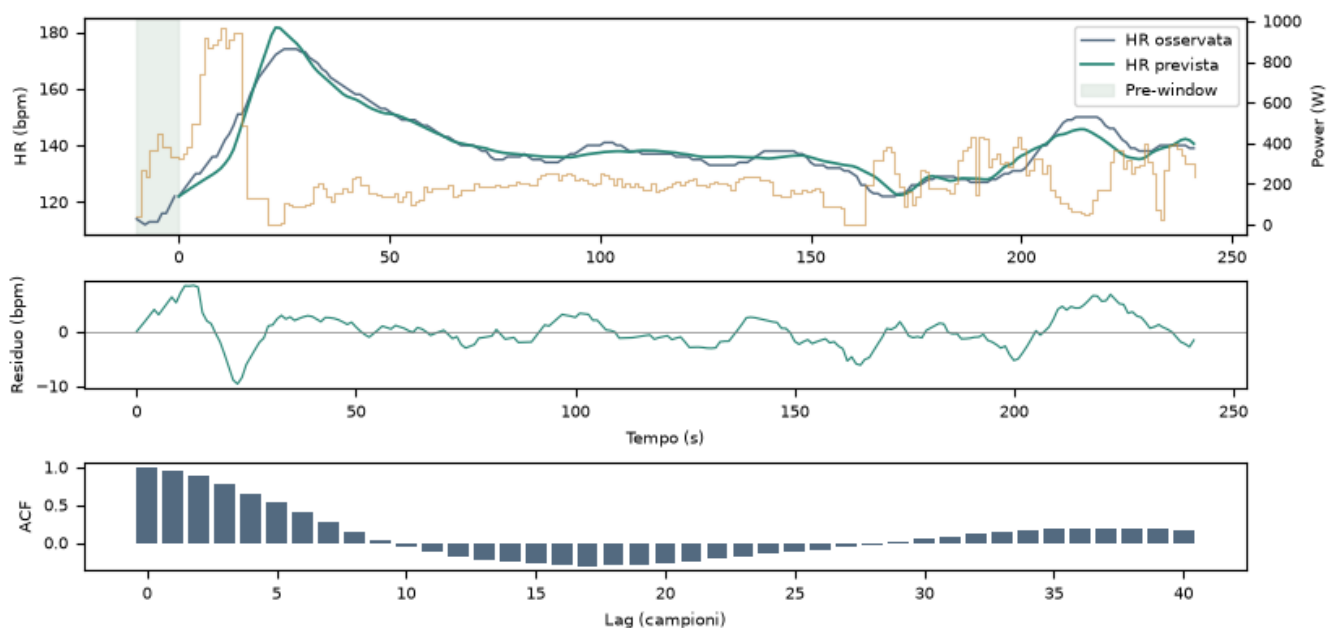
ID: 3156c25d43f4239bdb596f18 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=315.2 W; HR0=122 bpm.

HR(0) osservata fissata; B=174.64 bpm stimato; x(0)=-52.636 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.9947	2.2899	0.19281	0.93737	1.1464
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.19281	2.9947	0.95588	0.52961	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.36258	0.015235	0.33272 / 0.39244	4.2017	no	no
L	7.5017	0.19886	7.1119 / 7.8914	2.6509	no	no
tau	50.913	2.4533	46.105 / 55.721	4.8186	no	no
B	174.64	1.4893	171.72 / 177.55	0.8528	no	no

rho=4.5862; Jacobian rank=4/4; cond=1276.1; accordo multistart=no.

Correlazioni: {"K_L": -0.4917247202368018, "K_tau": 0.9502953906562079, "K_B": 0.9872344502534929, "L_tau": -0.48162420297888175, "L_B": -0.5056489514434682, "tau_B": 0.9410473098403985}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 2170.3 / 32647

Griglia: 42441 coppie L/tau; SSE multistart originale: 2170.3; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_018-019.csv

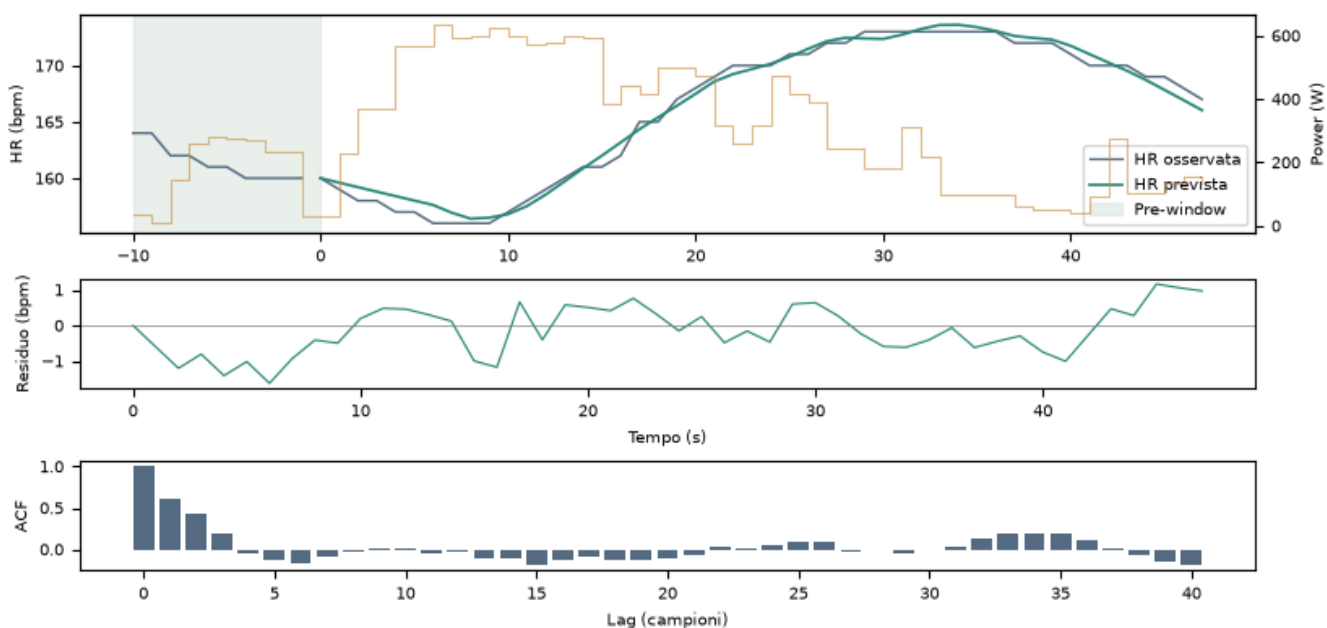
ID: 34527474ec4cefb8690d541 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=177.3 W; HR0=160 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=100 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.68847	0.58543	-0.14777	0.98771	1.0118
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.14777	0.67954	0.60913	-0.13525	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.90878	0.99973	-1.0507 / 2.8682	110.01	no	no
L	6.4392	0.33726	5.7781 / 7.1002	5.2377	no	no
tau	249.44	275.05	-289.66 / 788.54	110.27	no	no
B	60	115.87	-167.1 / 287.1	193.11	si	si

rho=0.16261; Jacobian rank=4/4; cond=58450; accordo multistart=no.

Correlazioni: {"K_L": -0.922054404084581, "K_tau": 0.9996591849692338, "K_B": -0.9998003698581939, "L_tau": -0.9232459250900031, "L_B": 0.922068951001164, "tau_B": -0.9989985734533031}

Multistart: 16 tentativi, 14 convergenti; SSE min/max: 22.752 / 651.68

Griglia: 42441 coppie L/tau; SSE multistart originale: 22.752; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_003-004.csv

ID: 38bf5e52e0c89cac13997cfc | Stato: fitted

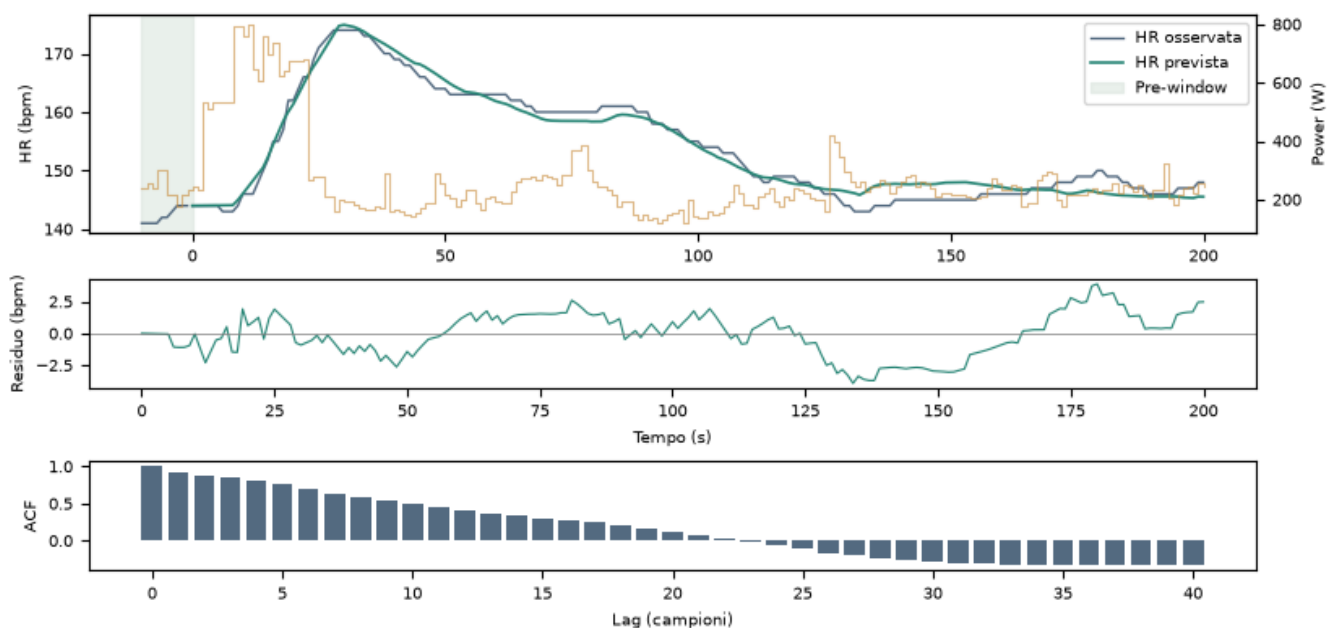
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=239.8 W; HR0=144 bpm.

HR(0) osservata fissata; B=144.98 bpm stimato; $x(0)=-0.98483$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.6873	1.3877	-0.04769	0.96551	0.99715

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.04769	1.6908	0.93202	0.7597



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.34552	0.0184	0.30946 / 0.38159	5.3253	no	no
L	6.251	0.25159	5.7579 / 6.7441	4.0248	no	no
tau	87.28	5.3418	76.81 / 97.75	6.1203	no	no
B	144.98	0.59418	143.82 / 146.15	0.40982	no	no

rho=2.2199; Jacobian rank=4/4; cond=1678.6; accordo multistart=no.

Correlazioni: {"K_L": -0.4694698273457047, "K_tau": 0.9787842300402829, "K_B": -0.9502995072305187, "L_tau": -0.4981912296268546, "L_B": 0.40750616587301586, "tau_B": -0.9303246771523583}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 572.22 / 15812

Griglia: 42441 coppie L/tau; SSE multistart originale: 572.22; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_076-077.csv

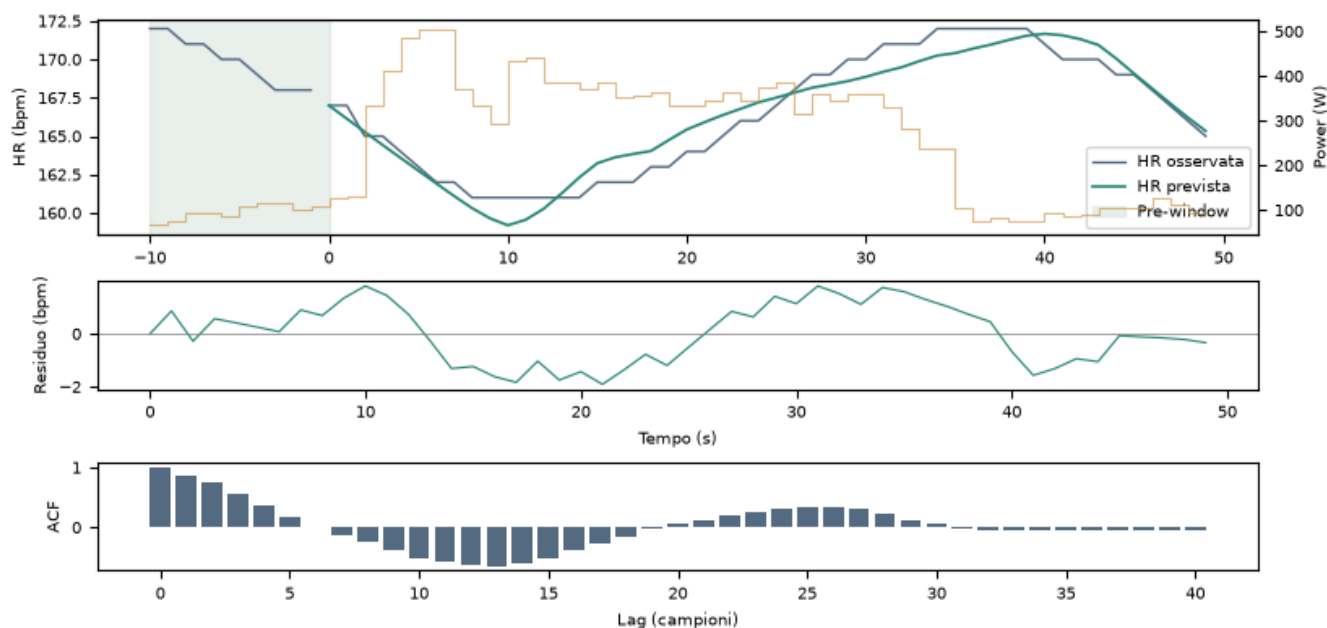
ID: 3d72a3ba651acda8c1791da7 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=96.5 W; HR0=167 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=107 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1036	0.95133	0.033939	0.91685	1.1342
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.033939	1.1143	0.86473		0.17846



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.56629	0.79211	-0.98624 / 2.1188	139.88	no	no
L	7.7967	0.52393	6.7698 / 8.8236	6.7199	no	no
tau	123.37	173.49	-216.68 / 463.42	140.63	no	no
B	60	148.61	-231.28 / 351.28	247.69	no	si

rho=0.33398; Jacobian rank=4/4; cond=55596; accordo multistart=no.

Correlazioni: {"K_L": -0.8942197512824597, "K_tau": 0.999366516350577, "K_B": -0.9999710379828403, "L_tau": -0.8878081016401383, "L_B": 0.8964150482126643, "tau_B": -0.9992141062222683}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 60.895 / 731.95

Griglia: 42441 coppie L/tau; SSE multistart originale: 60.895; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_006-007.csv

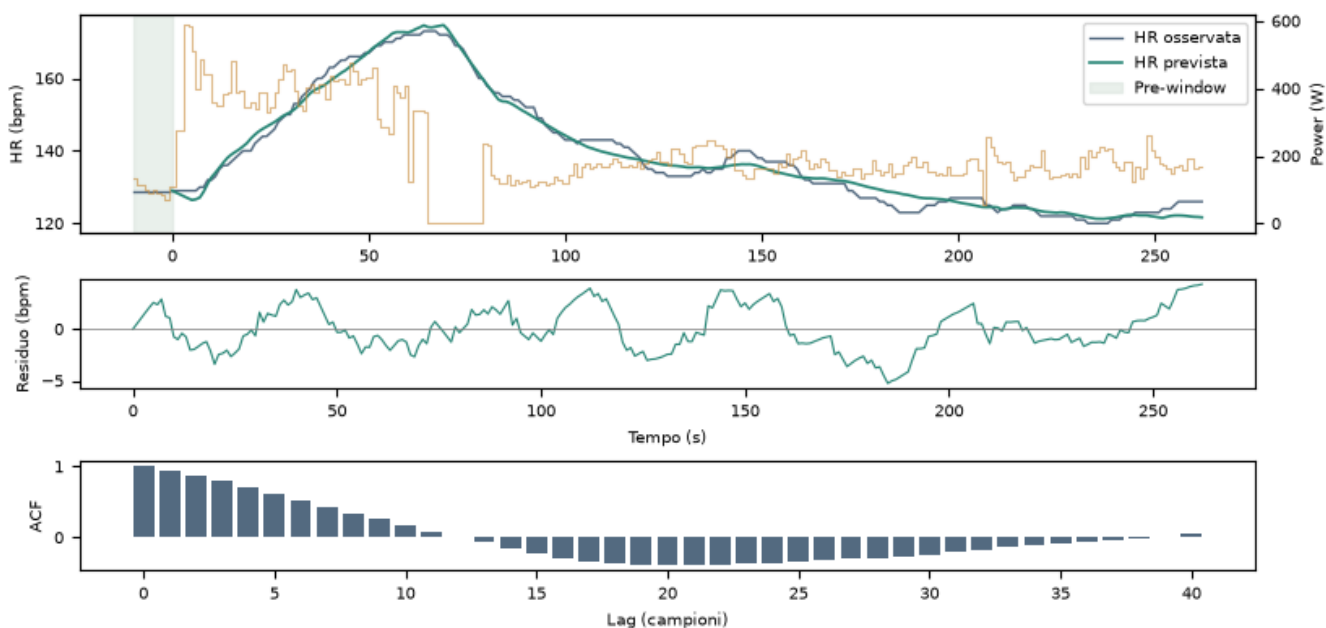
ID: 3f32a678f8f61184bbaa1dc0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=100.1 W; HR0=129 bpm.

HR(0) osservata fissata; B=84.573 bpm stimato; x(0)=44.427 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.1015	1.7529	0.02657	0.98222	1.0084
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.02657	2.1053	0.93878	0.62055	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.4617	0.010929	0.44028 / 0.48312	2.3671	no	no
L	4.3011	0.2261	3.858 / 4.7443	5.2568	no	no
tau	83.551	2.612	78.431 / 88.67	3.1262	no	no
B	84.573	1.4019	81.825 / 87.321	1.6577	no	no

rho=3.0843; Jacobian rank=4/4; cond=2101.4; accordo multistart=no.

Correlazioni: {"K_L": -0.4141937304147623, "K_tau": 0.9450011643537646, "K_B": -0.9915598407834862, "L_tau": -0.51395810887052, "L_B": 0.4199927052659565, "tau_B": -0.9473165334545126}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1161.5 / 64546

Griglia: 42441 coppie L/tau; SSE multistart originale: 1161.5; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_013-014.csv

ID: 4300d93cc4bc4e3f7e8a0770 | Stato: fitted

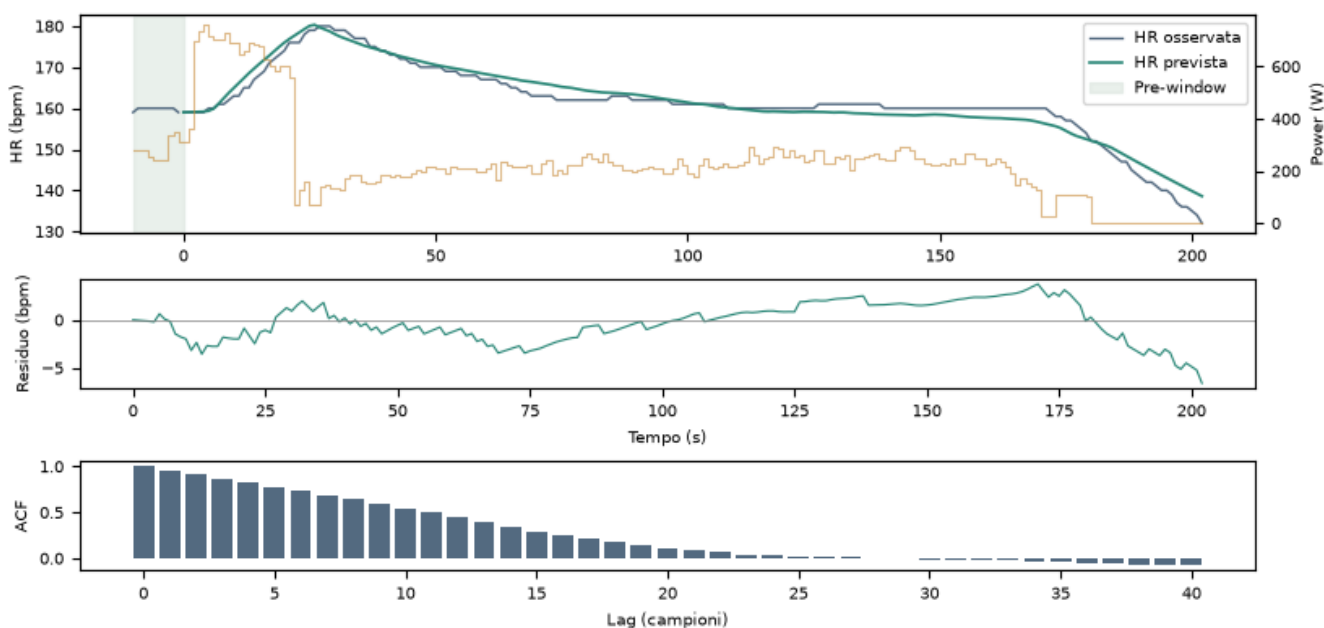
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=281.7 W; HR0=159 bpm.

HR(0) osservata fissata; B=166.21 bpm stimato; x(0)=-7.2136 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0446	1.6858	-0.24098	0.94907	0.86919

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.24098	2.0353	0.94606	0.78228



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.47628	0.064525	0.34981 / 0.60275	13.548	no	no
L	3.6879	0.38127	2.9406 / 4.4352	10.338	no	no
tau	170.87	26.248	119.43 / 222.32	15.361	no	no
B	166.21	0.38492	165.46 / 166.97	0.23158	no	no

rho=1.1606; Jacobian rank=4/4; cond=3778.8; accordo multistart=no.

Correlazioni: {"K_L": -0.15649957691161173, "K_tau": 0.9928839625954322, "K_B": 0.2956902971946243, "L_tau": -0.18383421147061257, "L_B": -0.4216296579970326, "tau_B": 0.2691333130369706}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 848.59 / 16632

Griglia: 42441 coppie L/tau; SSE multistart originale: 848.59; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_007-008.csv

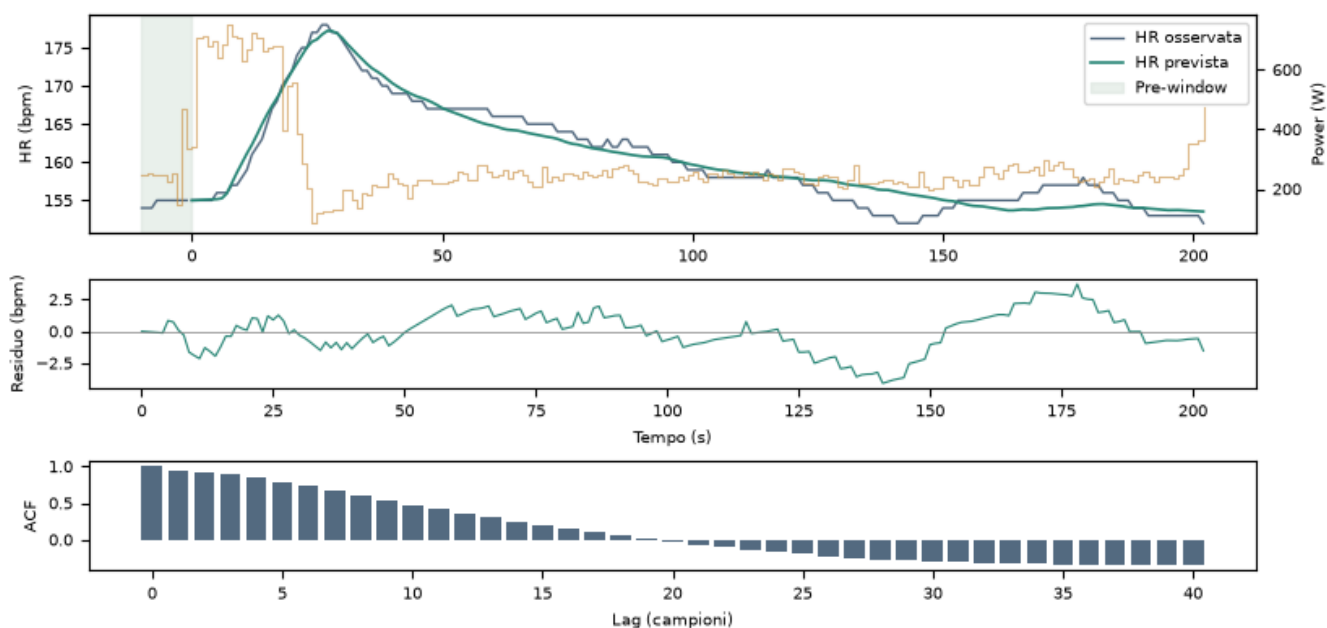
ID: 435427565eac77eef857fb21 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=267.4 W; HR0=155 bpm.

HR(0) osservata fissata; B=157.42 bpm stimato; x(0)=-2.4219 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.5667	1.2413	-0.018496	0.94458	0.90961
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.018496	1.5705	0.94997	0.78378	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23091	0.012543	0.20632 / 0.25549	5.4321	no	no
L	5.6581	0.31302	5.0446 / 6.2716	5.5322	no	no
tau	74.388	4.9469	64.693 / 84.084	6.6501	no	no
B	157.42	0.28698	156.86 / 157.98	0.1823	no	no

rho=2.6394; Jacobian rank=4/4; cond=1526.9; accordo multistart=no.

Correlazioni: {"K_L": -0.40517658084708996, "K_tau": 0.96394168027526, "K_B": -0.8240877322568634, "L_tau": -0.4322761611806583, "L_B": 0.253778370164689, "tau_B": -0.8176133368881334}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 498.29 / 8833.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 498.29; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_015-016.csv

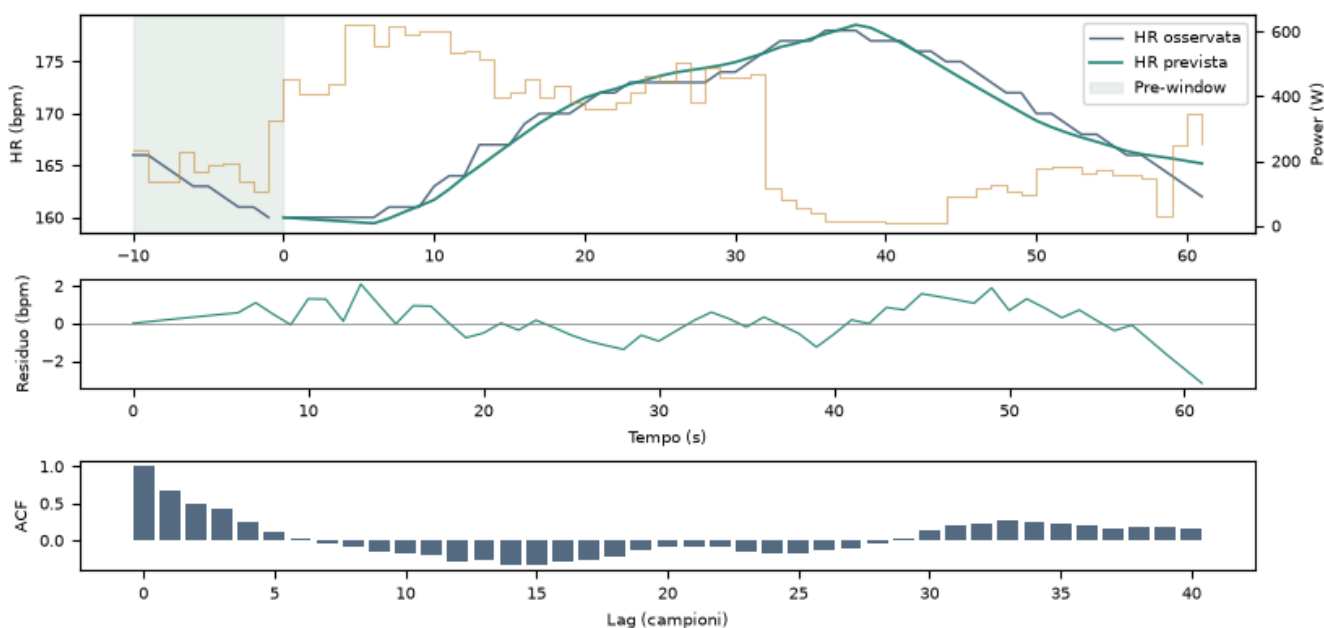
ID: 467e3d3507204e47a1d06103 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=185.5 W; HR0=160 bpm.

HR(0) osservata fissata; B=154.05 bpm stimato; x(0)=5.9525 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.96766	0.72838	0.10111	0.97237	1.0617
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.10111	0.97022	0.67883	0.11543	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.18429	0.056125	0.074286 / 0.29429	30.454	no	no
L	6.315	0.47002	5.3937 / 7.2362	7.443	no	no
tau	60.331	17.89	25.266 / 95.395	29.653	no	no
B	154.05	4.8531	144.54 / 163.56	3.1504	no	no

rho=0.90642; Jacobian rank=4/4; cond=10593; accordo multistart=no.

Correlazioni: {"K_L": -0.872757011974183, "K_tau": 0.9961627458392447, "K_B": -0.9948584855138813, "L_tau": -0.8907076028735877, "L_B": 0.8474274061725762, "tau_B": -0.9841287331876654}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 58.055 / 1078.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 58.261; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_062-063.csv

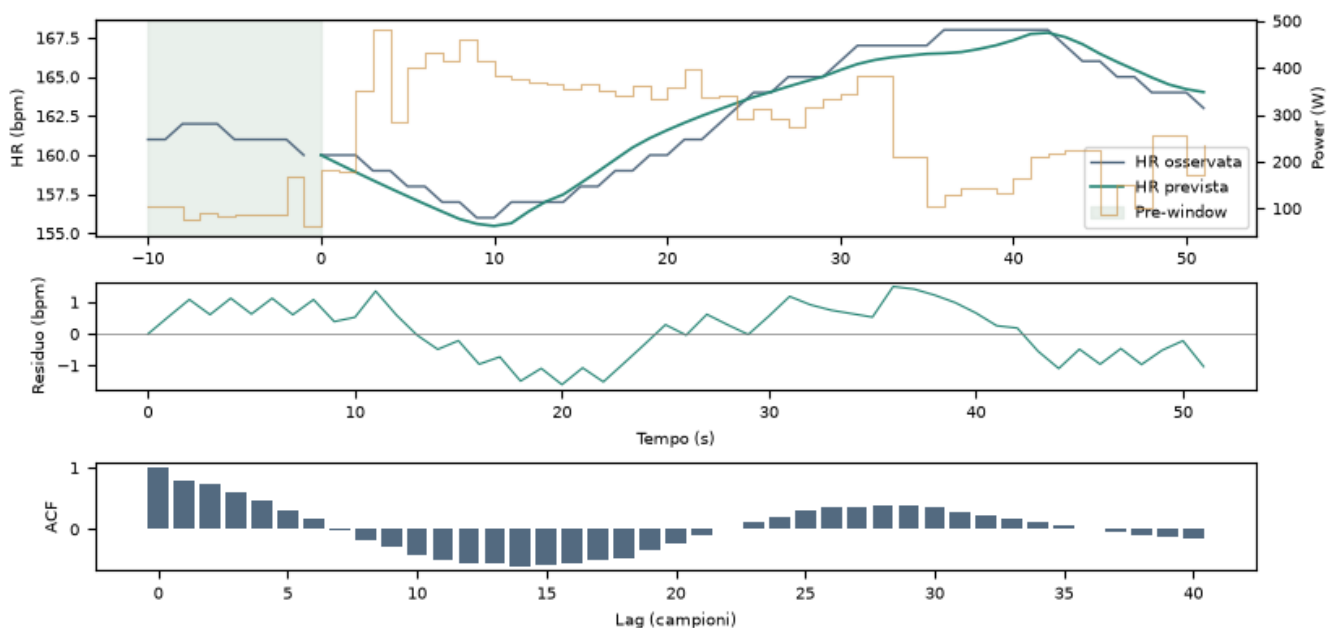
ID: 47eea1ca7bd657074778883b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=95.7 W; HR0=160 bpm.

HR(0) osservata fissata; B=132.99 bpm stimato; x(0)=27.014 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.85059	0.73762	0.096404	0.95547	1.0284
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.096404	0.85336	0.80281	0.31513	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.18771	0.062815	0.064595 / 0.31083	33.463	no	no
L	8.5222	0.52729	7.4887 / 9.5557	6.1873	no	no
tau	48.838	16.809	15.891 / 81.784	34.419	no	no
B	132.99	9.8753	113.63 / 152.34	7.4258	no	no

rho=0.86978; Jacobian rank=4/4; cond=13030; accordo multistart=no.

Correlazioni: {"K_L": -0.8897353839030228, "K_tau": 0.9893632839767252, "K_B": -0.9985270910961273, "L_tau": -0.8672244519223713, "L_B": 0.900627379534499, "tau_B": -0.9818716279938824}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 37.623 / 412.93

Griglia: 42441 coppie L/tau; SSE multistart originale: 37.623; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_017-018.csv

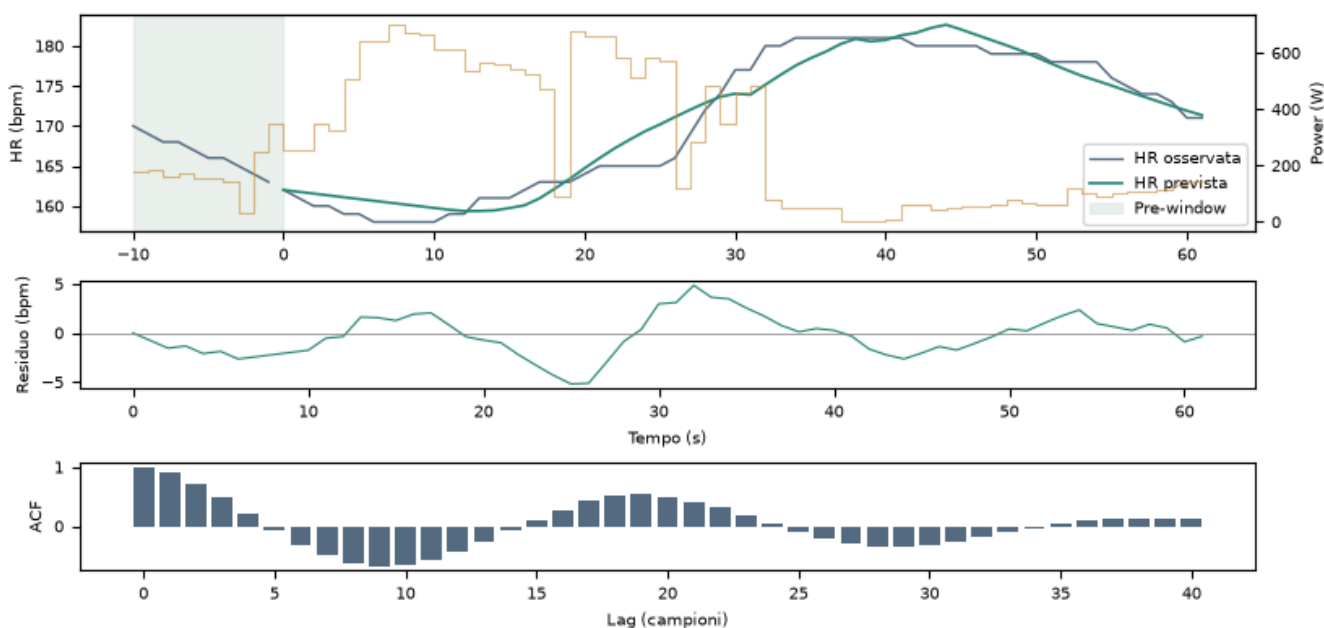
ID: 48e72ea2c9db441feda919d0 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=178 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0791	1.6579	-0.28804	0.9421	1.0121
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.28804	2.0759	0.89937	-0.06065	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	1.3709	3.2094	-4.9195 / 7.6613	234.11	no	no
L	11.789	0.70527	10.407 / 13.172	5.9824	no	no
tau	449.29	1051.5	-1611.6 / 2510.1	234.02	no	no
B	60	256.04	-441.84 / 561.84	426.74	si	si

rho=0.10953; Jacobian rank=4/4; cond=55632; accordo multistart=no.

Correlazioni: {"K_L": -0.8679776499391167, "K_tau": 0.9997059861744098, "K_B": -0.9994211528884323, "L_tau": -0.8691513823334497, "L_B": 0.8673110064980538, "tau_B": -0.998382663009771}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 268.02 / 1722.6

Griglia: 42441 coppie L/tau; SSE multistart originale: 268.02; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuels, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_013-014.csv

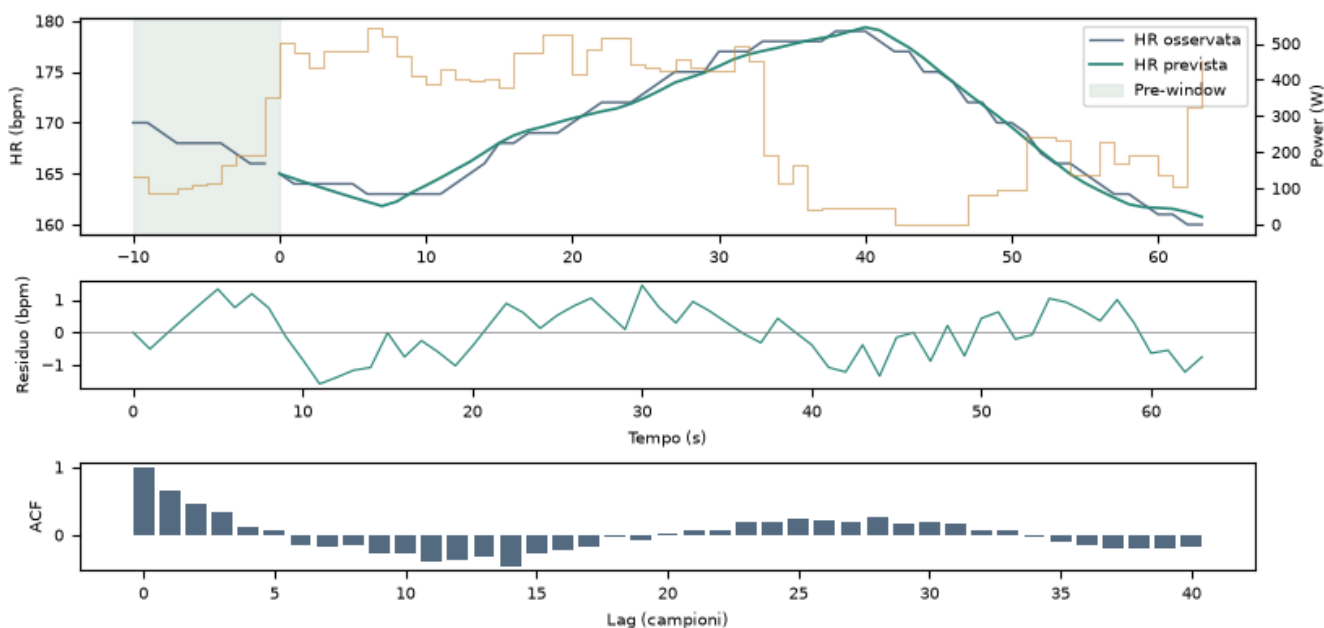
ID: 49069ab3343d75d7654ee1ec | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=153.3 W; HR0=165 bpm.

HR(0) osservata fissata; B=144.47 bpm stimato; x(0)=20.533 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.75583	0.63107	0.011593	0.98385	0.98049
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.011593	0.76172	0.65231	0.058786	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.16405	0.029831	0.10558 / 0.22251	18.184	no	no
L	7.3792	0.31987	6.7522 / 8.0061	4.3347	no	no
tau	41.573	7.9536	25.983 / 57.162	19.132	no	no
B	144.47	4.7102	135.24 / 153.7	3.2604	no	no

rho=1.3379; Jacobian rank=4/4; cond=9694.5; accordo multistart=no.

Correlazioni: {"K_L": -0.8960508548966035, "K_tau": 0.9951778317759562, "K_B": -0.9991422081483201, "L_tau": -0.9117515270986112, "L_B": 0.8922153596425204, "tau_B": -0.9929146302683579}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 36.562 / 1898

Griglia: 42441 coppie L/tau; SSE multistart originale: 37.283; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260523080618__laps_008-009.csv

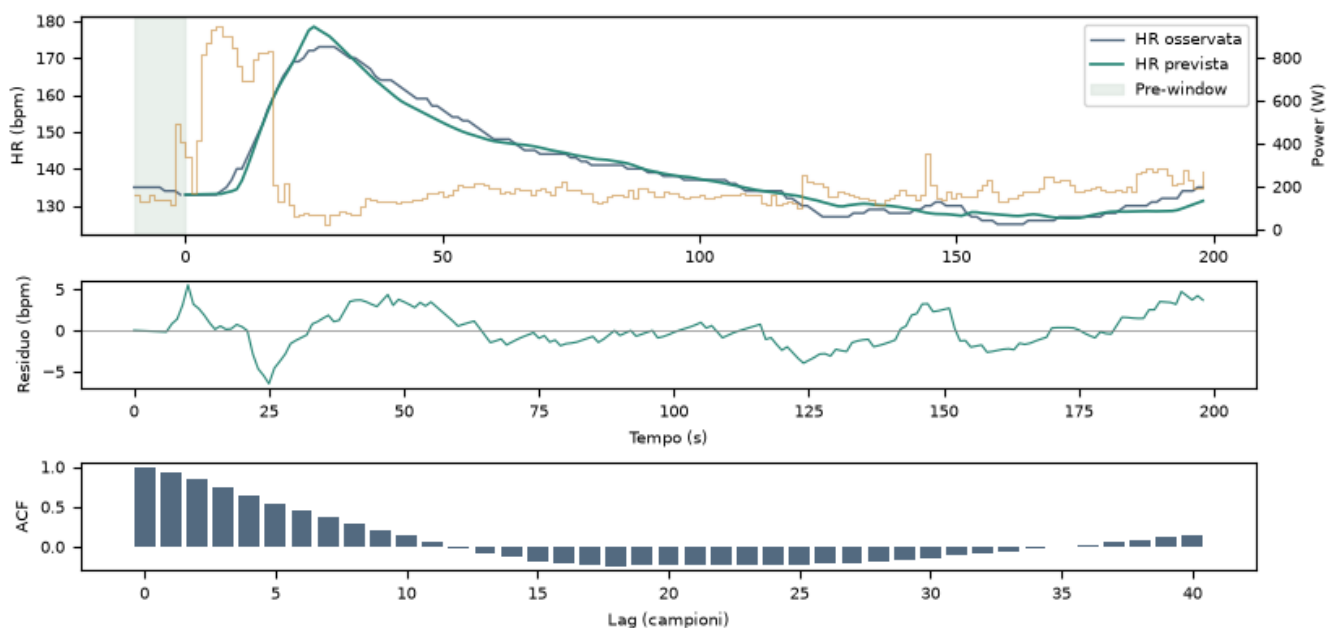
ID: 4aa1b5bbb6dd5a0951c1e32 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=201.7 W; HR0=133 bpm.

HR(0) osservata fissata; B=134.87 bpm stimato; x(0)=-1.8681 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.1149	1.6564	0.081811	0.97648	1.0798
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.081811	2.1186	0.93789	0.54596	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.33979	0.0097457	0.32069 / 0.35889	2.8681	no	no
L	7.4885	0.17593	7.1437 / 7.8334	2.3494	no	no
tau	60.663	2.2178	56.316 / 65.009	3.656	no	no
B	134.87	0.30515	134.27 / 135.47	0.22626	no	no

rho=3.1405; Jacobian rank=4/4; cond=656.4; accordo multistart=no.

Correlazioni: {"K_L": -0.39958364701417104, "K_tau": 0.9358143185821108, "K_B": -0.7267334627960237, "L_tau": -0.43372376404171475, "L_B": 0.24922621909065837, "tau_B": -0.7395494196135128}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 890.06 / 37591

Griglia: 42441 coppie L/tau; SSE multistart originale: 890.06; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_072-073.csv

ID: 4b3e2f428247d572c53644aa | Stato: fitted

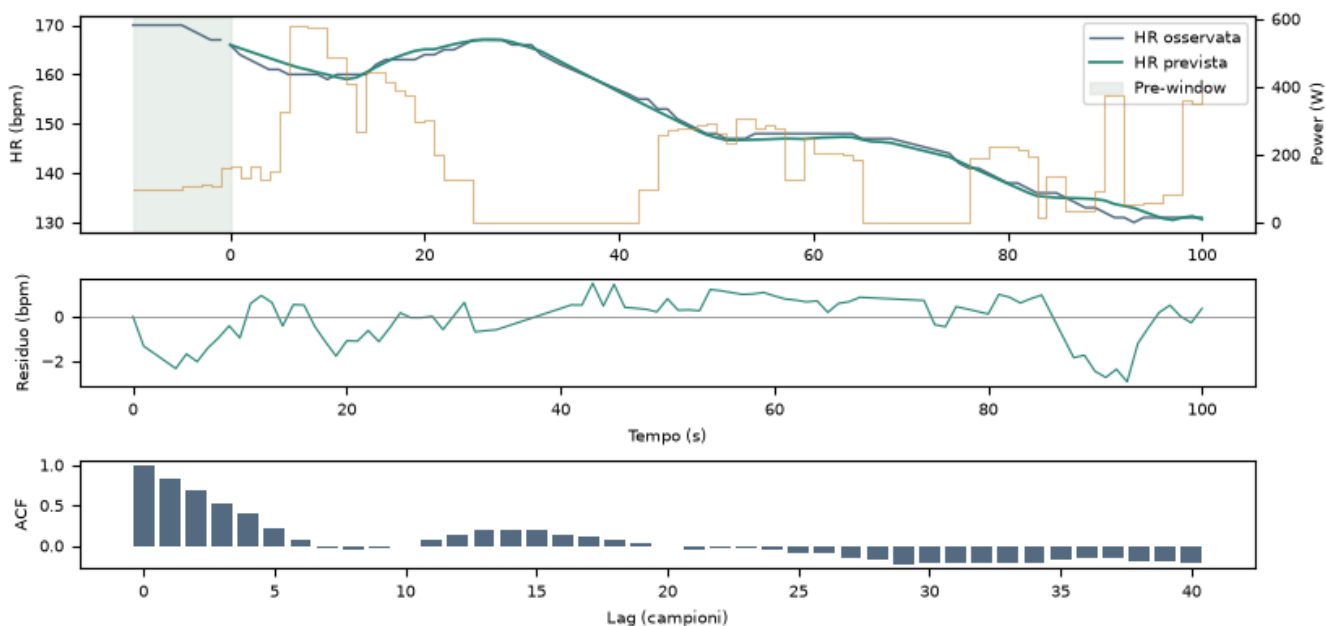
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=108.1 W; HR0=166 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=106 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.045	0.83003	-0.13181	0.99245	0.98805

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.13181	1.0424	0.82426	0.22422



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.55904	0.0584	0.44458 / 0.67351	10.446	no	no
L	6.8182	0.19269	6.4405 / 7.1959	2.8261	no	no
tau	156.87	19.632	118.39 / 195.35	12.515	no	no
B	60	10.927	38.583 / 81.417	18.212	si	si

rho=0.59399; Jacobian rank=4/4; cond=4645.5; accordo multistart=no.

Correlazioni: {"K_L": 0.08198505922206495, "K_tau": 0.9733392640875612, "K_B": -0.9892882840701708, "L_tau": 0.028595552214026137, "L_B": -0.04521014037478921, "tau_B": -0.9954021019358369}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 98.279 / 1917.6

Griglia: 42441 coppie L/tau; SSE multistart originale: 98.279; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, recording_gaps, irregular_sampling, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250715165154__laps_008-009.csv

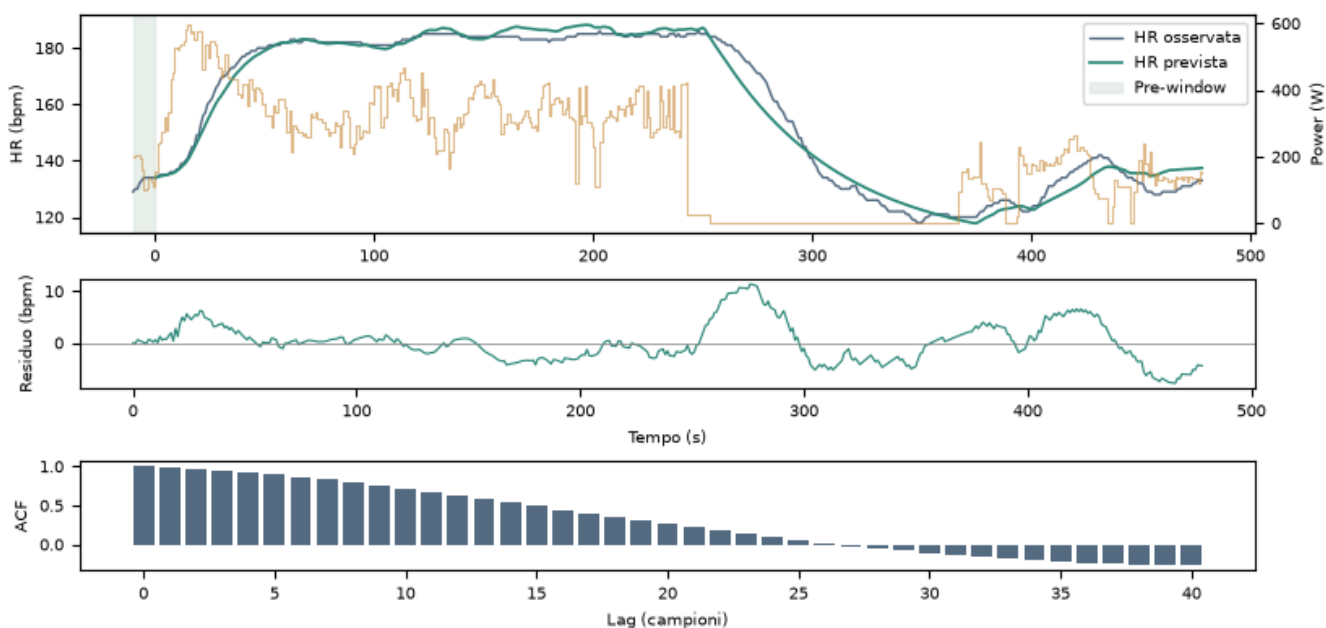
ID: 4c896ae2c98d387828b17b0b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=153.6 W; HR0=134 bpm.

HR(0) osservata fissata; B=145.29 bpm stimato; x(0)=-11.292 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.7877	2.8683	0.15386	0.97877	1.0354
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.15386	3.7885	0.98507	0.89787	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.23739	0.0024358	0.23262 / 0.24217	1.0261	no	no
L	7.48	0.46207	6.5744 / 8.3857	6.1775	no	no
tau	56.587	1.1423	54.348 / 58.826	2.0187	no	no
B	145.29	0.25241	144.8 / 145.79	0.17373	no	no

rho=8.3149; Jacobian rank=4/4; cond=866.92; accordo multistart=no.

Correlazioni: {"K_L": -0.4742960822895889, "K_tau": 0.7791908397233451, "K_B": -0.6394580689398369, "L_tau": -0.6545388823028621, "L_B": 0.23671306681528548, "tau_B": -0.5233771892593235}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 6871.9 / 3.172e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 6871.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_064-065.csv

ID: 4cec3ef57a53ef93c97b68a2 | Stato: fitted

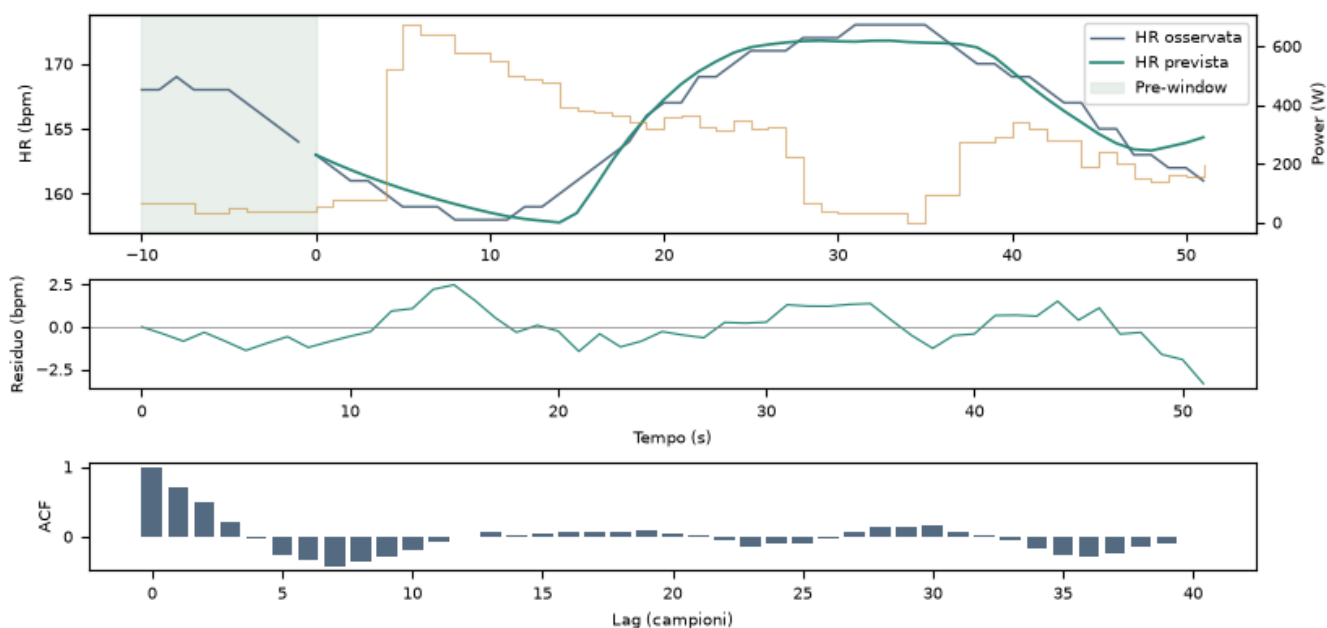
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=47.2 W; HR0=163 bpm.

HR(0) osservata fissata; B=154.27 bpm stimato; x(0)=8.7325 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.0938	0.88256	-0.055661	0.95394	0.93278

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.055661	1.1031	0.70165	-0.25661



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.058799	0.0060794	0.046884 / 0.070715	10.339	no	no
L	10.536	0.34918	9.8512 / 11.22	3.3143	no	no
tau	14.06	1.7114	10.706 / 17.415	12.172	no	no
B	154.27	1.2563	151.81 / 156.73	0.81435	no	no

rho=2.8779; Jacobian rank=4/4; cond=2521.9; accordo multistart=no.

Correlazioni: {"K_L": -0.7955513341276679, "K_tau": 0.9346351530655697, "K_B": -0.9851025665537616, "L_tau": -0.8135922201092884, "L_B": 0.792272504935888, "tau_B": -0.9039665686003406}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 62.216 / 1009.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 62.216; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260501094312__laps_016-017.csv

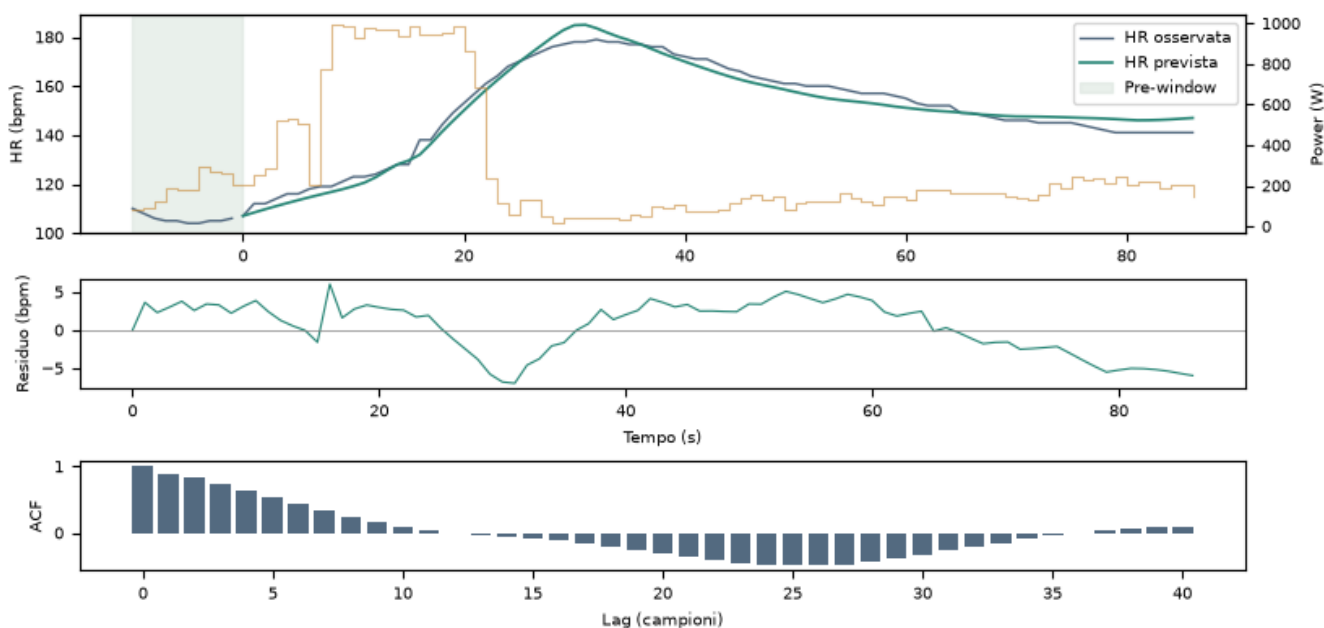
ID: 4e46b2308f9432598cbf72ca | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=186.9 W; HR0=107 bpm.

HR(0) osservata fissata; B=147.19 bpm stimato; x(0)=-40.192 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.4031	2.9976	0.43523	0.96816	1.083
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.43523	3.3947	0.89705	0.54423	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.16204	0.016219	0.13025 / 0.19383	10.009	no	no
L	8.4774	0.38172	7.7292 / 9.2255	4.5028	no	no
tau	28.665	2.61	23.549 / 33.78	9.1053	no	no
B	147.19	0.88602	145.46 / 148.93	0.60195	no	no

rho=2.7045; Jacobian rank=4/4; cond=999.92; accordo multistart=no.

Correlazioni: {"K_L": -0.7613181727413009, "K_tau": 0.9744380281611169, "K_B": -0.7177213486483007, "L_tau": -0.7971393703562756, "L_B": 0.4100483675546141, "tau_B": -0.6136245577410498}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1007.6 / 15445

Griglia: 42441 coppie L/tau; SSE multistart originale: 1007.6; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_006-007.csv

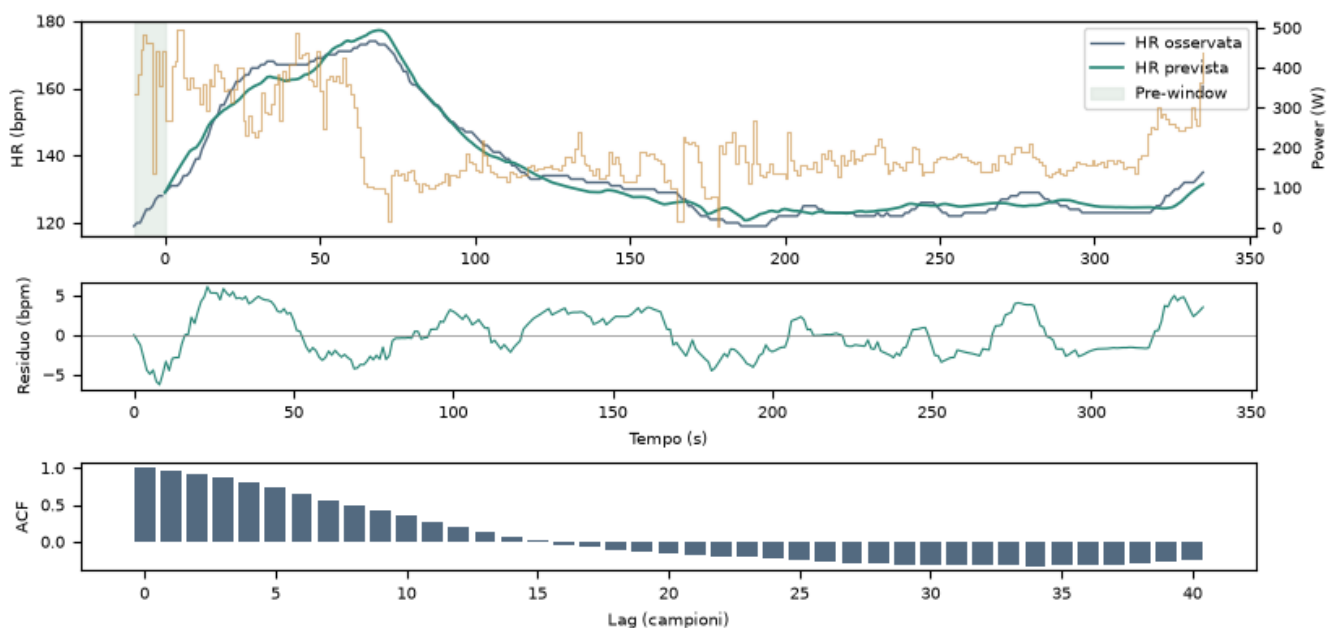
ID: 52ada156ddb27babe7514d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=395.1 W; HR0=129 bpm.

HR(0) osservata fissata; B=200 bpm stimato; x(0)=-71 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.6459	2.2907	0.087943	0.97608	1.0272
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.087943	2.6483	0.96559	0.72826	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.32403	0.0049479	0.31434 / 0.33373	1.527	no	no
L	8.264	0.36068	7.557 / 8.9709	4.3645	no	no
tau	42.199	0.92812	40.38 / 44.018	2.1994	no	no
B	200	1.0216	198 / 202	0.51081	si	si

rho=7.7428; Jacobian rank=4/4; cond=1728.6; accordo multistart=no.

Correlazioni: {"K_L": -0.6433493088625689, "K_tau": 0.8017550527508506, "K_B": 0.9855116296880355, "L_tau": -0.5350080480279399, "L_B": -0.675658034254848, "tau_B": 0.7738410353067712}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 2352.2 / 93347

Griglia: 42441 coppie L/tau; SSE multistart originale: 2352.2; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_026-027.csv

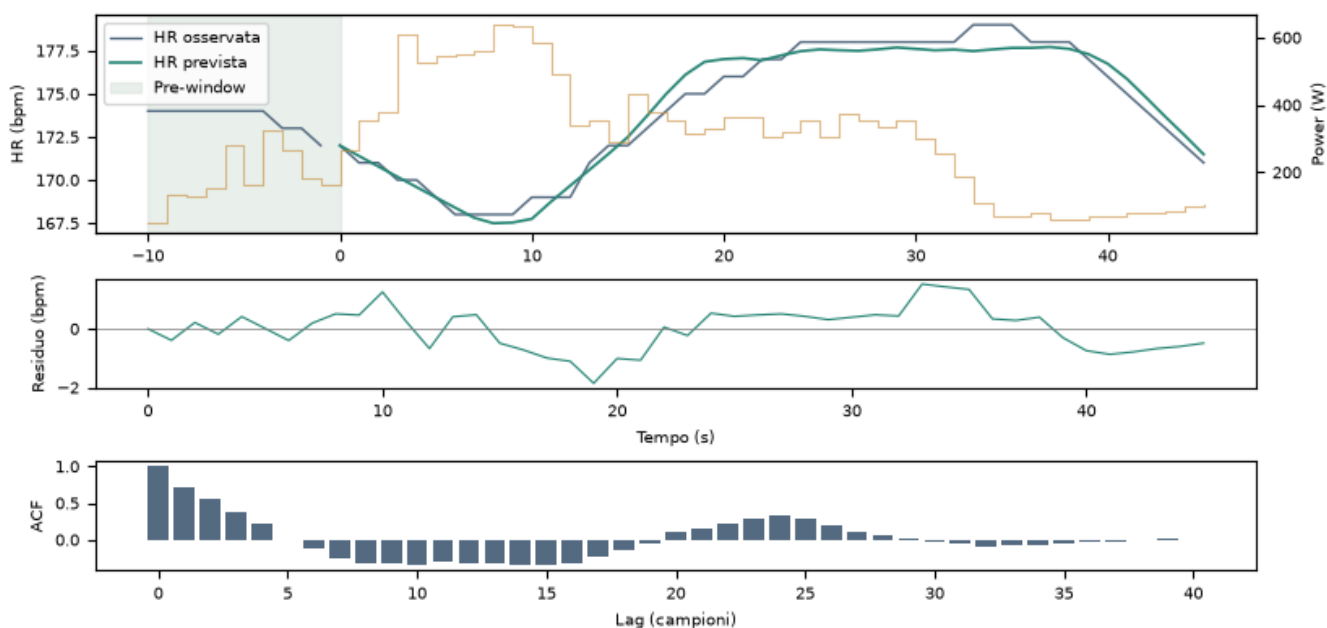
ID: 57b47931dfdd06086ce2e322 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=184.5 W; HR0=172 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=112 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.7157	0.58988	-0.00019838	0.96306	0.92946
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.00019838	0.72361	0.72419	0.0056137	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.77502	1.0437	-1.2707 / 2.8207	134.67	no	no
L	7.2195	0.32634	6.5799 / 7.8591	4.5203	no	no
tau	183.51	248.64	-303.82 / 670.84	135.49	no	no
B	60	153.33	-240.52 / 360.52	255.55	si	si

rho=0.20587; Jacobian rank=4/4; cond=61127; accordo multistart=no.

Correlazioni: {"K_L": -0.8940059571094067, "K_tau": 0.9995907399739852, "K_B": -0.9999664933889776, "L_tau": -0.8949275218784022, "L_B": 0.8948713984312224, "tau_B": -0.9994011824034179}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.562 / 402.55

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.562; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_055-056.csv

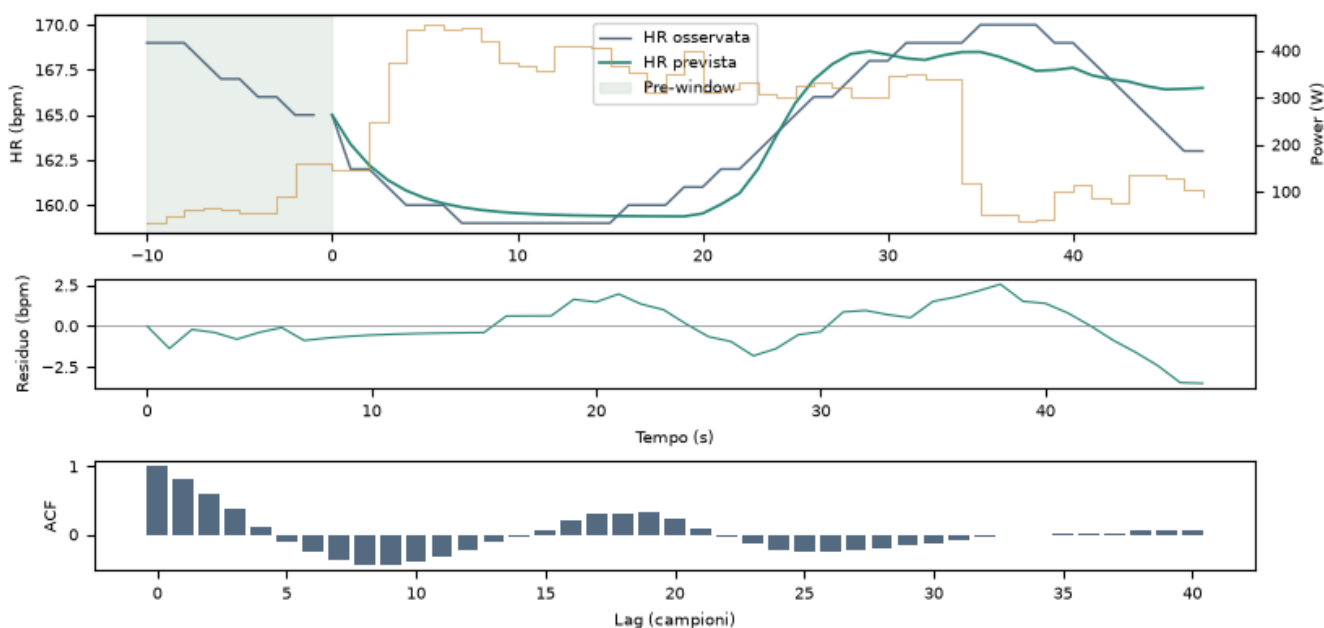
ID: 5d4622ec4d3d0e05a5299850 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=78.7 W; HR0=165 bpm.

HR(0) osservata fissata; B=159.37 bpm stimato; x(0)=5.6345 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.309	1.0369	-0.035576	0.88895	0.83241
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.035576	1.3224	0.81563	-0.09839	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.02886	0.0019506	0.025037 / 0.032683	6.759	no	no
L	19.744	0.52249	18.72 / 20.768	2.6463	no	no
tau	2.932	0.68592	1.5876 / 4.2764	23.394	no	no
B	159.37	0.39154	158.6 / 160.13	0.24569	no	no

rho=9.296; Jacobian rank=4/4; cond=846.27; accordo multistart=no.

Correlazioni: {"K_L": -0.5096049619844532, "K_tau": 0.6210914592629232, "K_B": -0.8306452103821158, "L_tau": -0.7256102500521453, "L_B": 0.5478889908408048, "tau_B": -0.5358817463150761}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 82.251 / 797.09

Griglia: 42441 coppie L/tau; SSE multistart originale: 82.254; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_043-044.csv

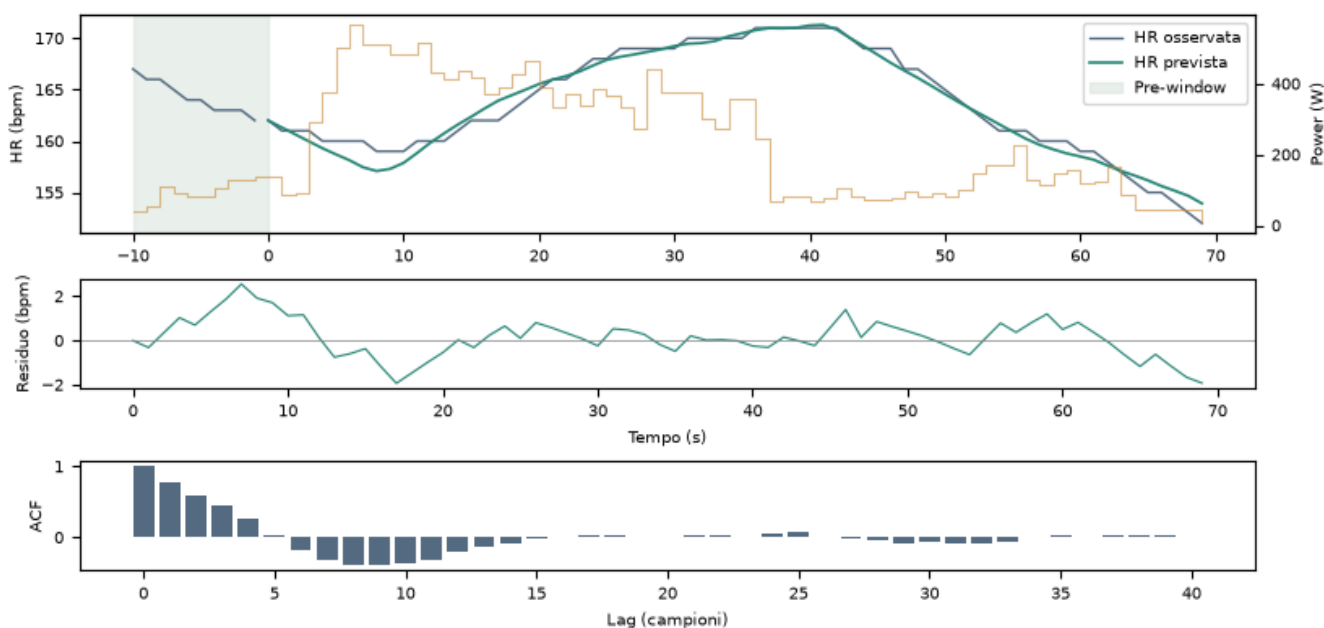
ID: 5e1214afe505fdf442c8ca93 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=96.3 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.87302	0.65858	0.12851	0.9715	0.91491
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.12851	0.86974	0.78258	0.030567	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.56053	0.38615	-0.19633 / 1.3174	68.891	no	no
L	4.595	0.31122	3.985 / 5.205	6.773	no	no
tau	149.15	104.49	-55.664 / 353.96	70.062	no	no
B	60	72.227	-81.565 / 201.56	120.38	si	si

rho=0.43182; Jacobian rank=4/4; cond=52532; accordo multistart=no.

Correlazioni: {"K_L": -0.8587672130634888, "K_tau": 0.999510291265812, "K_B": -0.9999766839245608, "L_tau": -0.8639325034904901, "L_B": 0.8596194668848877, "tau_B": -0.999481436471091}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 53.351 / 1795.1

Griglia: 42441 coppie L/tau; SSE multistart originale: 53.351; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260919065417__laps_004-005.csv

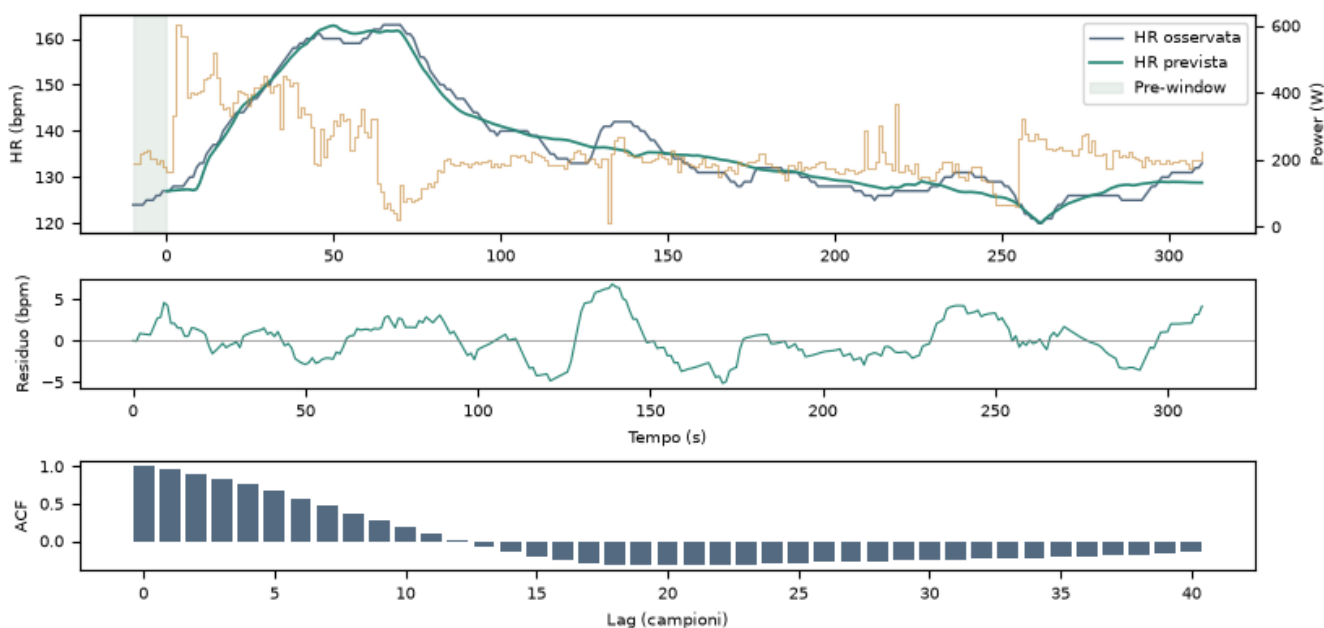
ID: 67b628ab6ed7b51e02b6a68f | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=203.2 W; HR0=127 bpm.

HR(0) osservata fissata; B=130.71 bpm stimato; x(0)=-3.7124 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.398	1.914	0.083822	0.9576	0.99284
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.083822	2.4004	0.96061	0.66754	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.34693	0.0082585	0.33074 / 0.36312	2.3804	no	no
L	6.5975	0.33553	5.9398 / 7.2551	5.0858	no	no
tau	62.655	2.0793	58.58 / 66.731	3.3186	no	no
B	130.71	0.23974	130.24 / 131.18	0.18341	no	no

rho=4.8424; Jacobian rank=4/4; cond=604.05; accordo multistart=no.

Correlazioni: {"K_L": -0.4383466765826009, "K_tau": 0.8964136529551033, "K_B": -0.7185267894203209, "L_tau": -0.5075922990676942, "L_B": 0.25851784895860025, "tau_B": -0.6855067423185321}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1788.4 / 41626

Griglia: 42441 coppie L/tau; SSE multistart originale: 1788.4; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250708171459__laps_004-005.csv

ID: 6f088b34503f7c77f7be1e0f | Stato: fitted

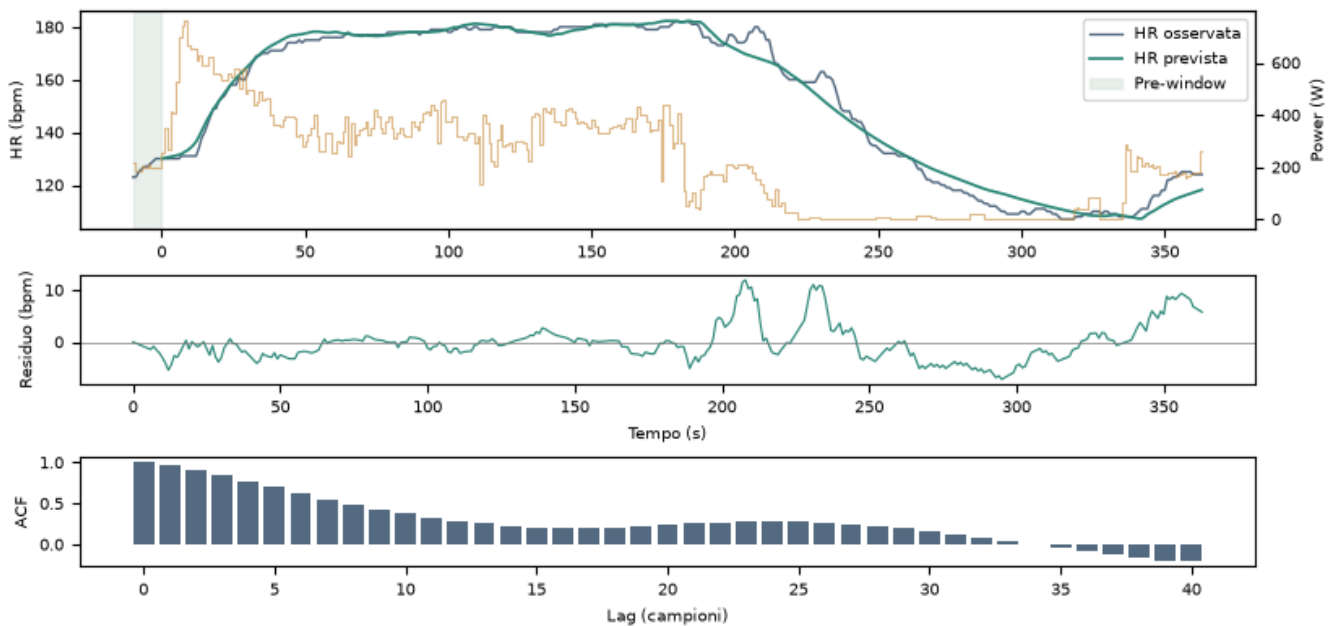
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=199.8 W; HR0=130 bpm.

HR(0) osservata fissata; B=145.09 bpm stimato; x(0)=-15.09 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.5507	2.5128	-0.19139	0.98394	1.0008

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.19139	3.5505	0.96161	0.6992



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.24573	0.0037239	0.23843 / 0.25303	1.5154	no	no
L	6.2919	0.50615	5.2998 / 7.2839	8.0446	no	no
tau	60.167	1.6356	56.961 / 63.373	2.7184	no	no
B	145.09	0.30868	144.48 / 145.69	0.21275	no	no

rho=5.9286; Jacobian rank=4/4; cond=1121.4; accordo multistart=no.

Correlazioni: {"K_L": -0.5953539354168322, "K_tau": 0.9067101017152399, "K_B": -0.6771121930166362, "L_tau": -0.7047063153602832, "L_B": 0.28355054641447147, "tau_B": -0.64749512176188}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 4589.2 / 2.8021e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 4589.2; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_047-048.csv

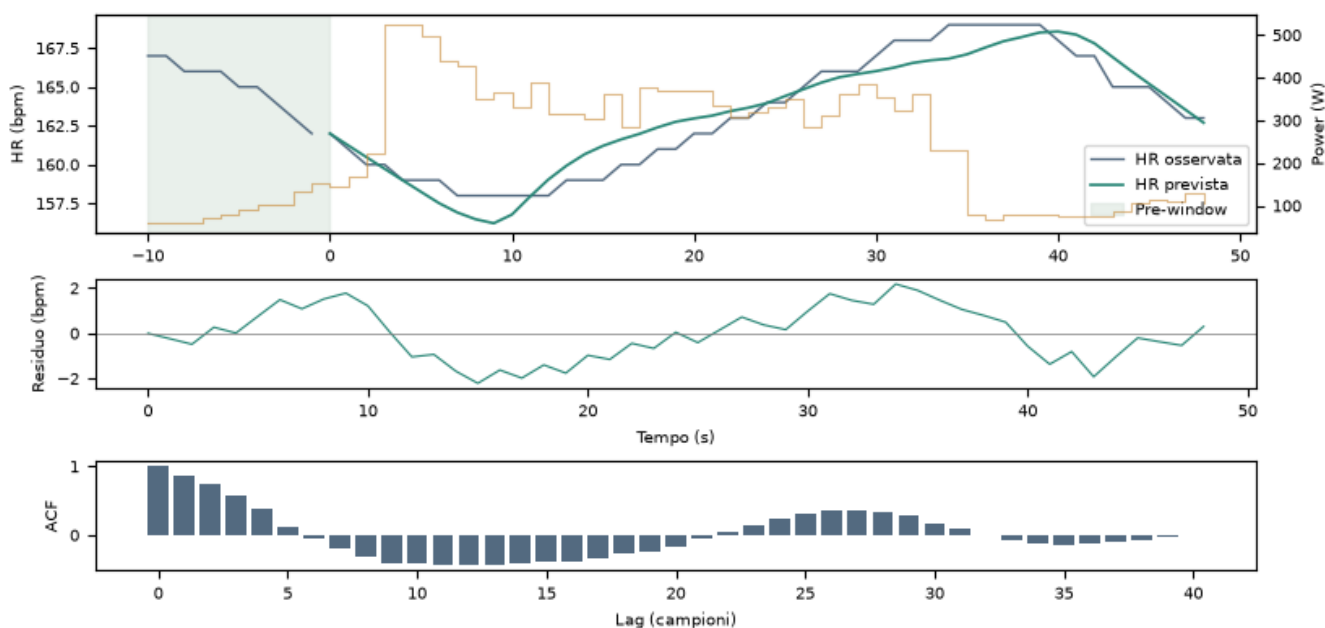
ID: 705e9102abe4cc2745feb9c7 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=91.3 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.1494	0.96088	-0.009831	0.90799	1.1223
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.009831	1.1613	0.86071		0.12831



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.58302	0.81703	-1.0184 / 2.1844	140.14	no	no
L	6.4606	0.50949	5.462 / 7.4592	7.8862	no	no
tau	133.38	188.68	-236.43 / 503.19	141.46	no	no
B	60	144.33	-222.88 / 342.88	240.54	si	si

rho=0.31144; Jacobian rank=4/4; cond=50920; accordo multistart=no.

Correlazioni: {"K_L": -0.8747693270544614, "K_tau": 0.9989560703913867, "K_B": -0.9999414245714775, "L_tau": -0.8682585080063723, "L_B": 0.8777646259961914, "tau_B": -0.9985614018198259}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 64.736 / 455.2

Griglia: 42441 coppie L/tau; SSE multistart originale: 64.736; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_008-009.csv

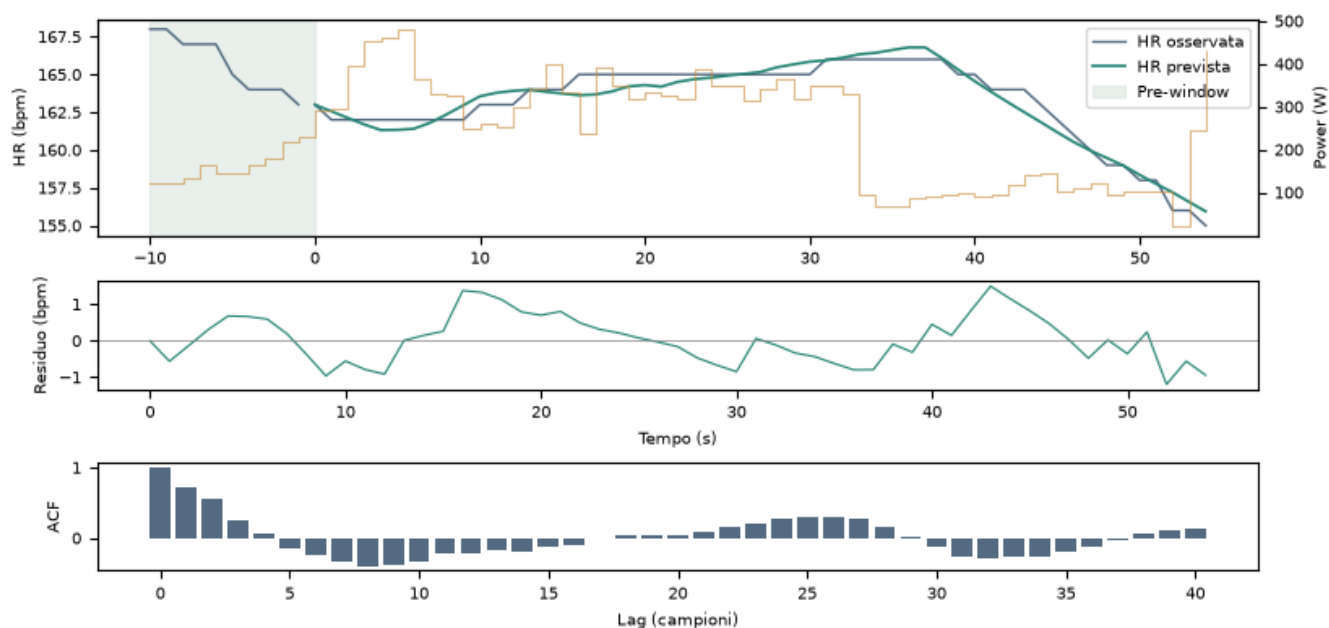
ID: 715861e52e9a32a967362366 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=161.7 W; HR0=163 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=103 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.65468	0.53207	0.042526	0.94272	0.98671
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.042526	0.65932	0.72786	-0.14098	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.80157	2.0459	-3.2084 / 4.8116	255.24	no	no
L	3.8498	0.3353	3.1926 / 4.507	8.7094	no	no
tau	233.9	604.95	-951.81 / 1419.6	258.64	no	no
B	60	265.12	-459.64 / 579.64	441.87	no	si

rho=0.21441; Jacobian rank=4/4; cond=1.1395e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.7817721128449447, "K_tau": 0.9998875963760885, "K_B": -0.9999956708176203, "L_tau": -0.7858005032573168, "L_B": 0.7822521951302888, "tau_B": -0.9998935227514644}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 23.574 / 407.84

Griglia: 42441 coppie L/tau; SSE multistart originale: 23.574; selezione: profile_grid.

Segnalazioni: near_bounds, autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_006-007.csv

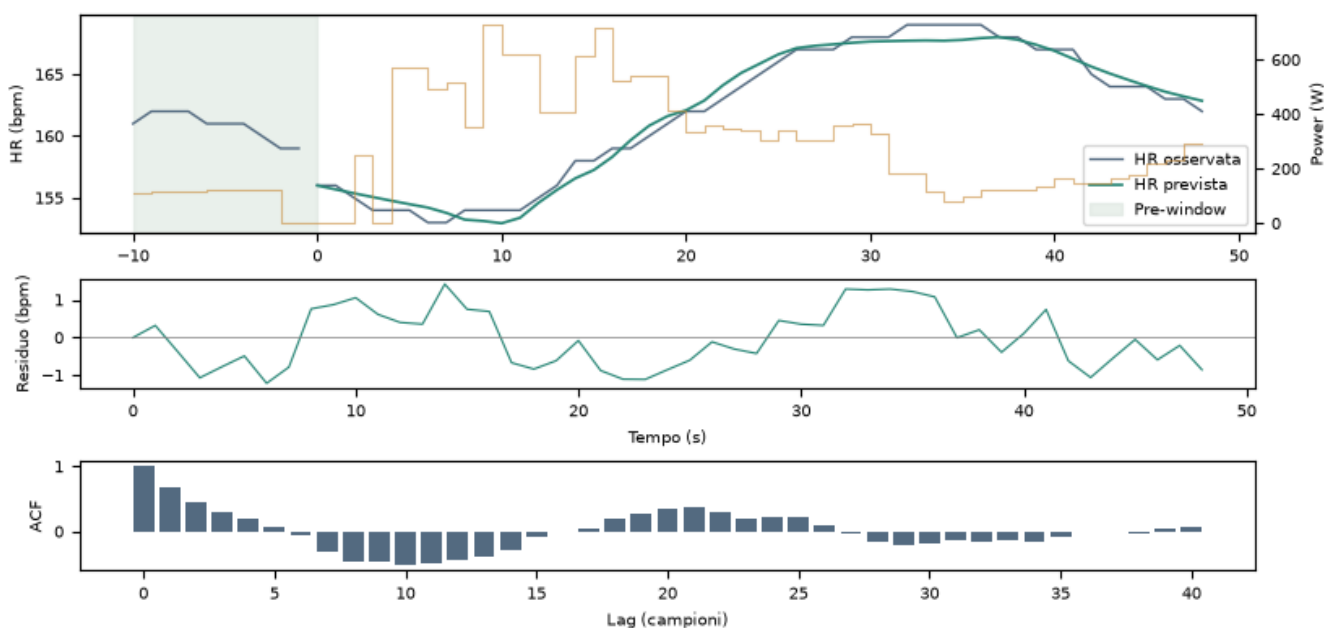
ID: 774d0fa4b92b786dd0448208 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=94.9 W; HR0=156 bpm.

HR(0) osservata fissata; B=145.17 bpm stimato; x(0)=10.832 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.75901	0.65591	-0.020246	0.98116	0.94134
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.020246	0.7666	0.68895	0.061808	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.10305	0.014582	0.074465 / 0.13163	14.151	no	no
L	6.4529	0.3264	5.8132 / 7.0927	5.0582	no	no
tau	33.237	4.9317	23.57 / 42.903	14.838	no	no
B	145.17	2.3665	140.53 / 149.81	1.6302	no	no

rho=1.25; Jacobian rank=4/4; cond=6198.4; accordo multistart=no.

Correlazioni: {"K_L": -0.8468576287972779, "K_tau": 0.9687057856823059, "K_B": -0.986223245152575, "L_tau": -0.8519259060460572, "L_B": 0.8327661406676274, "tau_B": -0.9219935158126612}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 28.229 / 453.95

Griglia: 42441 coppie L/tau; SSE multistart originale: 28.229; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_008-010.csv

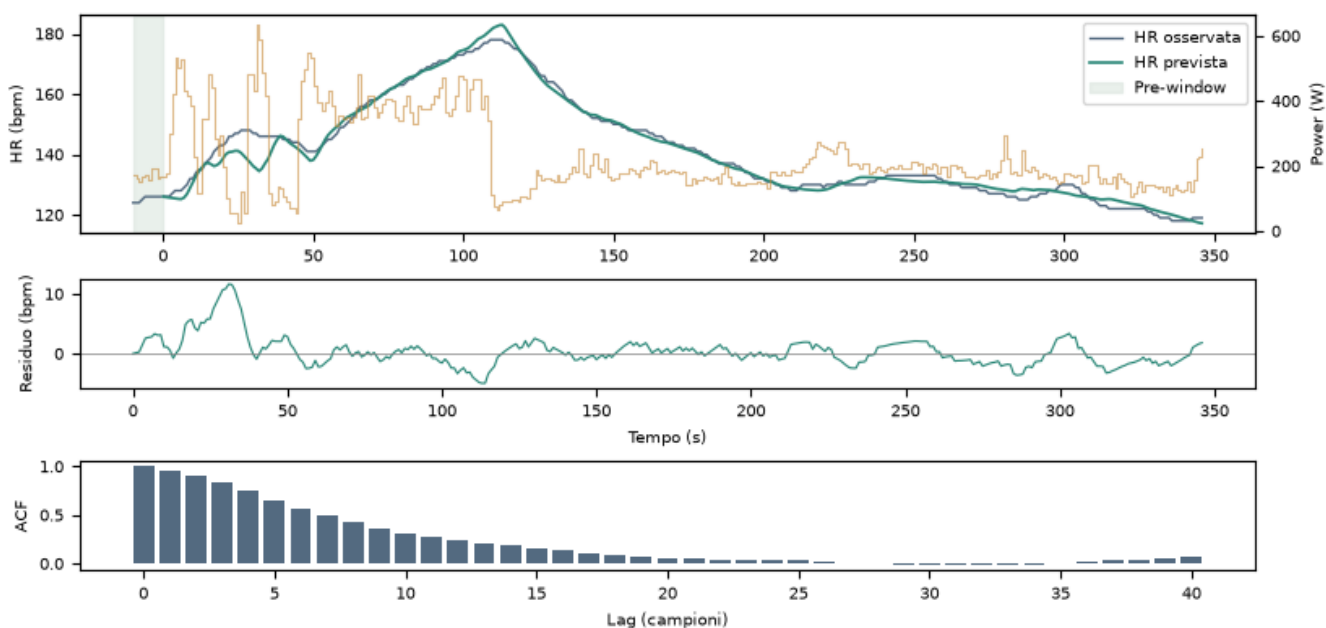
ID: 7da2631ee651e8807b7c0d7f | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=169.9 W; HR0=126 bpm.

HR(0) osservata fissata; B=116.91 bpm stimato; x(0)=9.0857 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.4577	1.6658	0.33499	0.97738	1.0962
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.33499	2.4382	0.96259	0.65714	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.39445	0.006024	0.38264 / 0.40626	1.5272	no	no
L	4.2808	0.22018	3.8493 / 4.7124	5.1434	no	no
tau	61.316	1.3381	58.693 / 63.938	2.1823	no	no
B	116.91	0.44419	116.04 / 117.78	0.37993	no	no

rho=5.5731; Jacobian rank=4/4; cond=742.14; accordo multistart=no.

Correlazioni: {"K_L": -0.3214797378140794, "K_tau": 0.8664918923813608, "K_B": -0.9359164344647449, "L_tau": -0.44688716738265594, "L_B": 0.2770605469974838, "tau_B": -0.8271367590842255}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 2095.9 / 90838

Griglia: 42441 coppie L/tau; SSE multistart originale: 2095.9; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_017-018.csv

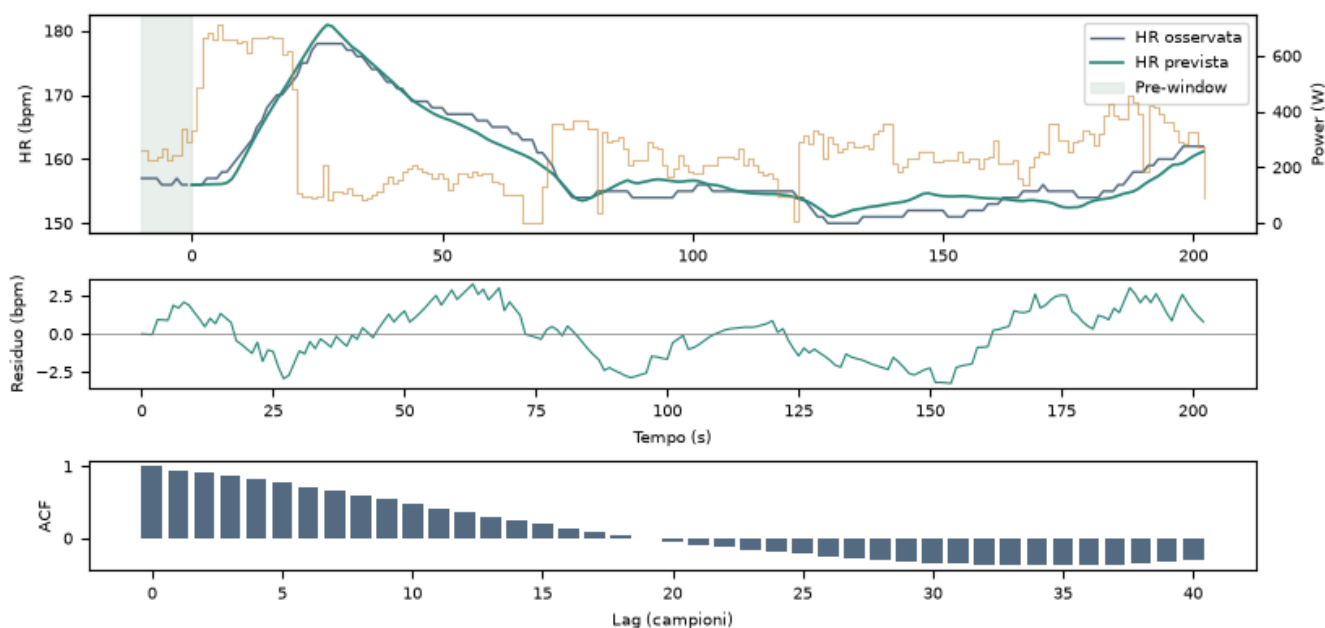
ID: 8019887eeae9da53fdc43a3c | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=256.9 W; HR0=156 bpm.

HR(0) osservata fissata; B=157.83 bpm stimato; $x(0)=-1.8344$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.6515	1.3946	0.036041	0.95431	1.0692
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.036041	1.6552	0.95282	0.7736	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.32623	0.017703	0.29153 / 0.36093	5.4265	no	no
L	6.4223	0.25122	5.9299 / 6.9147	3.9118	no	no
tau	91.417	5.6867	80.271 / 102.56	6.2206	no	no
B	157.83	0.22246	157.4 / 158.27	0.14094	no	no

rho=2.1394; Jacobian rank=4/4; cond=1211.2; accordo multistart=no.

Correlazioni: {"K_L": -0.4775068861740743, "K_tau": 0.9641033066044823, "K_B": -0.5517204436549158, "L_tau": -0.498209965142998, "L_B": 0.27342634475283506, "tau_B": -0.5795245118219894}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 553.68 / 13822

Griglia: 42441 coppie L/tau; SSE multistart originale: 553.68; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250715165154__laps_011-012.csv

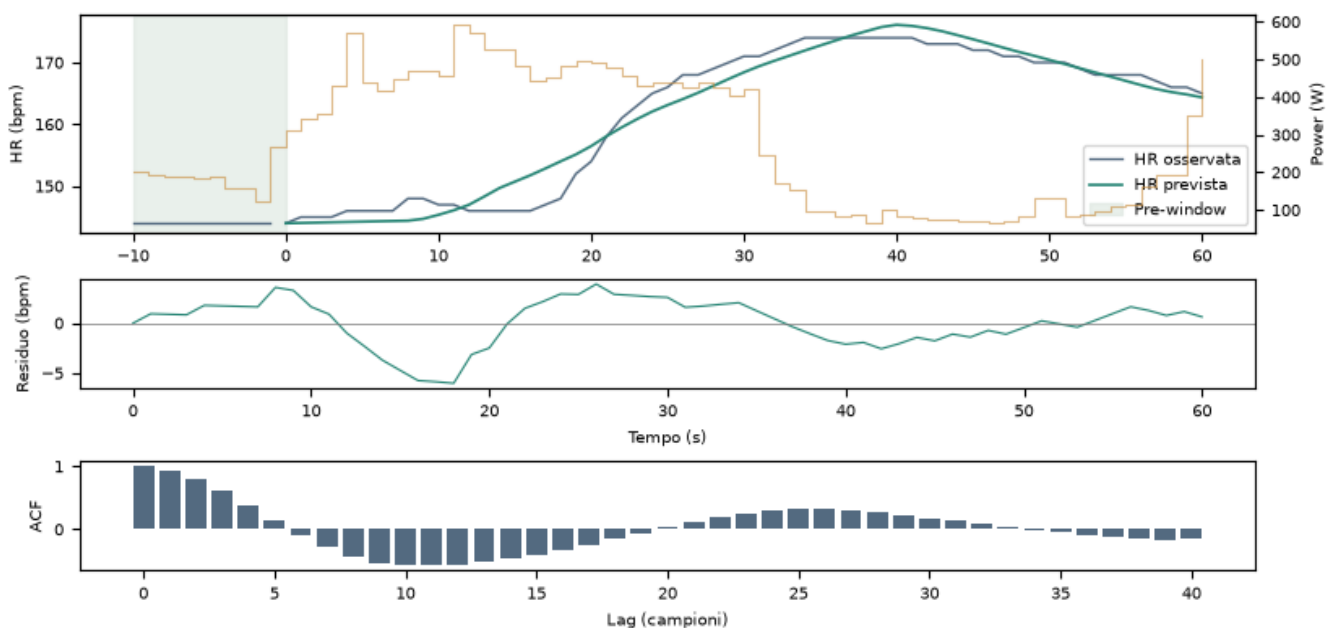
ID: 811df6313aef0591fb62ef23 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=185.1 W; HR0=144 bpm.

HR(0) osservata fissata; B=149.2 bpm stimato; x(0)=-5.2017 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.3206	1.8753	0.064711	0.9582	1.0701
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.064711	2.3389	0.92603	0.14531	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.40555	0.25236	-0.089085 / 0.90018	62.228	no	no
L	8.5292	1.0946	6.3839 / 10.675	12.833	no	no
tau	96.092	58.564	-18.694 / 210.88	60.946	no	no
B	149.2	8.0403	133.44 / 164.96	5.3888	no	no

rho=0.53564; Jacobian rank=4/4; cond=9252.9; accordo multistart=no.

Correlazioni: {"K_L": -0.9197151799184659, "K_tau": 0.9956431710924786, "K_B": -0.9152642617439868, "L_tau": -0.9328210129629863, "L_B": 0.775976288674427, "tau_B": -0.8758015612210681}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 328.49 / 1934.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 328.49; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_070-071.csv

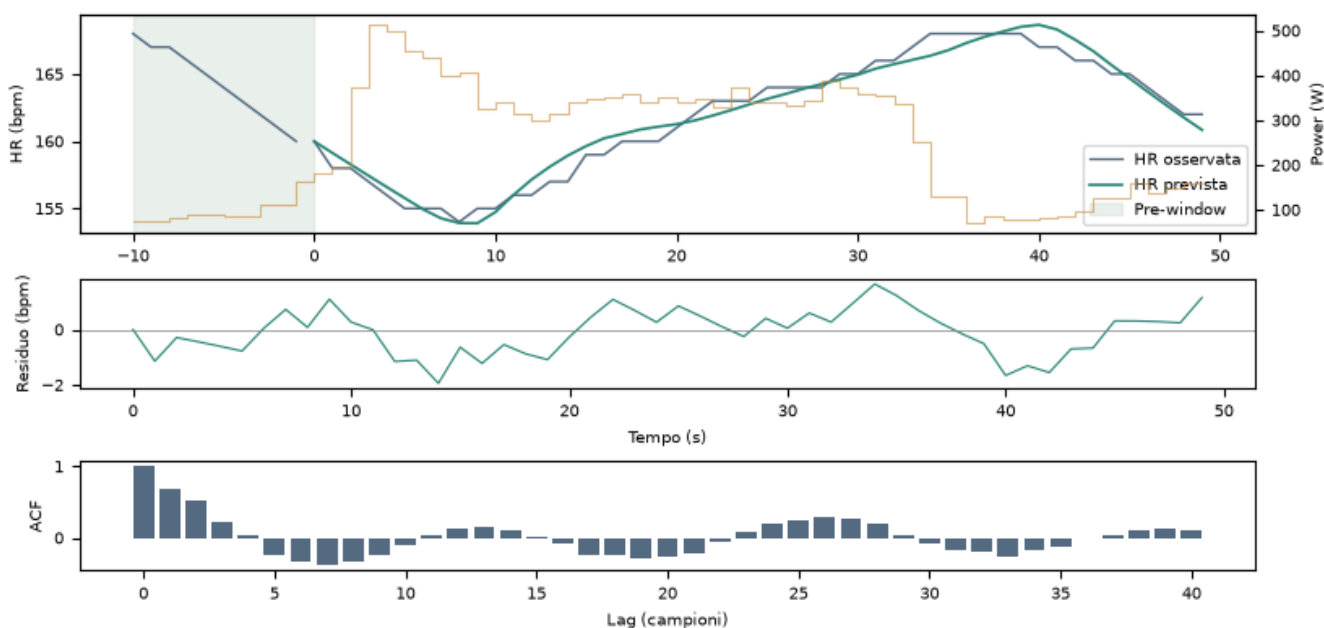
ID: 817afcbd82486c444de90a4d | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=97 W; HR0=160 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=100 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.82246	0.66821	-0.087203	0.96525	1.0537
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.087203	0.82613	0.68284		-0.23126



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.59094	0.44424	-0.27977 / 1.4617	75.175	no	no
L	6.6832	0.33528	6.026 / 7.3404	5.0168	no	no
tau	115.81	87.682	-56.048 / 287.67	75.713	no	no
B	60	76.461	-89.863 / 209.86	127.43	si	si

rho=0.3654; Jacobian rank=4/4; cond=40058; accordo multistart=no.

Correlazioni: {"K_L": -0.8949360912797508, "K_tau": 0.9986421341498402, "K_B": -0.999919205909154, "L_tau": -0.8888347395751354, "L_B": 0.897818365818478, "tau_B": -0.9980913284375609}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 33.822 / 555.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 33.822; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_008-009.csv

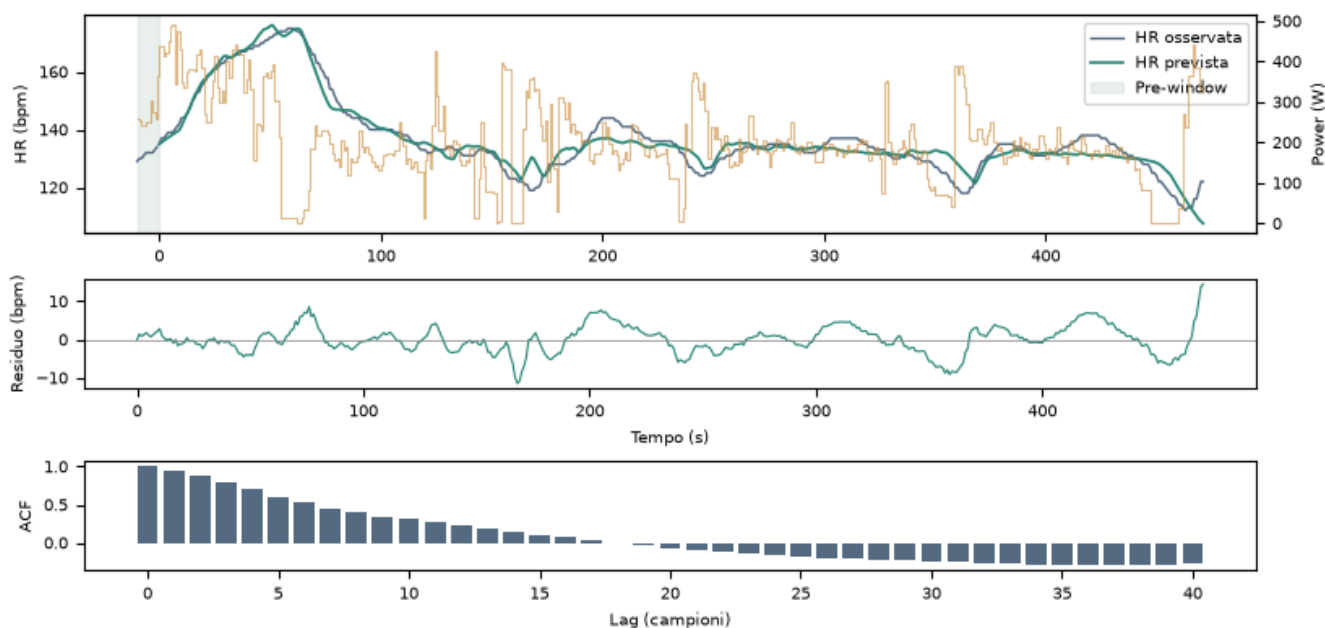
ID: 89905a574f644b7823b2fd7e | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=270.4 W; HR0=135 bpm.

HR(0) osservata fissata; B=160.73 bpm stimato; x(0)=-25.727 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.5939	2.7223	0.0034202	0.92731	1.0901
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.0034202	3.5977	0.95031	0.60851	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.35998	0.0071704	0.34592 / 0.37403	1.9919	no	no
L	9.3036	0.25227	8.8092 / 9.7981	2.7115	no	no
tau	43.756	1.3403	41.129 / 46.383	3.0631	no	no
B	160.73	0.50607	159.74 / 161.72	0.31486	no	no

rho=10.552; Jacobian rank=4/4; cond=626.45; accordo multistart=no.

Correlazioni: {"K_L": -0.2858655651190091, "K_tau": 0.7659846227396795, "K_B": 0.9308777357737256, "L_tau": -0.36978189850644405, "L_B": -0.31371574816875514, "tau_B": 0.6738782619564395}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 6096.3 / 83867

Griglia: 42441 coppie L/tau; SSE multistart originale: 6096.3; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260523080618__laps_010-011.csv

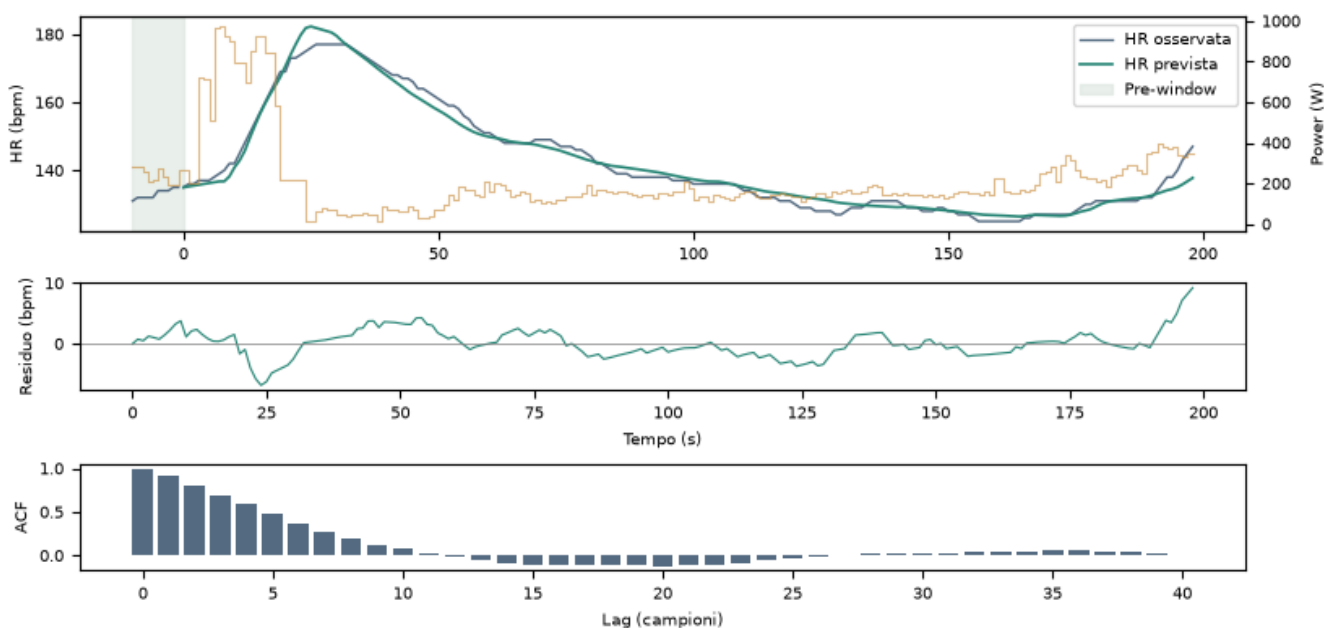
ID: 8a6768c86f26302179877d94 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=240.8 W; HR0=135 bpm.

HR(0) osservata fissata; B=155.97 bpm stimato; $x(0)=-20.974$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.2702	1.6954	0.071836	0.97854	1.074
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.071836	2.2748	0.907	0.48404	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.41852	0.014773	0.38956 / 0.44747	3.5298	no	no
L	5.4661	0.20111	5.0719 / 5.8603	3.6793	no	no
tau	82.19	3.4631	75.402 / 88.978	4.2135	no	no
B	155.97	0.4676	155.06 / 156.89	0.2998	no	no

rho=2.3425; Jacobian rank=4/4; cond=1042.6; accordo multistart=no.

Correlazioni: {"K_L": -0.5089608632533034, "K_tau": 0.9613651368916859, "K_B": 0.840614257813919, "L_tau": -0.5141917632492619, "L_B": -0.482440105041678, "tau_B": 0.7704838847238458}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1025.6 / 47575

Griglia: 42441 coppie L/tau; SSE multistart originale: 1025.6; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260523080618__laps_002-003.csv

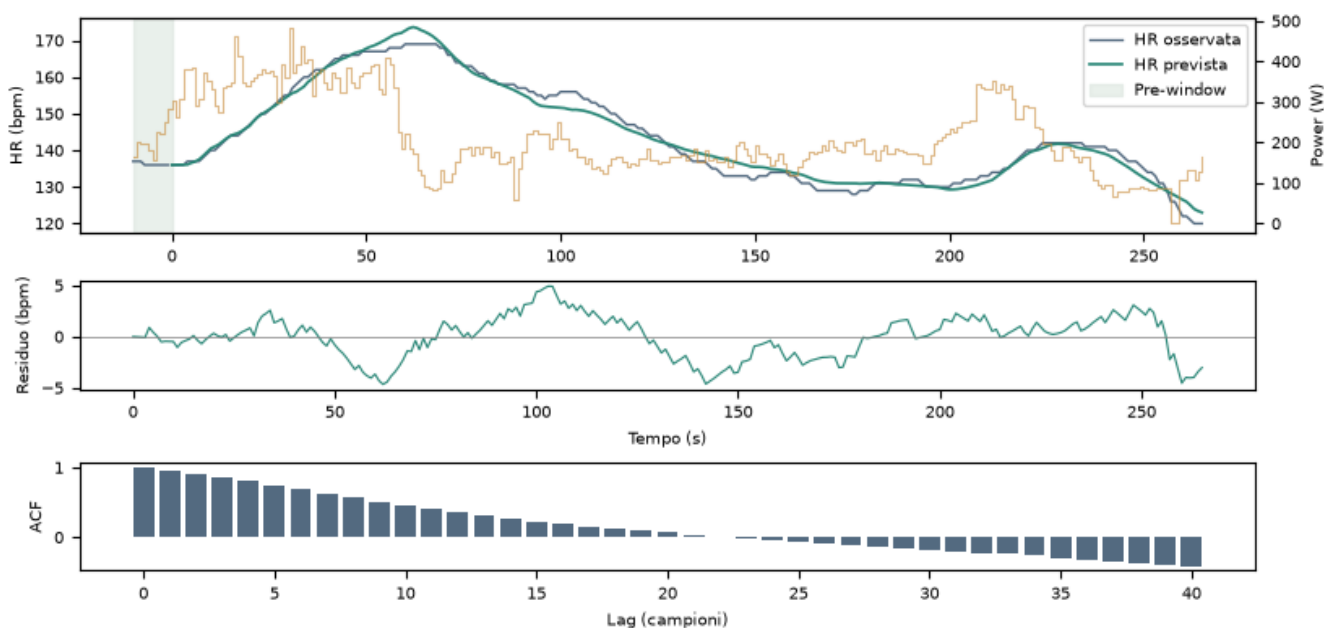
ID: 8c09ad54632d44d9ded0189b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=208.3 W; HR0=136 bpm.

HR(0) osservata fissata; B=138.02 bpm stimato; x(0)=-2.0168 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0387	1.6149	-0.017379	0.97661	1.0333
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.017379	2.0424	0.95749	0.74747	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.44218	0.010339	0.42191 / 0.46244	2.3382	no	no
L	4.2034	0.29564	3.624 / 4.7829	7.0332	no	no
tau	74.801	2.2997	70.294 / 79.309	3.0745	no	no
B	138.02	0.25324	137.52 / 138.51	0.18348	no	no

rho=3.4865; Jacobian rank=4/4; cond=623.54; accordo multistart=no.

Correlazioni: {"K_L": -0.4120949790124069, "K_tau": 0.9272099810965385, "K_B": -0.7373665093616951, "L_tau": -0.49422419591158556, "L_B": 0.17689414963865296, "tau_B": -0.7156652054201131}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1105.5 / 46829

Griglia: 42441 coppie L/tau; SSE multistart originale: 1105.5; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_028-029.csv

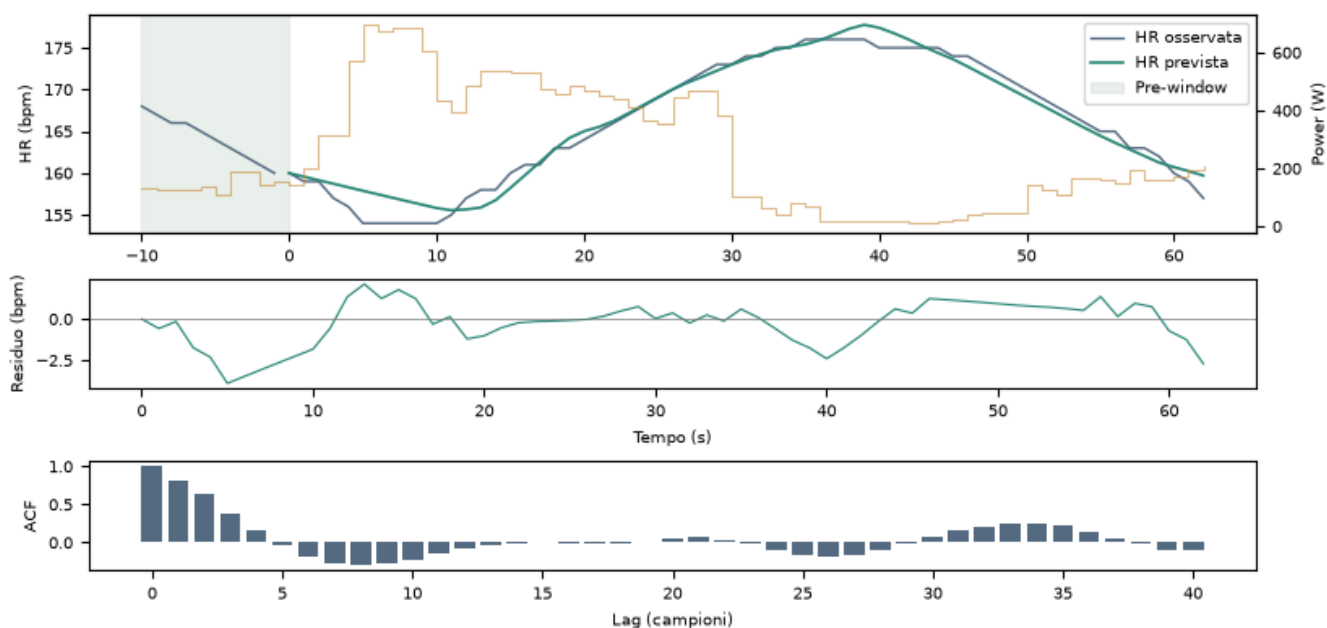
ID: 964197ddc49501314e2f4456 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=144 W; HR0=160 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=100 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.3462	1.011	-0.25857	0.96712	1.0084
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.25857	1.3317	0.82209	-0.028526	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.85712	1.0697	-1.2394 / 2.9537	124.8	no	no
L	9.2956	0.5197	8.277 / 10.314	5.5909	no	no
tau	234.26	292.34	-338.72 / 807.24	124.79	no	no
B	60	132.39	-199.49 / 319.49	220.65	si	si

rho=0.22498; Jacobian rank=4/4; cond=46970; accordo multistart=no.

Correlazioni: {"K_L": -0.9131088702143279, "K_tau": 0.9997199362739148, "K_B": -0.9999058201114297, "L_tau": -0.915835206112435, "L_B": 0.9123791684415195, "tau_B": -0.9993944588340389}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 114.17 / 2258.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 114.17; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_084-085.csv

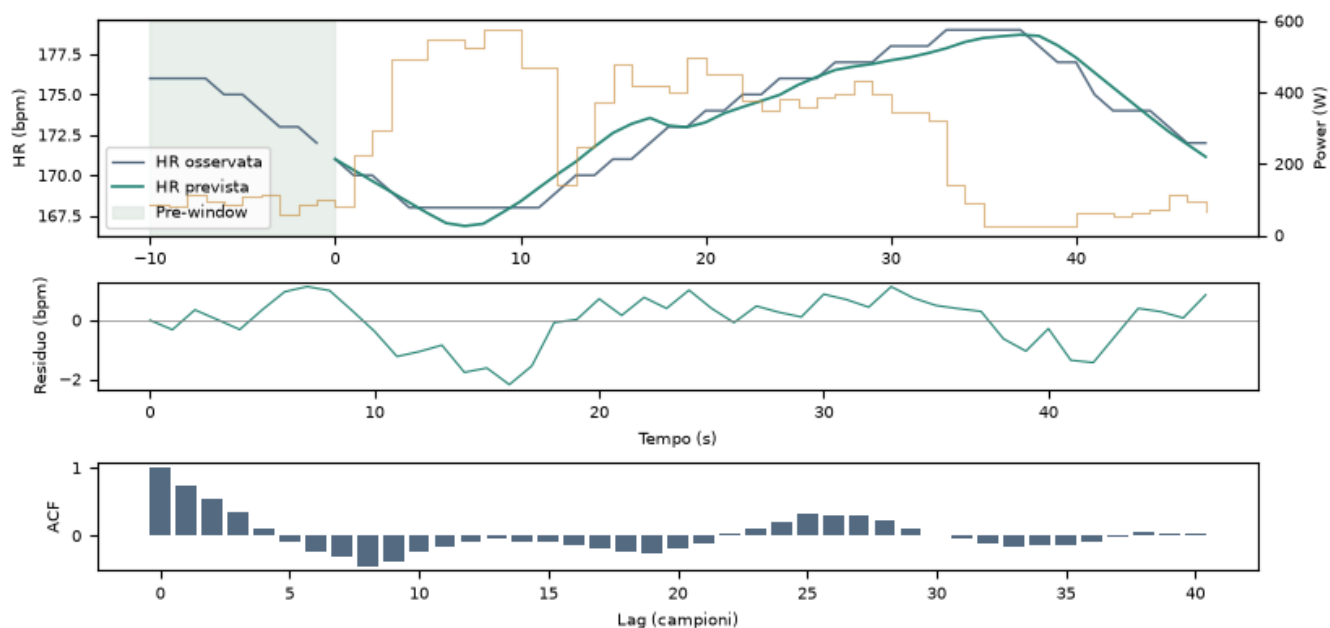
ID: 9a66bc064c74937bc5b9c2e9 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=91.8 W; HR0=171 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=111 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.83599	0.66503	-0.031346	0.95051	1.0764
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.031346	0.84425	0.74358	-0.098185	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.54162	0.72908	-0.88738 / 1.9706	134.61	no	no
L	4.8267	0.39022	4.0618 / 5.5915	8.0847	no	no
tau	163.47	220.11	-267.94 / 594.88	134.64	no	no
B	60	152.56	-239.01 / 359.01	254.26	no	si

rho=0.25798; Jacobian rank=4/4; cond=79717; accordo multistart=no.

Correlazioni: {"K_L": -0.9031294467670188, "K_tau": 0.9993856738787376, "K_B": -0.9999583009977268, "L_tau": -0.9024683457944594, "L_B": 0.904344086626086, "tau_B": -0.9991092592827114}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 33.546 / 380.69

Griglia: 42441 coppie L/tau; SSE multistart originale: 33.546; selezione: multistart.

Segnalazioni: near_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_002-003.csv

ID: 9c7bee85ac62ed78098de523 | Stato: fitted

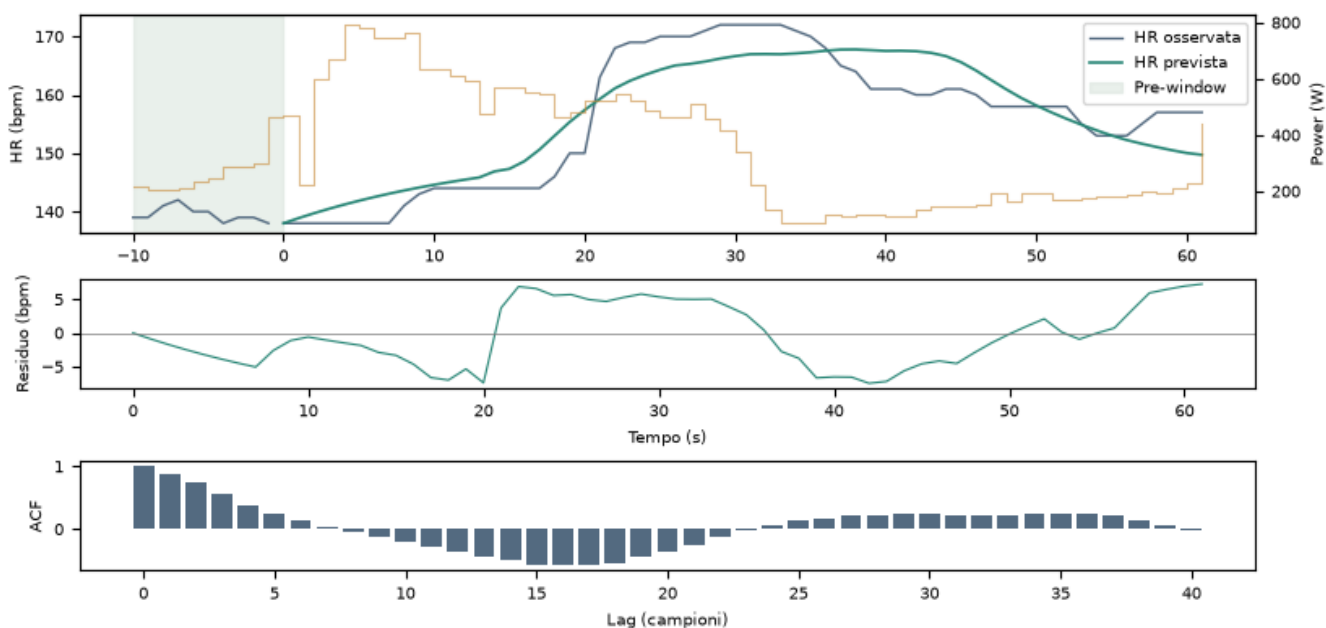
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=266.3 W; HR0=138 bpm.

HR(0) osservata fissata; B=151.39 bpm stimato; x(0)=-13.389 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
4.5338	3.9292	-0.357	0.83812	0.87568

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.357	4.5567	0.88813	0.2385



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.072407	0.01221	0.048476 / 0.096339	16.863	no	no
L	13.31	1.2334	10.893 / 15.728	9.2667	no	no
tau	14.738	2.9873	8.8827 / 20.593	20.27	no	no
B	151.39	1.3835	148.68 / 154.1	0.91385	no	no

rho=3.2359; Jacobian rank=4/4; cond=806.59; accordo multistart=no.

Correlazioni: {"K_L": -0.6017082601092764, "K_tau": 0.8183024404115065, "K_B": -0.6837258474546946, "L_tau": -0.8187591325668436, "L_B": 0.09594599257725656, "tau_B": -0.2779123776768696}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1274.5 / 3714.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 1324.7; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_010-011.csv

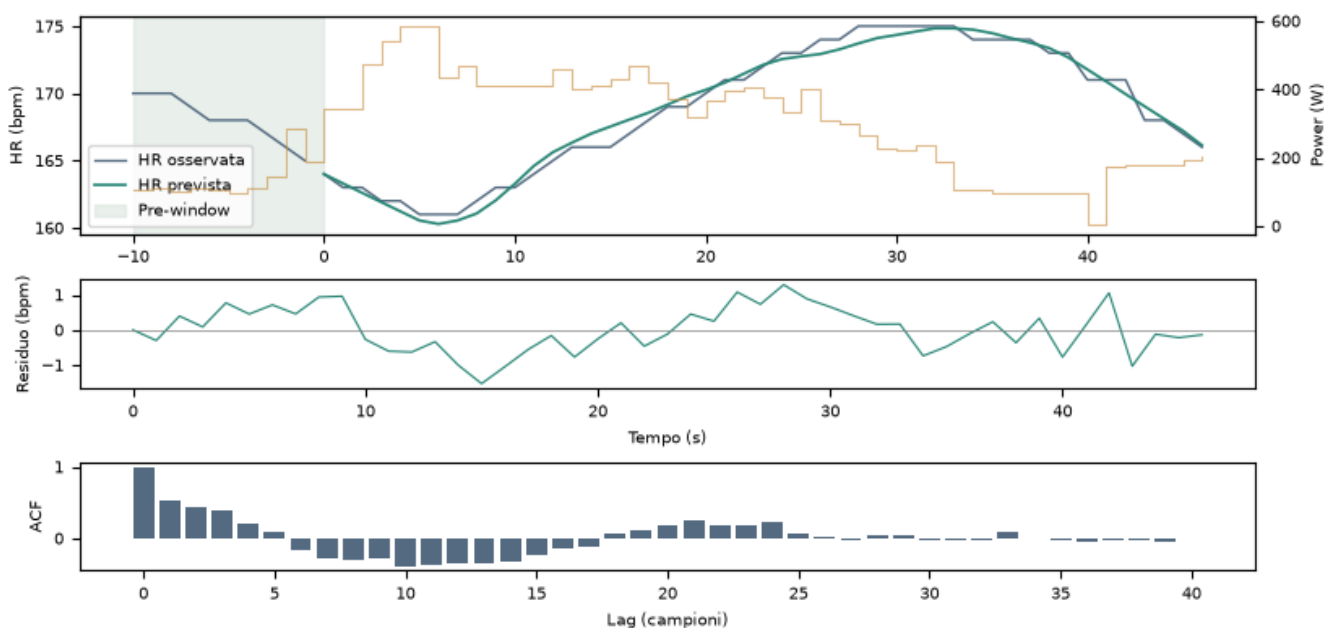
ID: 9d8b0d46644044c3e684759b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=135.8 W; HR0=164 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=104 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.6392	0.52757	0.016778	0.9818	1.0392
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.016778	0.64589	0.53193	0.093806	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.6716	0.61331	-0.53048 / 1.8737	91.32	no	no
L	5.5545	0.32882	4.9101 / 6.199	5.9198	no	no
tau	148.12	135.14	-116.75 / 412.99	91.238	no	no
B	60	99.326	-134.68 / 254.68	165.54	si	si

rho=0.27306; Jacobian rank=4/4; cond=58671; accordo multistart=no.

Correlazioni: {"K_L": -0.9350257813120031, "K_tau": 0.9994313837587985, "K_B": -0.9999317199899055, "L_tau": -0.937391651315833, "L_B": 0.935079678980622, "tau_B": -0.999040050238114}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 19.203 / 475.53

Griglia: 42441 coppie L/tau; SSE multistart originale: 19.203; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250625105513__laps_023-024.csv

ID: 9d9868f8bc0dfed8b631f5cf | Stato: fitted

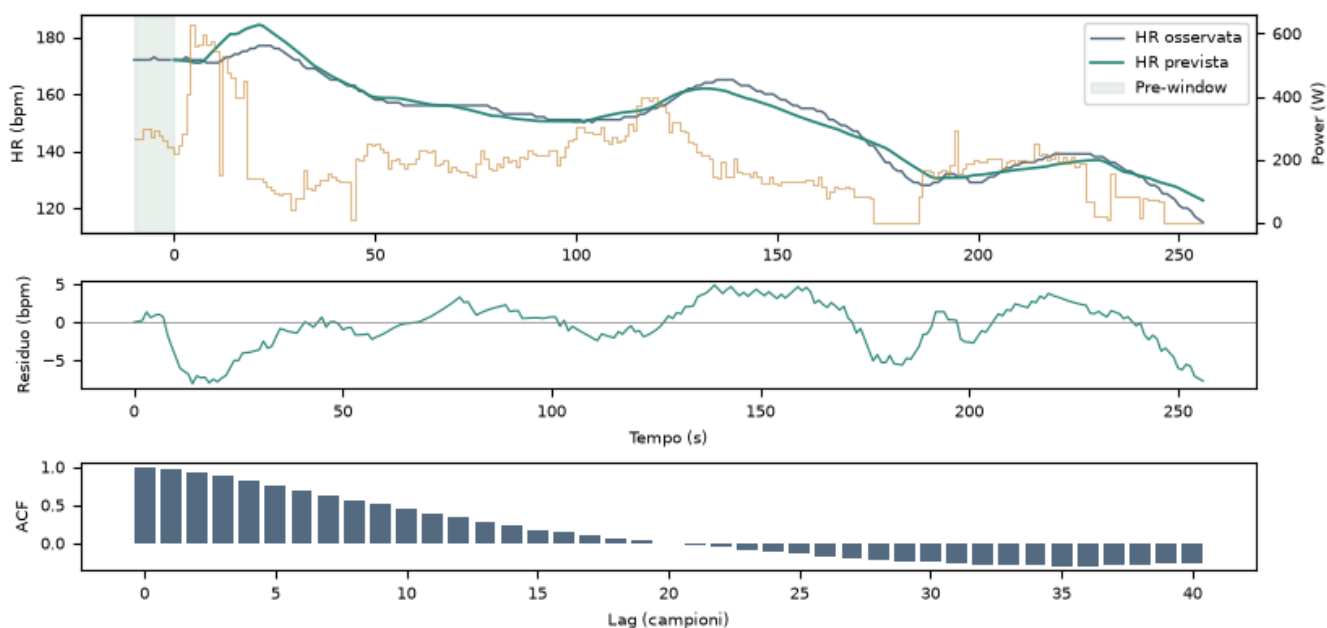
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=270.4 W; HR0=172 bpm.

HR(0) osservata fissata; B=165.42 bpm stimato; x(0)=6.585 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.0677	2.4062	-0.36025	0.95834	0.99396

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.36025	3.0524	0.96445	0.76052



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.29756	0.0073429	0.28317 / 0.31195	2.4677	no	no
L	3.423	0.37461	2.6888 / 4.1572	10.944	no	no
tau	55.806	2.3981	51.106 / 60.506	4.2972	no	no
B	165.42	0.48955	164.46 / 166.37	0.29595	no	no

rho=4.526; Jacobian rank=4/4; cond=878.06; accordo multistart=no.

Correlazioni: {"K_L": -0.13359493721595572, "K_tau": 0.6438039354737201, "K_B": 0.5762961811011085, "L_tau": -0.2500743870104717, "L_B": -0.10247545157904524, "tau_B": -0.11906723152617384}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 2418.6 / 15330

Griglia: 42441 coppie L/tau; SSE multistart originale: 2418.6; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250625105513__laps_013-016.csv

ID: 9da9a54d1702e65f30769703 | Stato: fitted

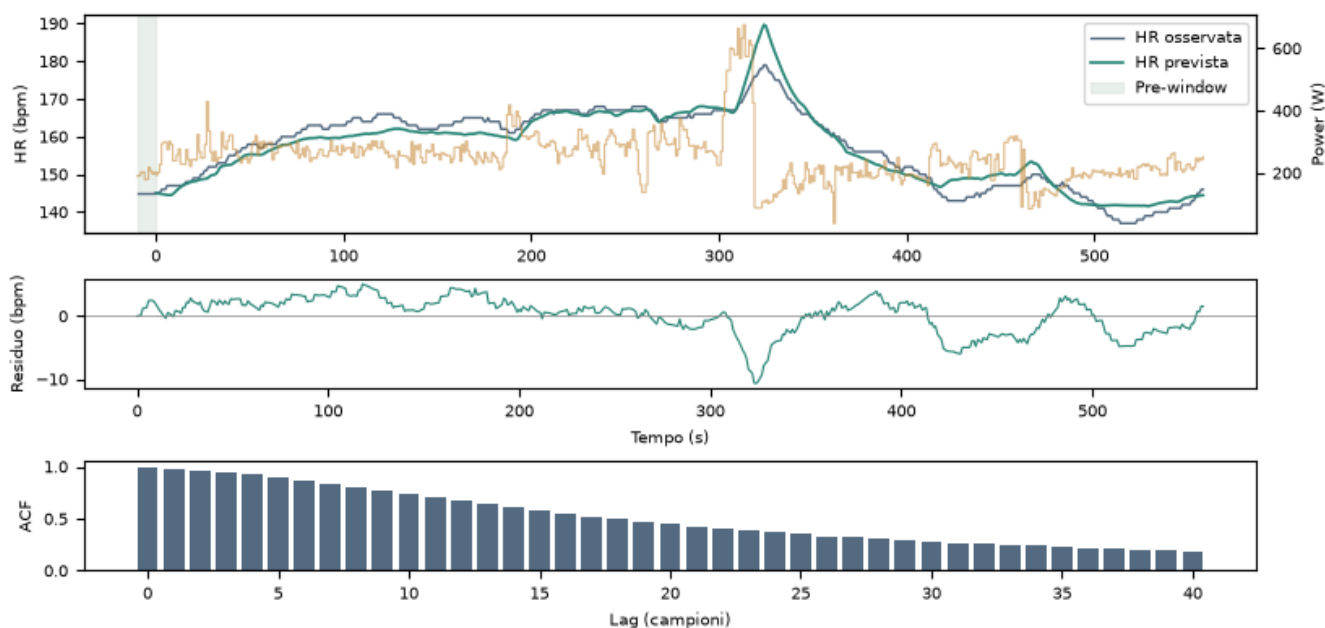
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=197 W; HR0=145 bpm.

HR(0) osservata fissata; B=141.05 bpm stimato; x(0)=3.9488 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.8496	2.2945	0.16513	0.92091	1.1447

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.16513	2.8473	0.98423	0.8963



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.29541	0.005512	0.28461 / 0.30621	1.8659	no	no
L	5.6583	0.30701	5.0566 / 6.2601	5.4258	no	no
tau	51.764	1.5734	48.68 / 54.848	3.0395	no	no
B	141.05	0.35506	140.36 / 141.75	0.25172	no	no

rho=10.67; Jacobian rank=4/4; cond=835.51; accordo multistart=no.

Correlazioni: {"K_L": -0.24581930496349577, "K_tau": 0.7419232872392403, "K_B": -0.9298002213870571, "L_tau": -0.36622016039285454, "L_B": 0.22492975461207937, "tau_B": -0.7041028245718226}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 4539.1 / 57237

Griglia: 42441 coppie L/tau; SSE multistart originale: 4539.1; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_086-087.csv

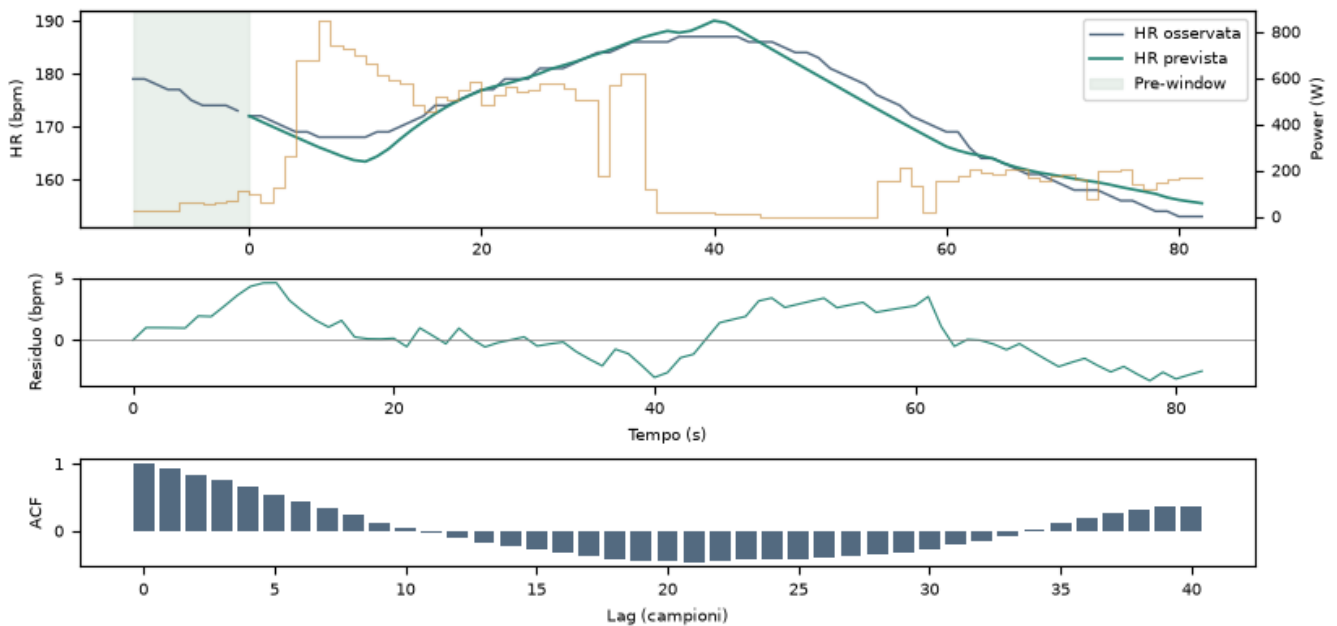
ID: a1ec45de9bf97f324b4258c9 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=53.1 W; HR0=172 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=112 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.0861	1.6959	0.42242	0.96008	1.0151
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.42242	2.0553	0.92325	0.54291	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.41863	0.17949	0.066821 / 0.77043	42.877	no	no
L	6.2826	0.35345	5.5898 / 6.9753	5.6258	no	no
tau	110.72	49.811	13.092 / 208.35	44.988	no	no
B	60	49.729	-37.47 / 157.47	82.882	si	si

rho=0.68386; Jacobian rank=4/4; cond=26683; accordo multistart=no.

Correlazioni: {"K_L": -0.7662865457750175, "K_tau": 0.998432941321063, "K_B": -0.9998839782064415, "L_tau": -0.7767005118098987, "L_B": 0.7688987503072647, "tau_B": -0.9985627023541782}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 361.2 / 8852.3

Griglia: 42441 coppie L/tau; SSE multistart originale: 361.2; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_012-013.csv

ID: a59f40db08815016b2c9a1b2 | Stato: fitted

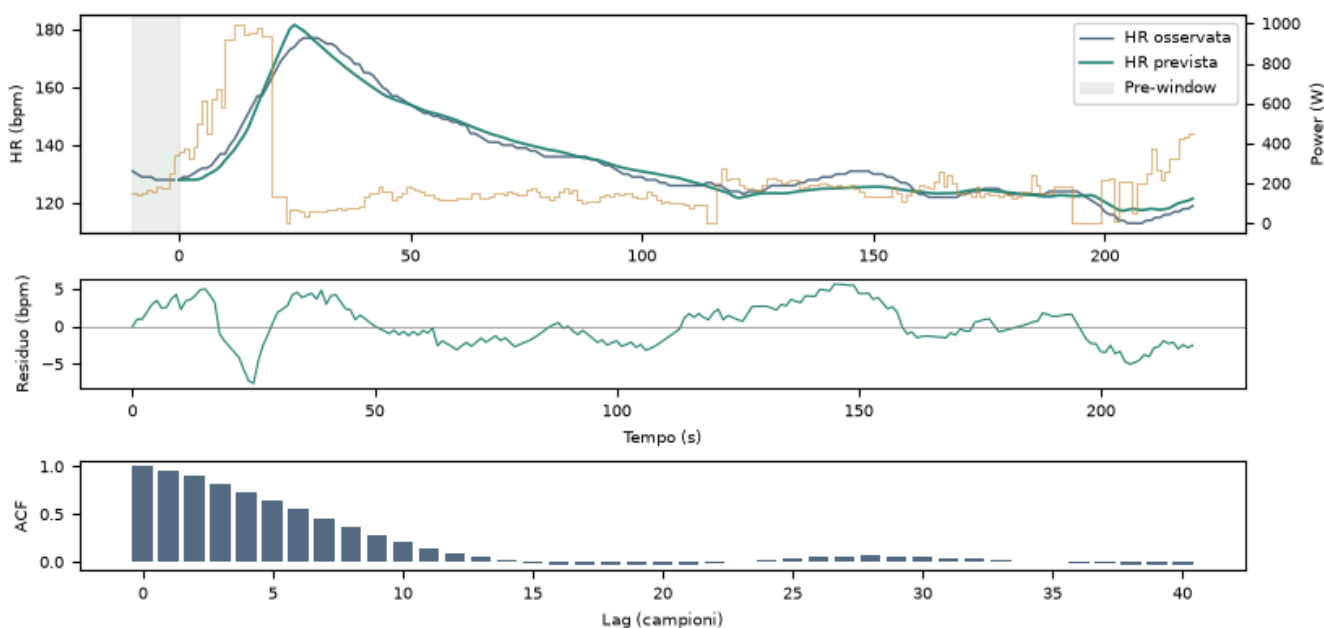
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=190.4 W; HR0=128 bpm.

HR(0) osservata fissata; B=129.58 bpm stimato; x(0)=-1.5751 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.6956	2.2468	0.15816	0.97275	1.0039

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.15816	2.6971	0.95607	0.64486



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.37699	0.010807	0.35581 / 0.39818	2.8666	no	no
L	4.5287	0.19524	4.1461 / 4.9114	4.3112	no	no
tau	66.774	2.4532	61.966 / 71.582	3.6738	no	no
B	129.58	0.34754	128.89 / 130.26	0.26822	no	no

rho=3.2119; Jacobian rank=4/4; cond=622.92; accordo multistart=no.

Correlazioni: {"K_L": -0.36962950149729734, "K_tau": 0.9299526222041814, "K_B": -0.6754565295750306, "L_tau": -0.40344479353780743, "L_B": 0.19602529984882663, "tau_B": -0.7032502299563512}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1598.6 / 58506

Griglia: 42441 coppie L/tau; SSE multistart originale: 1598.6; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_022-023.csv

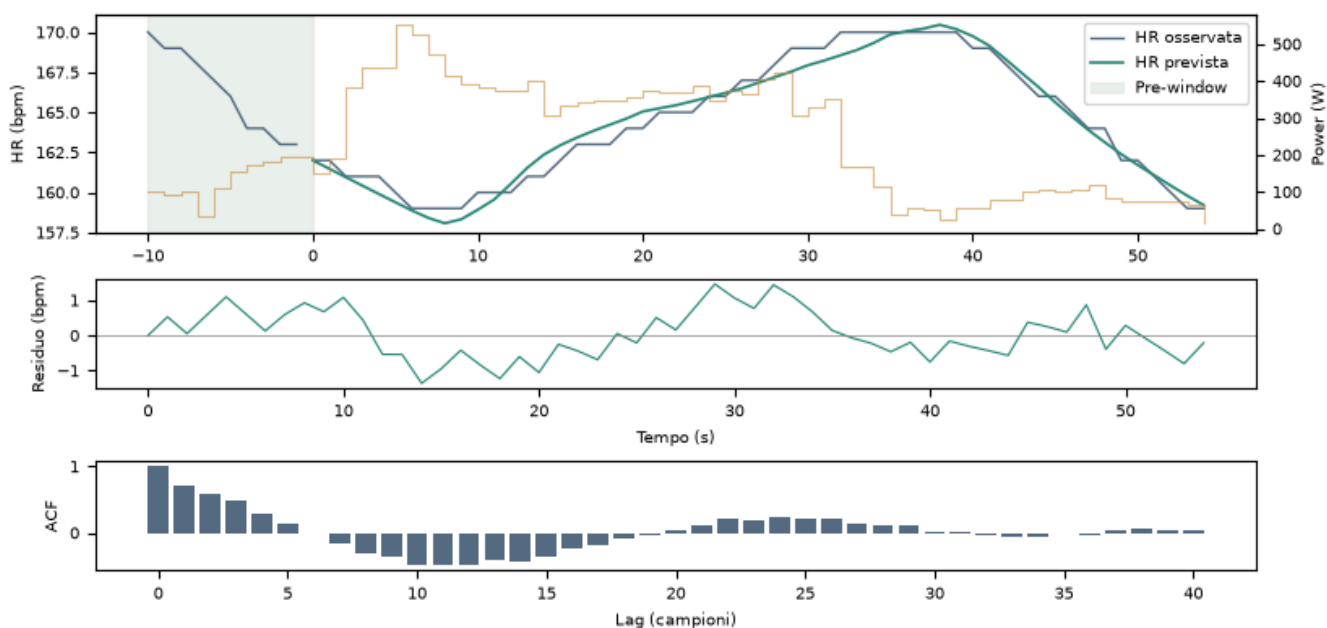
ID: a72221d63f18f9fef41b38f6 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=134 W; HR0=162 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=102 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.67764	0.56185	0.047608	0.96807	1.1262
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.047608	0.68219	0.70521		0.13663



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.73728	0.90427	-1.0351 / 2.5096	122.65	no	no
L	6.2524	0.37126	5.5247 / 6.9801	5.9379	no	no
tau	192.54	237.26	-272.49 / 657.57	123.23	no	no
B	60	128.06	-190.99 / 310.99	213.43	si	si

rho=0.24799; Jacobian rank=4/4; cond=69988; accordo multistart=no.

Correlazioni: {"K_L": -0.9160721005265231, "K_tau": 0.9997193396263057, "K_B": -0.9999795968766282, "L_tau": -0.9184890101201695, "L_B": 0.9163209576546499, "tau_B": -0.9996158056742634}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 25.256 / 605.79

Griglia: 42441 coppie L/tau; SSE multistart originale: 25.256; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_004-005.csv

ID: a7b44580dd83fe773f75f1f6 | Stato: fitted

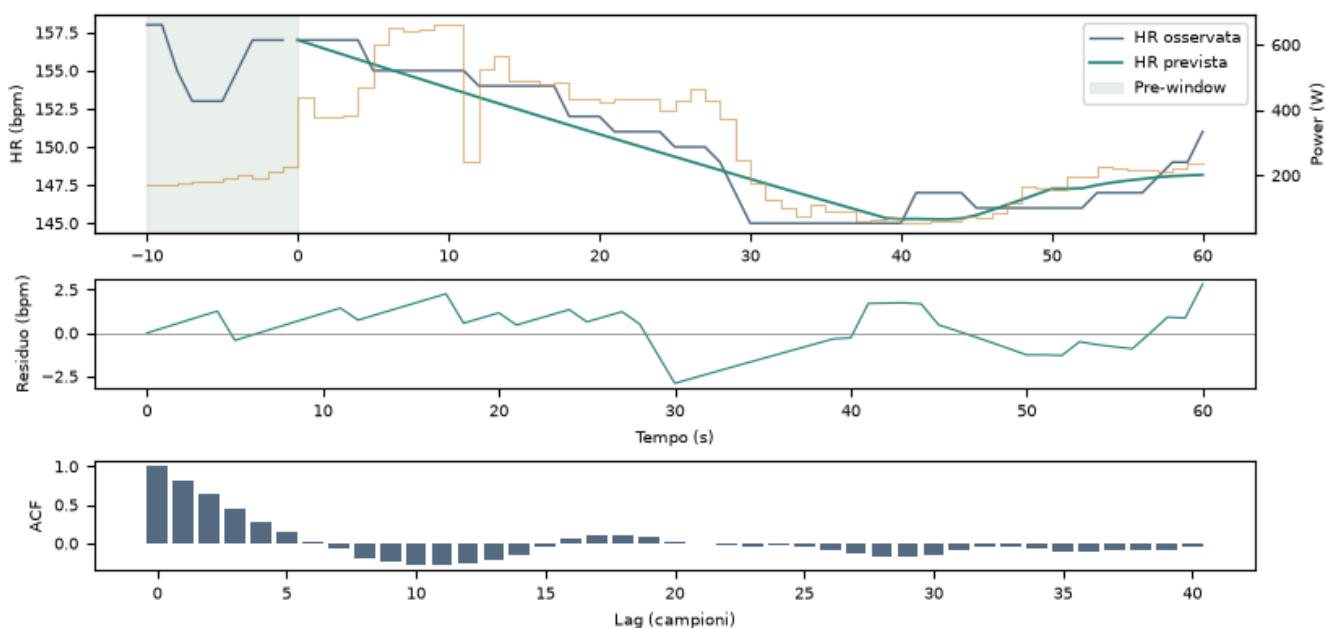
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=190.4 W; HR0=157 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=97 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.255	1.0571	0.18826	0.90587	0.9788

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.18826	1.2511	0.81067	0.15134



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.41857	0.95209	-1.4475 / 2.2847	227.46	no	no
L	39.367	1.3672	36.687 / 42.047	3.4731	no	no
tau	304.55	660.8	-990.62 / 1599.7	216.98	no	no
B	60	200.05	-332.09 / 452.09	333.41	si	si

rho=0.067748; Jacobian rank=4/4; cond=30418; accordo multistart=no.

Correlazioni: {"K_L": -0.4987780512008351, "K_tau": 0.9991288414299151, "K_B": -0.999255276526424, "L_tau": -0.5160428035230931, "L_B": 0.5212661916590638, "tau_B": -0.9999080270472162}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 96.082 / 214.82

Griglia: 42441 coppie L/tau; SSE multistart originale: 96.082; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250715165154__laps_004-005.csv

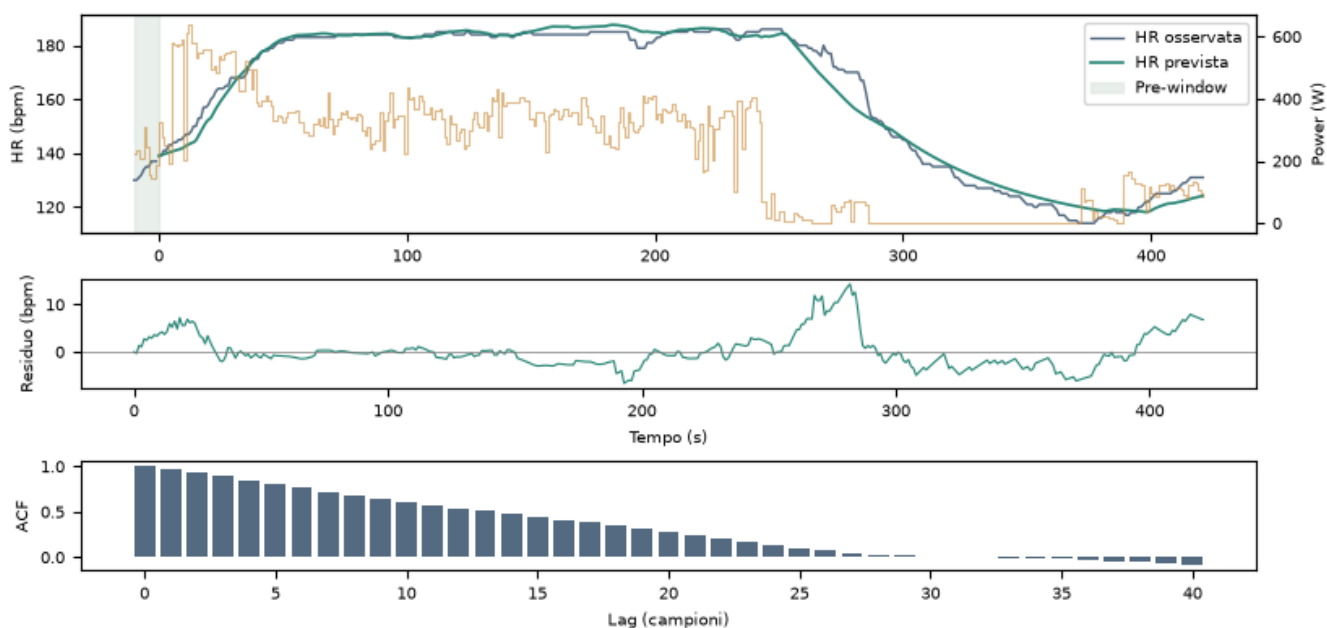
ID: a8e22f657d5002d3f9d9b849 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=206.9 W; HR0=139 bpm.

HR(0) osservata fissata; B=156.57 bpm stimato; x(0)=-17.57 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.6529	2.5606	0.17013	0.98057	0.96551
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.17013	3.6532	0.97152	0.80862	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.22319	0.002885	0.21754 / 0.22885	1.2926	no	no
L	9.6124	0.54026	8.5535 / 10.671	5.6205	no	no
tau	55.105	1.4272	52.308 / 57.903	2.59	no	no
B	156.57	0.2503	156.08 / 157.06	0.15987	no	no

rho=7.4655; Jacobian rank=4/4; cond=1036.8; accordo multistart=no.

Correlazioni: {"K_L": -0.5020999284722072, "K_tau": 0.8573812438974536, "K_B": -0.5499682997733589, "L_tau": -0.6613684227813739, "L_B": 0.12796125969956906, "tau_B": -0.5065885036092386}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 5630.9 / 2.8508e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 5630.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250715165154__laps_002-003.csv

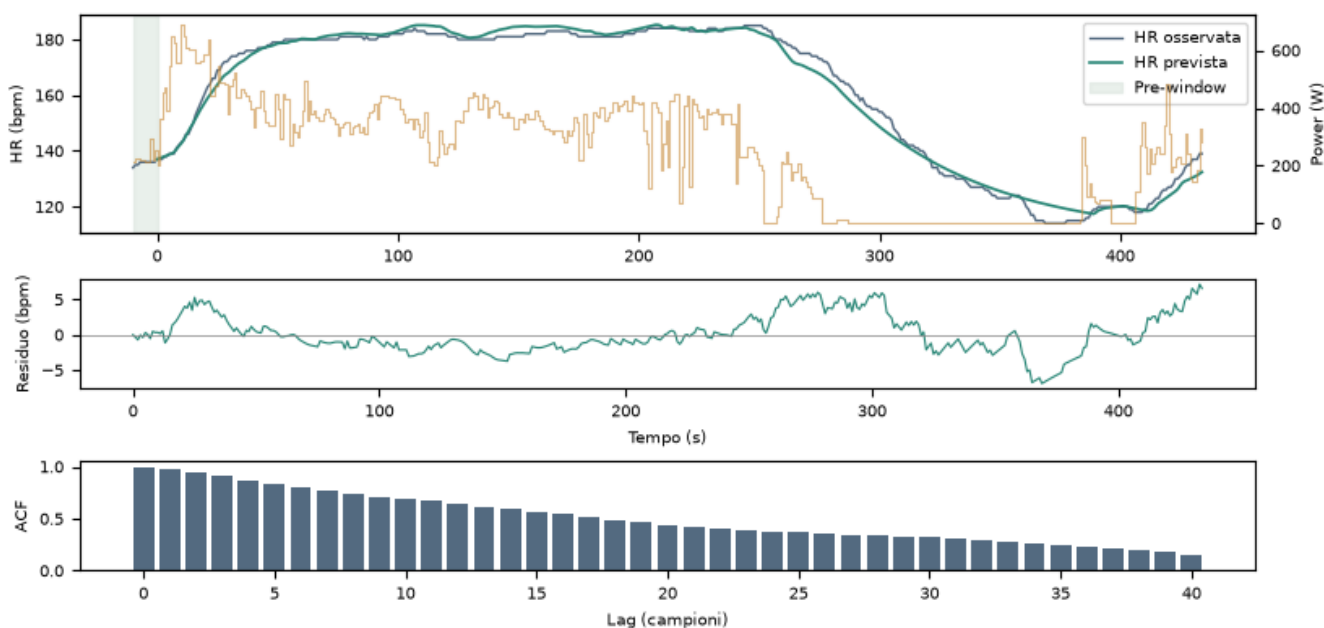
ID: aa444adc93a9b5093410ff88 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=229.3 W; HR0=137 bpm.

HR(0) osservata fissata; B=157.21 bpm stimato; x(0)=-20.211 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.7257	2.1128	0.072246	0.98848	0.95577
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.072246	2.7279	0.96927	0.83606	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21028	0.0020708	0.20622 / 0.21434	0.98477	no	no
L	5.4194	0.40159	4.6323 / 6.2065	7.4102	no	no
tau	57.019	1.1015	54.86 / 59.178	1.9318	no	no
B	157.21	0.17366	156.87 / 157.55	0.11046	no	no

rho=7.5164; Jacobian rank=4/4; cond=1118.3; accordo multistart=no.

Correlazioni: {"K_L": -0.5494940608158349, "K_tau": 0.8651693569210163, "K_B": -0.482916655067838, "L_tau": -0.6783710118970372, "L_B": 0.14600490825682735, "tau_B": -0.46399569641186}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 3231.8 / 2.7537e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 3231.8; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_039-040.csv

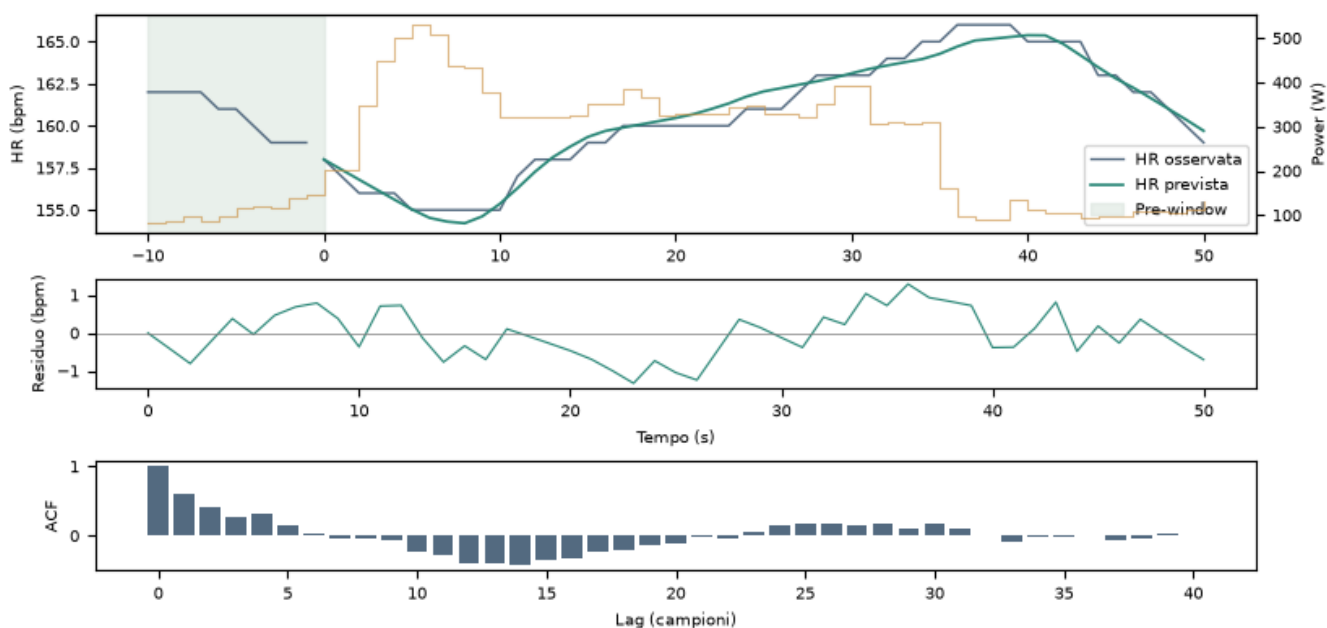
ID: ad3ebb337dccf39878288de5 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=109.1 W; HR0=158 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=98 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.62068	0.51992	-0.032717	0.9679	1.0157
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.032717	0.62598	0.61281	0.14561	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.61722	0.60112	-0.56097 / 1.7954	97.391	no	no
L	5.7762	0.34241	5.105 / 6.4473	5.928	no	no
tau	162.81	158.12	-147.11 / 472.73	97.122	no	no
B	60	97.986	-132.05 / 252.05	163.31	si	si

rho=0.27163; Jacobian rank=4/4; cond=64513; accordo multistart=no.

Correlazioni: {"K_L": -0.9049264003010334, "K_tau": 0.999107412558976, "K_B": -0.9999344204506837, "L_tau": -0.9026698655438055, "L_B": 0.9067185393897871, "tau_B": -0.9986539850015778}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 19.647 / 276.65

Griglia: 42441 coppie L/tau; SSE multistart originale: 19.647; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250708171459__laps_017-018.csv

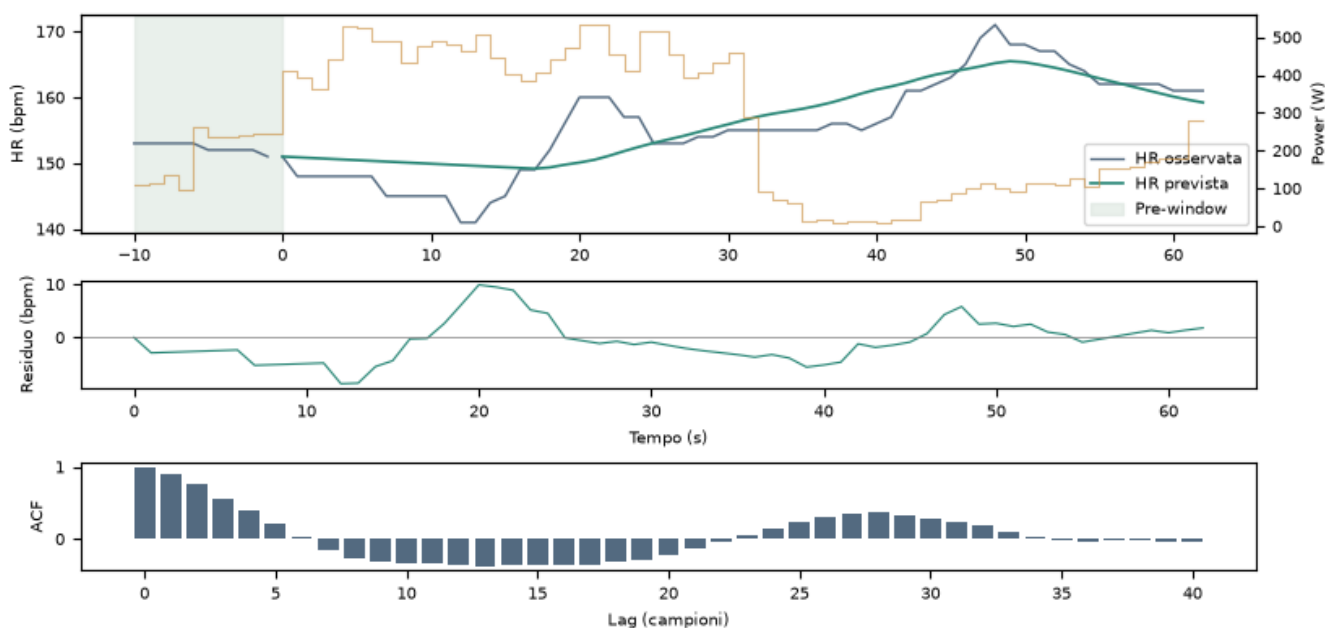
ID: b0c97b23215c4ef228ad5192 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=191.9 W; HR0=151 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=91 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.9375	3.0722	-0.6718	0.71871	0.54369
residual_mean	residual_sd	ACF lag 1		ACF lag 5
-0.6718	3.9109	0.89916		0.20684



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	2.0211	30.205	-57.181 / 61.223	1494.5	no	no
L	17.445	2.6969	12.159 / 22.731	15.46	no	no
tau	833.49	12557	-23777 / 25444	1506.5	no	no
B	60	1398.8	-2681.7 / 2801.7	2331.4	si	si

rho=0.053456; Jacobian rank=4/4; cond=1.4003e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.9096516753278882, "K_tau": 0.9999363437568431, "K_B": -0.9997410385648823, "L_tau": -0.9093467199589139, "L_B": 0.9111651422622961, "tau_B": -0.9994429535574486}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 976.75 / 1620.2

Griglia: 42441 coppie L/tau; SSE multistart originale: 976.75; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_011-012.csv

ID: b157a96f16e7f7750809b8dd | Stato: fitted

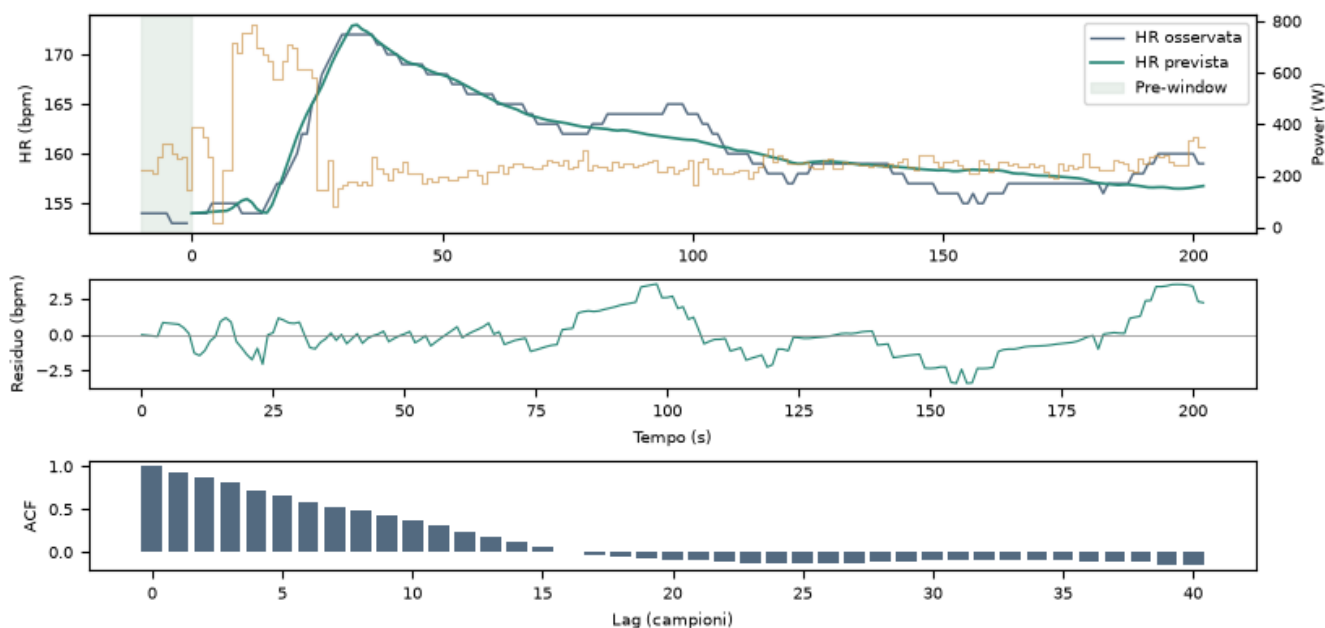
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=256.4 W; HR0=154 bpm.

HR(0) osservata fissata; B=158.32 bpm stimato; x(0)=-4.3204 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.4611	1.0789	-6.0604e-05	0.90831	1.0544

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-6.0604e-05	1.4647	0.93617	0.65562



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.32457	0.041865	0.24252 / 0.40662	12.898	no	no
L	7.4298	0.32342	6.7959 / 8.0637	4.353	no	no
tau	112.46	15.407	82.261 / 142.66	13.7	no	no
B	158.32	0.47174	157.4 / 159.24	0.29797	no	no

rho=1.7301; Jacobian rank=4/4; cond=3133.5; accordo multistart=no.

Correlazioni: {"K_L": -0.456296666357622, "K_tau": 0.9911725204061805, "K_B": -0.9223996381545184, "L_tau": -0.46896951921944435, "L_B": 0.36784263746143864, "tau_B": -0.9041552740917587}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 433.36 / 4263.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 433.36; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260321102409__laps_004-005.csv

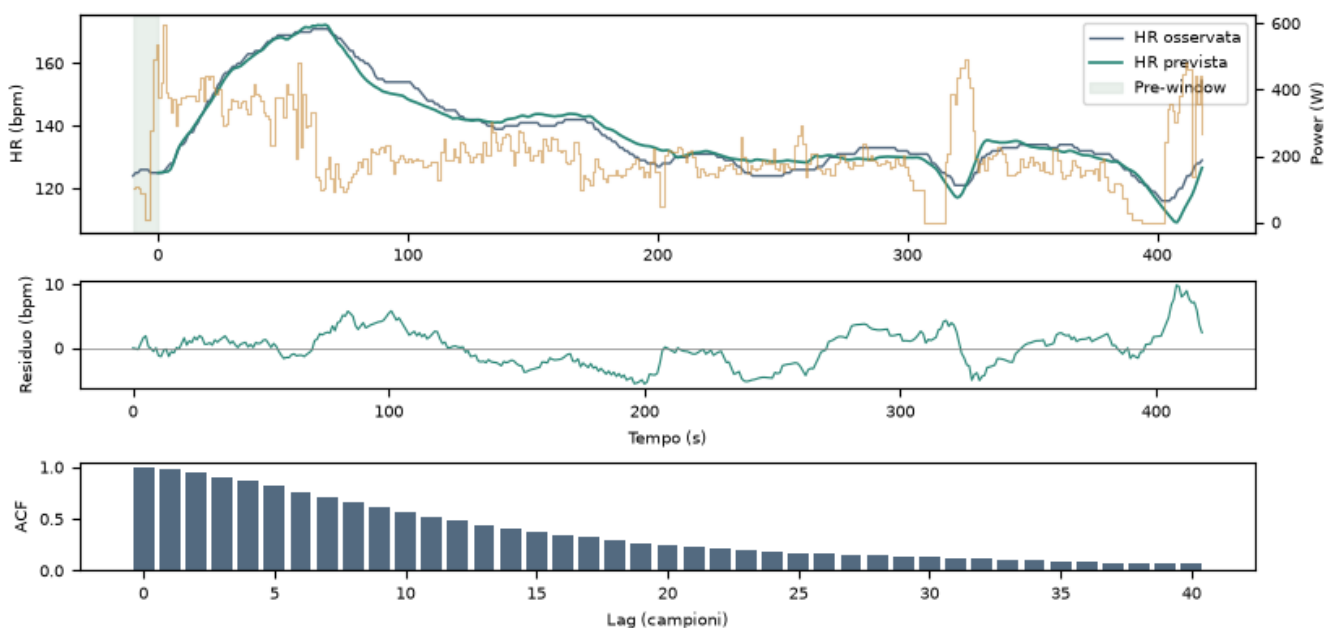
ID: b174858c39c7288152d48311 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=182.5 W; HR0=125 bpm.

HR(0) osservata fissata; B=129.97 bpm stimato; x(0)=-4.9748 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.8694	2.2928	0.065311	0.95792	1.149
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.065311	2.872	0.97646	0.81466	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.33854	0.0053353	0.32809 / 0.349	1.576	no	no
L	4.6038	0.22498	4.1628 / 5.0447	4.8868	no	no
tau	45.45	1.1473	43.201 / 47.699	2.5243	no	no
B	129.97	0.21361	129.56 / 130.39	0.16435	no	no

rho=9.0956; Jacobian rank=4/4; cond=402.68; accordo multistart=no.

Correlazioni: {"K_L": -0.3128835104222865, "K_tau": 0.8066245144118598, "K_B": -0.6828090591976214, "L_tau": -0.4120568872865266, "L_B": 0.180797696836591, "tau_B": -0.6085283493474268}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 3449.7 / 81084

Griglia: 42441 coppie L/tau; SSE multistart originale: 3449.7; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250715165154__laps_006-007.csv

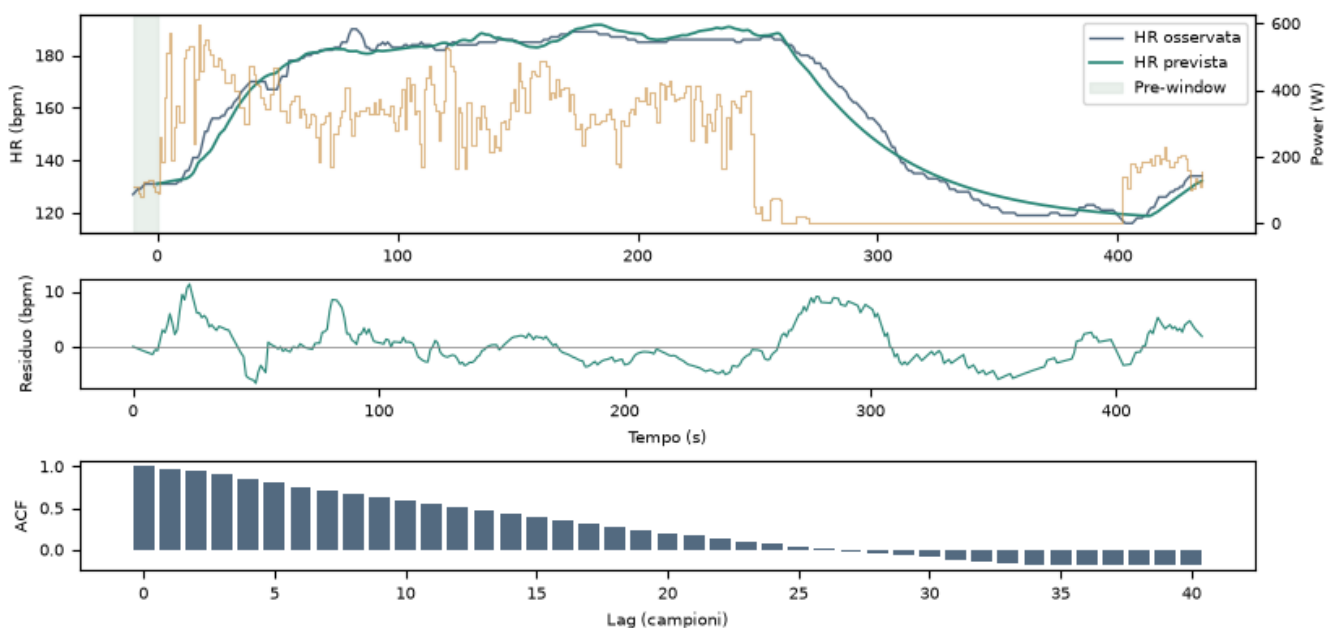
ID: b17c31f0a310602b50825ad1 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=111.3 W; HR0=131 bpm.

HR(0) osservata fissata; B=139.68 bpm stimato; x(0)=-8.6837 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.6782	2.9098	0.17225	0.98141	0.98468
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.17225	3.6784	0.97475	0.80576	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21033	0.0022433	0.20594 / 0.21473	1.0666	no	no
L	11.442	0.45543	10.549 / 12.334	3.9804	no	no
tau	44.974	1.0545	42.907 / 47.041	2.3447	no	no
B	139.68	0.3395	139.02 / 140.35	0.24305	no	no

rho=9.4179; Jacobian rank=4/4; cond=1064.6; accordo multistart=no.

Correlazioni: {"K_L": -0.505961578332629, "K_tau": 0.7894495216701777, "K_B": -0.8192087703408627, "L_tau": -0.7031962358480441, "L_B": 0.380182232933752, "tau_B": -0.6697326736865524}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 5898.8 / 3.0469e+05

Griglia: 42441 coppie L/tau; SSE multistart originale: 5910.9; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_004-005.csv

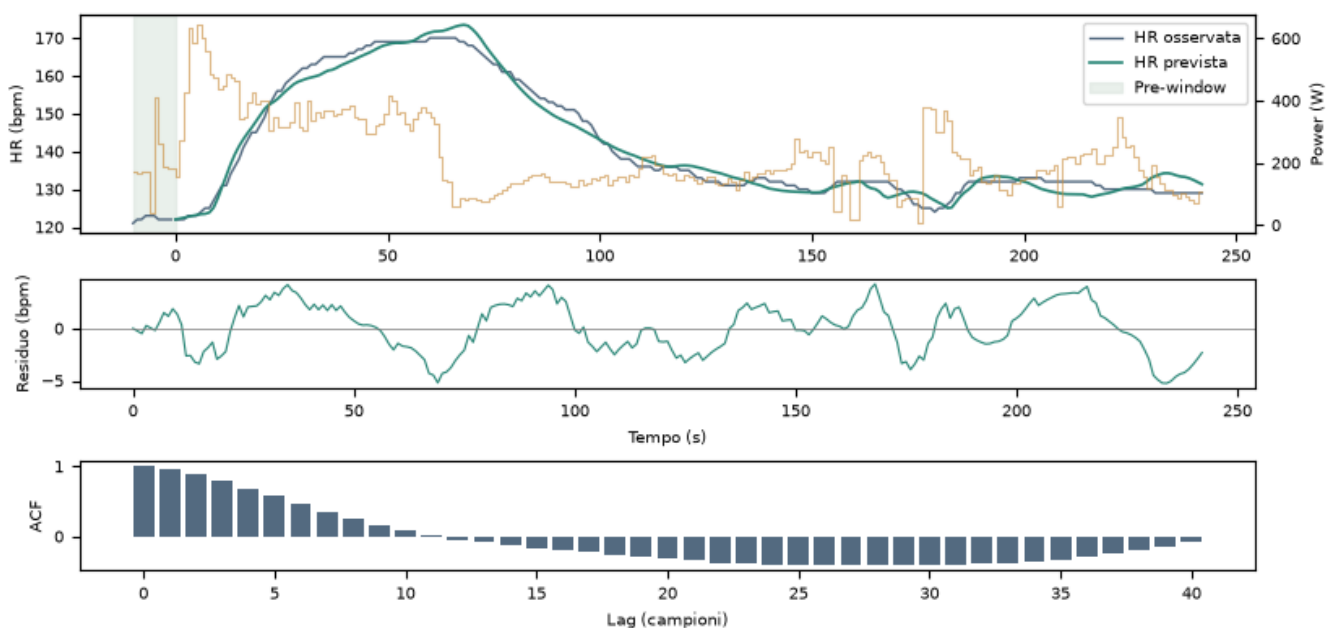
ID: b1b4978930221b9f8bb43aba | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=194.8 W; HR0=122 bpm.

HR(0) osservata fissata; B=133.9 bpm stimato; x(0)=-11.898 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.3238	1.9604	0.043716	0.97664	1.0722
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.043716	2.3282	0.95276	0.57554	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.33316	0.0067981	0.31983 / 0.34648	2.0405	no	no
L	6.5099	0.24969	6.0205 / 6.9993	3.8355	no	no
tau	47.567	1.2988	45.021 / 50.112	2.7305	no	no
B	133.9	0.25228	133.4 / 134.39	0.18841	no	no

rho=4.9507; Jacobian rank=4/4; cond=499.96; accordo multistart=no.

Correlazioni: {"K_L": -0.4994278509332096, "K_tau": 0.9078745180132344, "K_B": -0.6974666155475733, "L_tau": -0.5734842647507302, "L_B": 0.26068813519803713, "tau_B": -0.6313946909534728}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1312.2 / 53363

Griglia: 42441 coppie L/tau; SSE multistart originale: 1333.8; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_024-025.csv

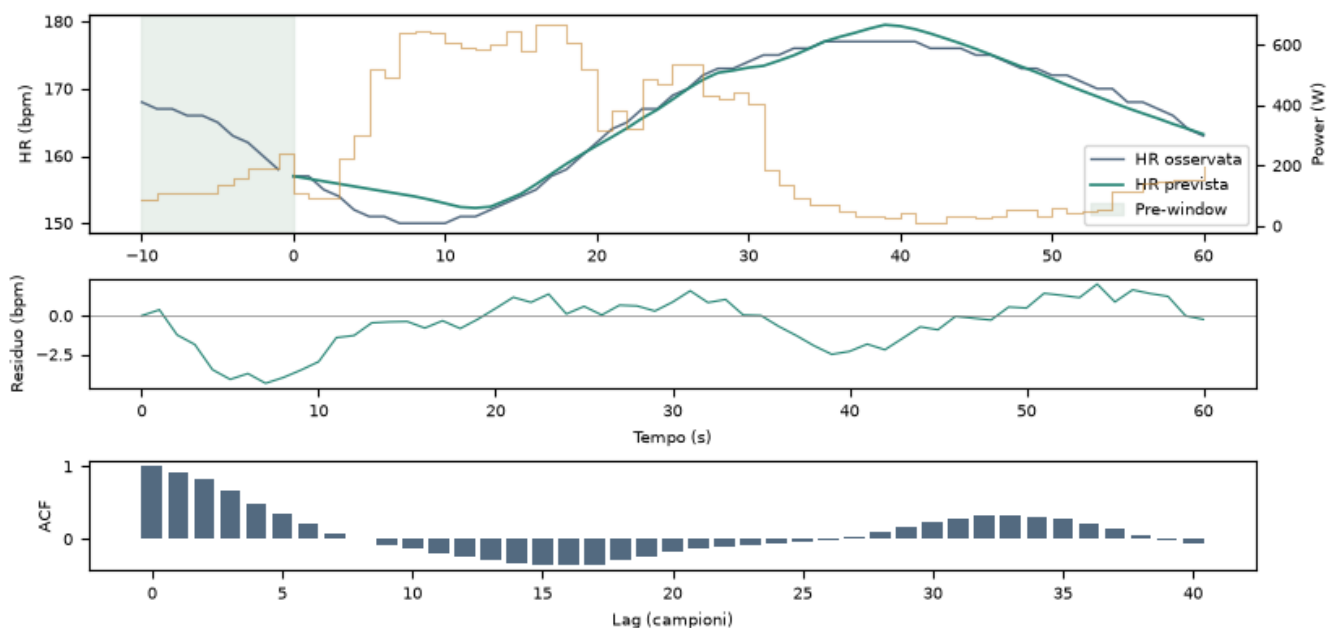
ID: c162ca94114fd93118d6d723 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=143.1 W; HR0=157 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=97 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.6524	1.2326	-0.46738	0.97018	1.0079
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.46738	1.5981	0.90784	0.34909	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.96371	1.1801	-1.3494 / 3.2768	122.46	no	no
L	8.1504	0.56768	7.0378 / 9.2631	6.965	no	no
tau	253.38	310.65	-355.5 / 862.26	122.6	no	no
B	60	128.71	-192.27 / 312.27	214.52	si	si

rho=0.20463; Jacobian rank=4/4; cond=38656; accordo multistart=no.

Correlazioni: {"K_L": -0.9110596500041687, "K_tau": 0.999590648433662, "K_B": -0.9996743809124705, "L_tau": -0.9137806075105863, "L_B": 0.90960541529196, "tau_B": -0.9986516763493415}

Multistart: 16 tentativi, 14 convergenti; SSE min/max: 166.57 / 2595.8

Griglia: 42441 coppie L/tau; SSE multistart originale: 166.57; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_022-023.csv

ID: c2b578d2d44afb5937d7392e | Stato: fitted

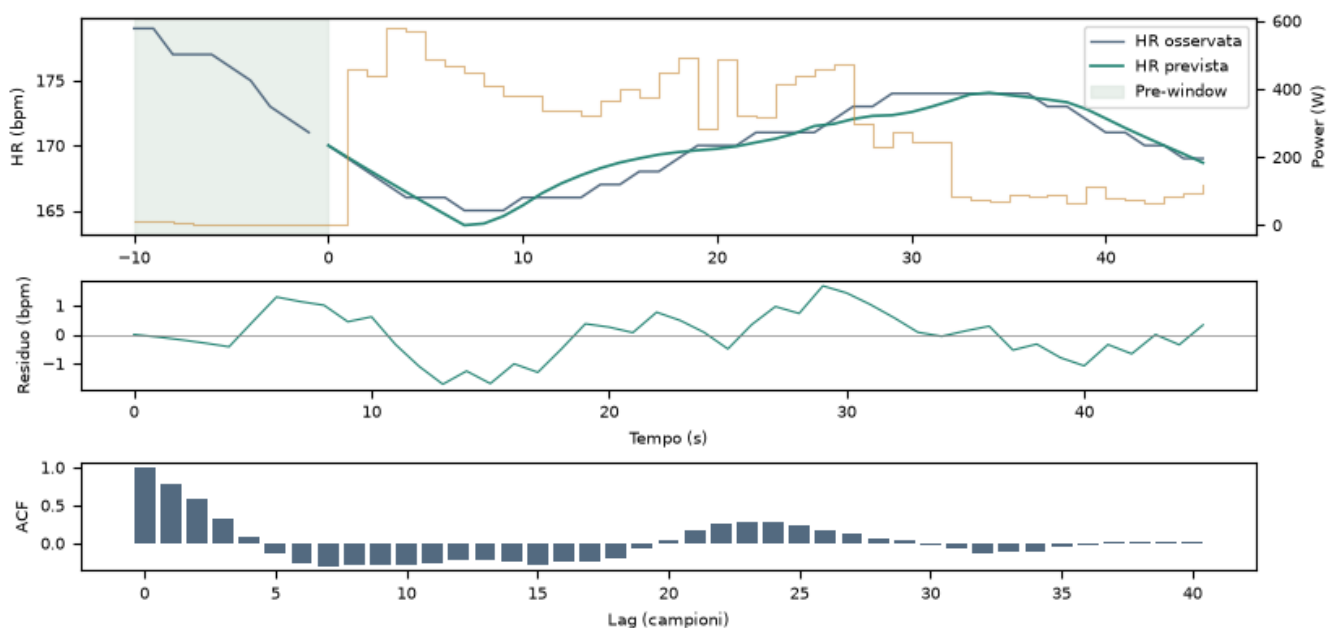
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=2.9 W; HR0=170 bpm.

HR(0) osservata fissata; B=108.12 bpm stimato; x(0)=61.876 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.79083	0.63029	-0.0013544	0.92925	1.1326

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.0013544	0.79957	0.77822	-0.12423



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.21304	0.15202	-0.084924 / 0.511	71.359	no	no
L	6.3352	0.38112	5.5882 / 7.0822	6.016	no	no
tau	67.143	48.388	-27.697 / 161.98	72.066	no	no
B	108.12	44.088	21.711 / 194.54	40.776	no	no

rho=0.57586; Jacobian rank=4/4; cond=50323; accordo multistart=no.

Correlazioni: {"K_L": -0.8793105827166371, "K_tau": 0.9977127023120576, "K_B": -0.9999214640433834, "L_tau": -0.8683326656459327, "L_B": 0.8829573528528912, "tau_B": -0.9972393452331737}

Multistart: 16 tentativi, 15 convergenti; SSE min/max: 28.769 / 370.62

Griglia: 42441 coppie L/tau; SSE multistart originale: 28.769; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_005-006.csv

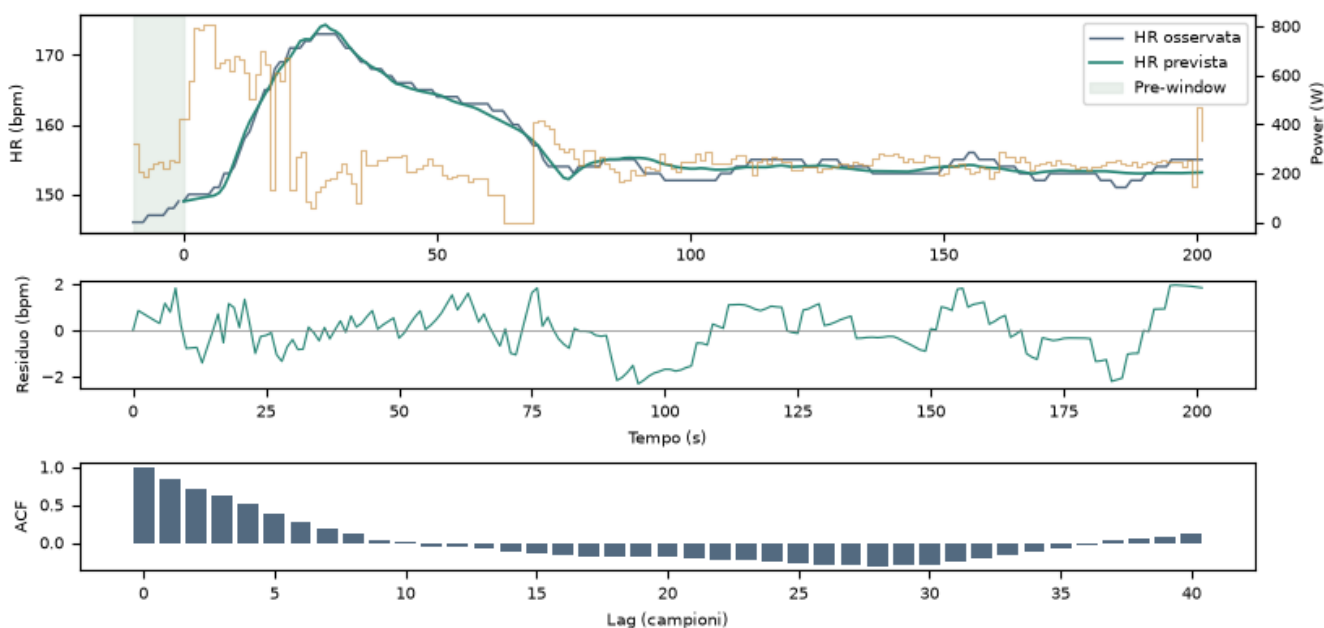
ID: c7ccd097cc8b797f41722582 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=256.2 W; HR0=149 bpm.

HR(0) osservata fissata; B=155.89 bpm stimato; $x(0)=-6.8938$ bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.97676	0.77555	0.017659	0.97518	1.0557
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.017659	0.97903	0.83808	0.39095	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.17217	0.0047474	0.16287 / 0.18148	2.7574	no	no
L	6.5712	0.1596	6.2583 / 6.884	2.4288	no	no
tau	46.966	1.578	43.873 / 50.059	3.3599	no	no
B	155.89	0.09582	155.71 / 156.08	0.061465	no	no

rho=4.1398; Jacobian rank=4/4; cond=1013.6; accordo multistart=no.

Correlazioni: {"K_L": -0.4581912634803313, "K_tau": 0.9432543344960167, "K_B": -0.43074209788070783, "L_tau": -0.4915628886711922, "L_B": 0.09867483753077412, "tau_B": -0.40940307783973073}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 192.72 / 7459.4

Griglia: 42441 coppie L/tau; SSE multistart originale: 192.72; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_008-009.csv

ID: c9f23a286f8476bb5c88043f | Stato: fitted

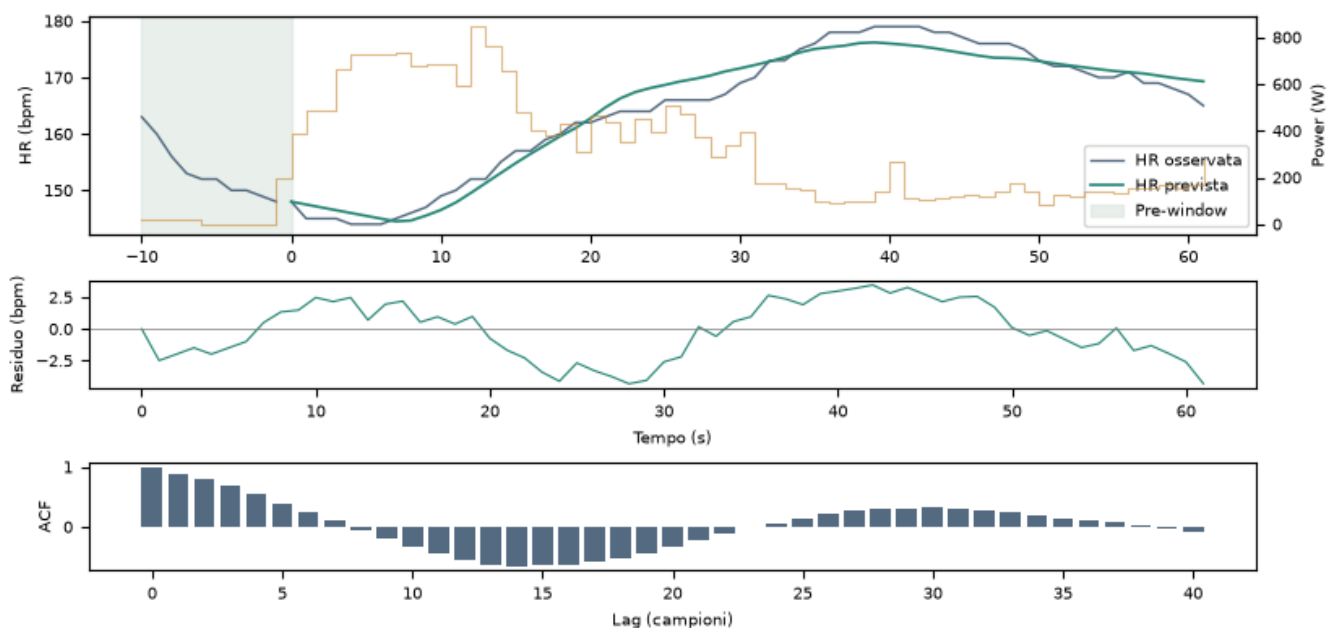
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=28.8 W; HR0=148 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=88 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.2336	1.9219	-0.079383	0.96128	0.90602

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.079383	2.2504	0.87777	0.39924



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.56702	0.60001	-0.609 / 1.7431	105.82	no	no
L	7.4499	0.77267	5.9355 / 8.9644	10.372	no	no
tau	172.61	175.63	-171.63 / 516.84	101.75	no	no
B	60	113.96	-163.37 / 283.37	189.94	si	si

rho=0.31024; Jacobian rank=4/4; cond=44745; accordo multistart=no.

Correlazioni: {"K_L": -0.920081389501601, "K_tau": 0.9987550947659674, "K_B": -0.9990956828072067, "L_tau": -0.9243155022426193, "L_B": 0.9176401735345575, "tau_B": -0.9958853571931232}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 309.31 / 1849.7

Griglia: 42441 coppie L/tau; SSE multistart originale: 309.31; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_041-042.csv

ID: cc388e0623bbd5035b607924 | Stato: fitted

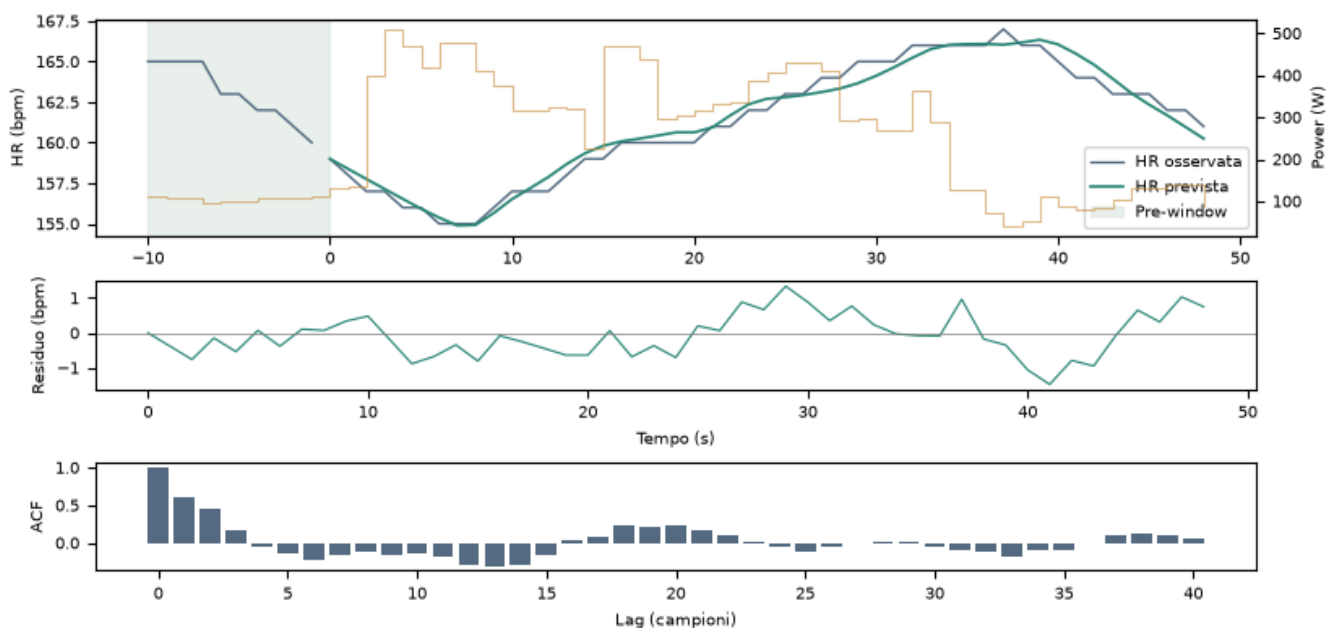
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=105 W; HR0=159 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=99 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.613	0.49346	-0.075191	0.97075	0.95456

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.075191	0.61467	0.60798	-0.12928



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.60259	0.53577	-0.44752 / 1.6527	88.912	no	no
L	5.486	0.293	4.9117 / 6.0603	5.3409	no	no
tau	158.65	141.78	-119.24 / 436.54	89.366	no	no
B	60	89.863	-116.13 / 236.13	149.77	si	si

rho=0.26797; Jacobian rank=4/4; cond=58305; accordo multistart=no.

Correlazioni: {"K_L": -0.8839411549725087, "K_tau": 0.9990700020113321, "K_B": -0.9999303475403867, "L_tau": -0.8815874848347575, "L_B": 0.8860286188389079, "tau_B": -0.998609909222829}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 18.412 / 303.76

Griglia: 42441 coppie L/tau; SSE multistart originale: 18.412; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_014-015.csv

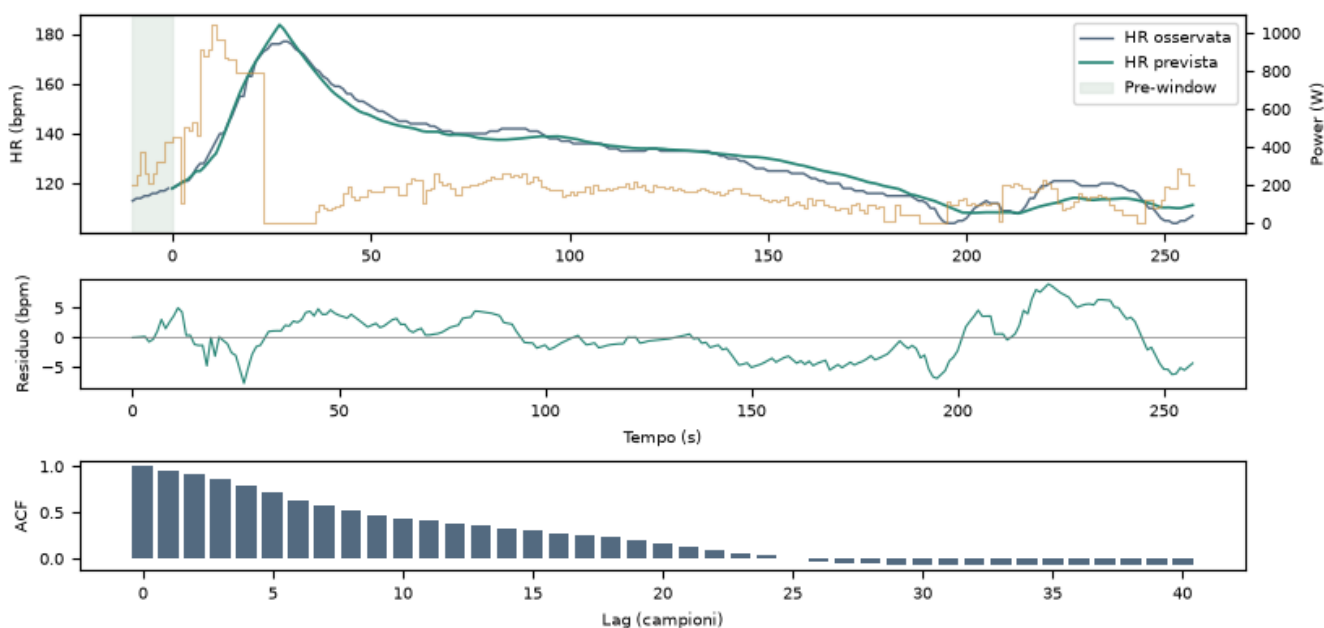
ID: d3beeaabe983fc8c3d8599df | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=307.3 W; HR0=118 bpm.

HR(0) osservata fissata; B=158.92 bpm stimato; x(0)=-40.923 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.5126	2.8525	0.10158	0.96112	1.0378
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.10158	3.518	0.95913	0.71538	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.24878	0.0057788	0.23745 / 0.26011	2.3229	no	no
L	4.1807	0.25082	3.6891 / 4.6723	5.9994	no	no
tau	42.024	1.4385	39.205 / 44.843	3.423	no	no
B	158.92	0.65819	157.63 / 160.21	0.41415	no	no

rho=6.0161; Jacobian rank=4/4; cond=900.76; accordo multistart=no.

Correlazioni: {"K_L": -0.3877395437933931, "K_tau": 0.8728802235631169, "K_B": 0.9141484609298814, "L_tau": -0.4027118295420138, "L_B": -0.4318528952626423, "tau_B": 0.758876786257612}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 3183.4 / 81280

Griglia: 42441 coppie L/tau; SSE multistart originale: 3183.4; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_058-059.csv

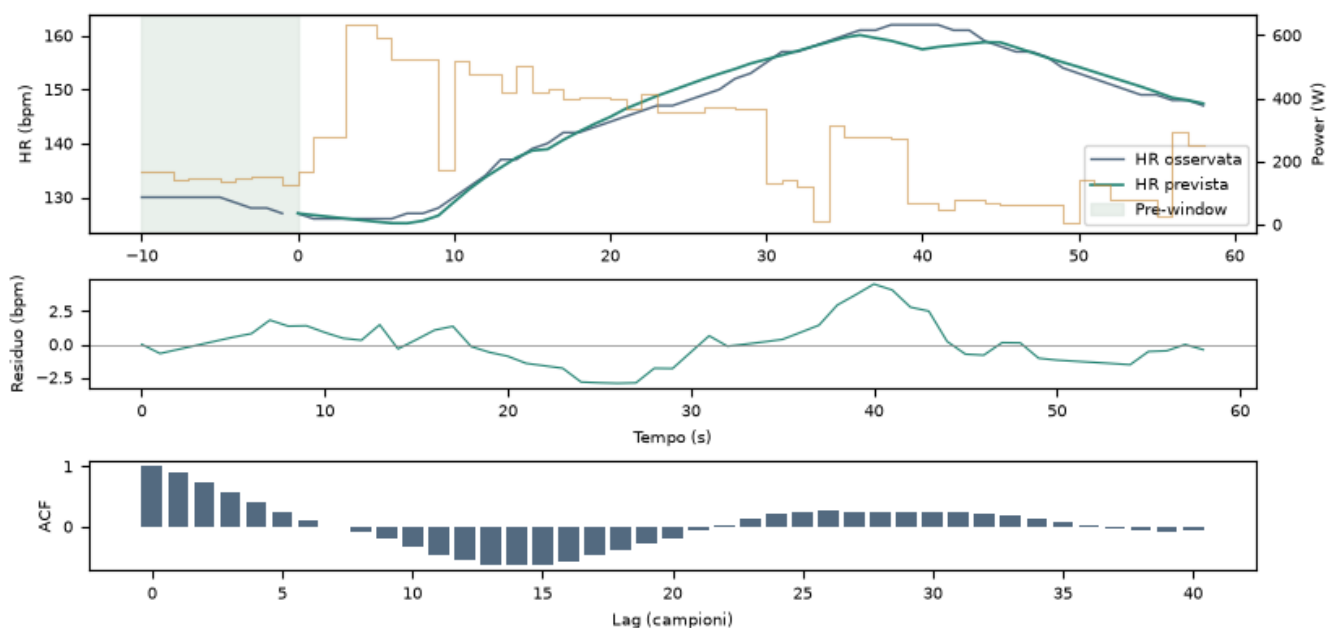
ID: d9c43da2a87aaf2001f63451 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=148.4 W; HR0=127 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=67 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.6228	1.2141	0.035493	0.98155	0.97019
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.035493	1.6363	0.89191	0.24506	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	1.2506	0.98816	-0.68618 / 3.1874	79.014	no	no
L	5.7483	0.40529	4.954 / 6.5427	7.0505	no	no
tau	214.82	165.07	-108.71 / 538.36	76.84	no	no
B	60	70.023	-77.245 / 197.25	116.71	si	si

rho=0.24323; Jacobian rank=4/4; cond=20997; accordo multistart=no.

Correlazioni: {"K_L": -0.8643787310480704, "K_tau": 0.999054286287673, "K_B": -0.9977277799941713, "L_tau": -0.8724624438511409, "L_B": 0.8515634004903091, "tau_B": -0.9940235949965981}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 155.37 / 1919.7

Griglia: 42441 coppie L/tau; SSE multistart originale: 155.37; selezione: multistart.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_006-007.csv

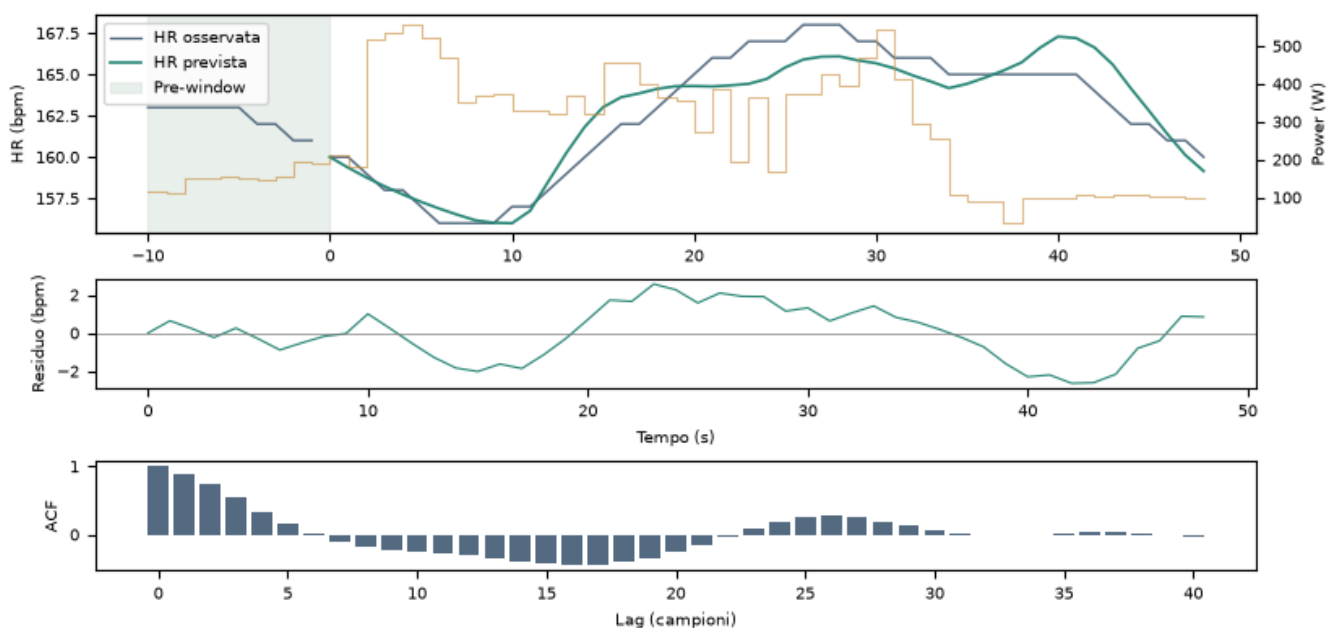
ID: dad73a96a82b268da0cc6e01 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=152.3 W; HR0=160 bpm.

HR(0) osservata fissata; B=152.62 bpm stimato; x(0)=7.375 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.381	1.1443	-0.010218	0.86496	0.94135
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.010218	1.3953	0.89996	0.15335	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.065657	0.0091048	0.047811 / 0.083502	13.867	no	no
L	8.5507	0.36597	7.8334 / 9.268	4.28	no	no
tau	10.926	1.9509	7.1023 / 14.75	17.856	no	no
B	152.62	1.4283	149.83 / 155.42	0.93582	no	no

rho=3.6106; Jacobian rank=4/4; cond=1662.6; accordo multistart=no.

Correlazioni: {"K_L": -0.6540127989771949, "K_tau": 0.8515025522366761, "K_B": -0.9808233486928031, "L_tau": -0.717332255411794, "L_B": 0.6270617335002809, "tau_B": -0.7934562026426106}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 93.449 / 451.79

Griglia: 42441 coppie L/tau; SSE multistart originale: 98.196; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/250604173939__laps_015-016.csv

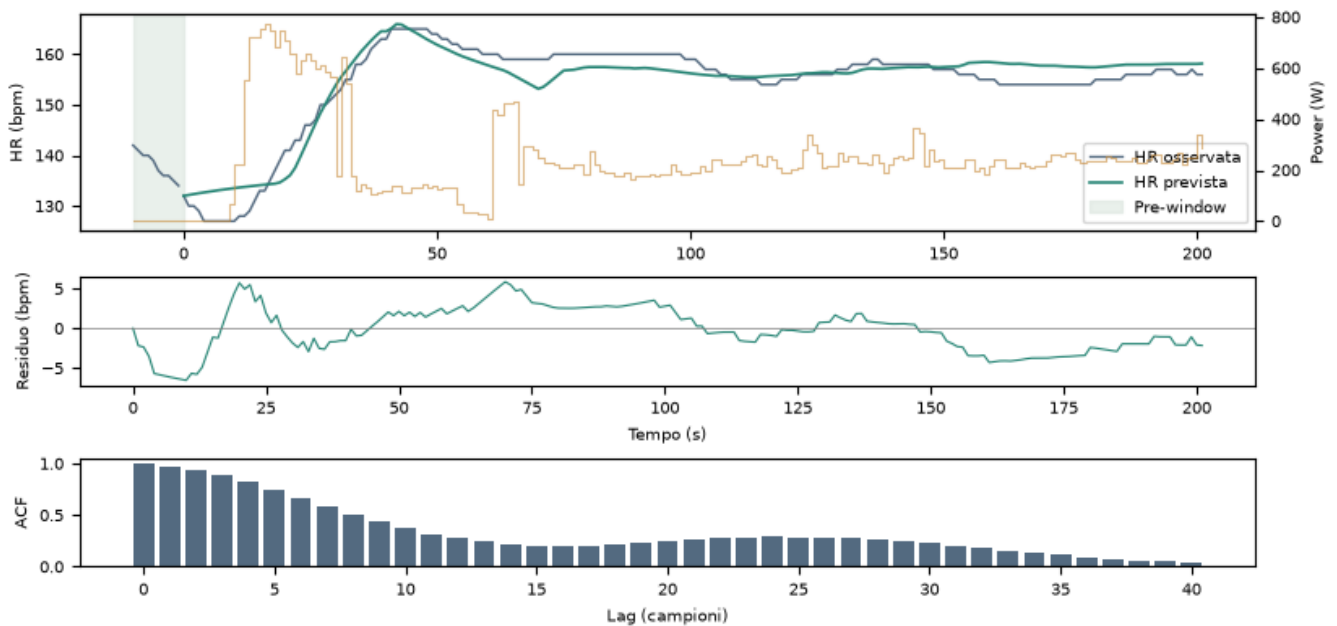
ID: e1f752e5efcb07b2d33a1d4b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=0 W; HR0=132 bpm.

HR(0) osservata fissata; B=137 bpm stimato; x(0)=-5.0001 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.793	2.3366	-0.16616	0.90353	0.89339
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.16616	2.795	0.96977	0.74627	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.089448	0.0068559	0.076011 / 0.10289	7.6647	no	no
L	9.3293	0.4869	8.375 / 10.284	5.219	no	no
tau	27.38	1.6901	24.067 / 30.693	6.1729	no	no
B	137	1.5164	134.03 / 139.97	1.1068	no	no

rho=7.0004; Jacobian rank=4/4; cond=2382.8; accordo multistart=no.

Correlazioni: {"K_L": -0.551918049570837, "K_tau": 0.811011894003373, "K_B": -0.9863256438397703, "L_tau": -0.6514676241499879, "L_B": 0.5329545232866869, "tau_B": -0.75750265186104}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1575.8 / 5332.3

Griglia: 42441 coppie L/tau; SSE multistart originale: 1575.8; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_006-007.csv

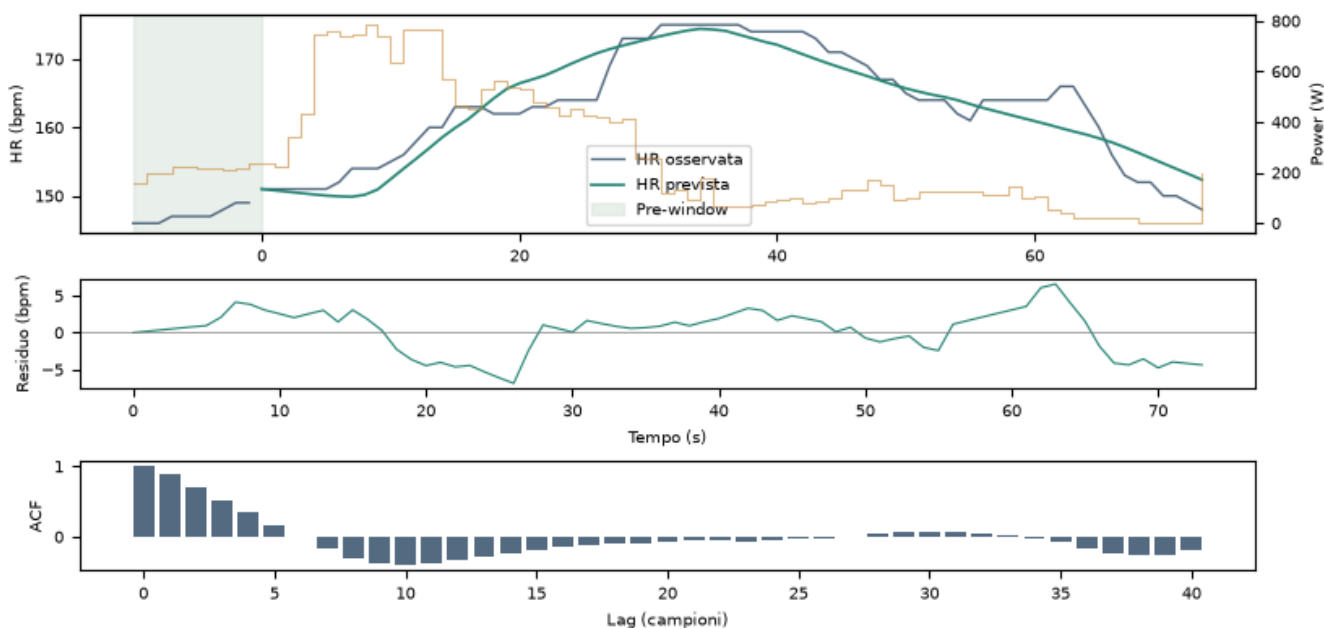
ID: e4d5945efc8258a4c905f544 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=210 W; HR0=151 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=91 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.9273	2.4089	0.17329	0.87005	0.90891
residual_mean	residual_sd	ACF lag 1		ACF lag 5
0.17329	2.9421	0.88302		0.1636



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	1.4687	6.4216	-11.118 / 14.055	437.23	no	no
L	4.7295	1.0357	2.6995 / 6.7595	21.899	no	no
tau	464.42	2025.3	-3505.1 / 4434	436.09	no	no
B	60	452.48	-826.86 / 946.86	754.14	si	si

rho=0.147; Jacobian rank=4/4; cond=84281; accordo multistart=no.

Correlazioni: {"K_L": -0.873369202523552, "K_tau": 0.9999508562284989, "K_B": -0.9998905727304964, "L_tau": -0.8755578960534033, "L_B": 0.8701257284056463, "tau_B": -0.9997256741160213}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 634.1 / 3482.6

Griglia: 42441 coppie L/tau; SSE multistart originale: 634.1; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250729173622__laps_020-021.csv

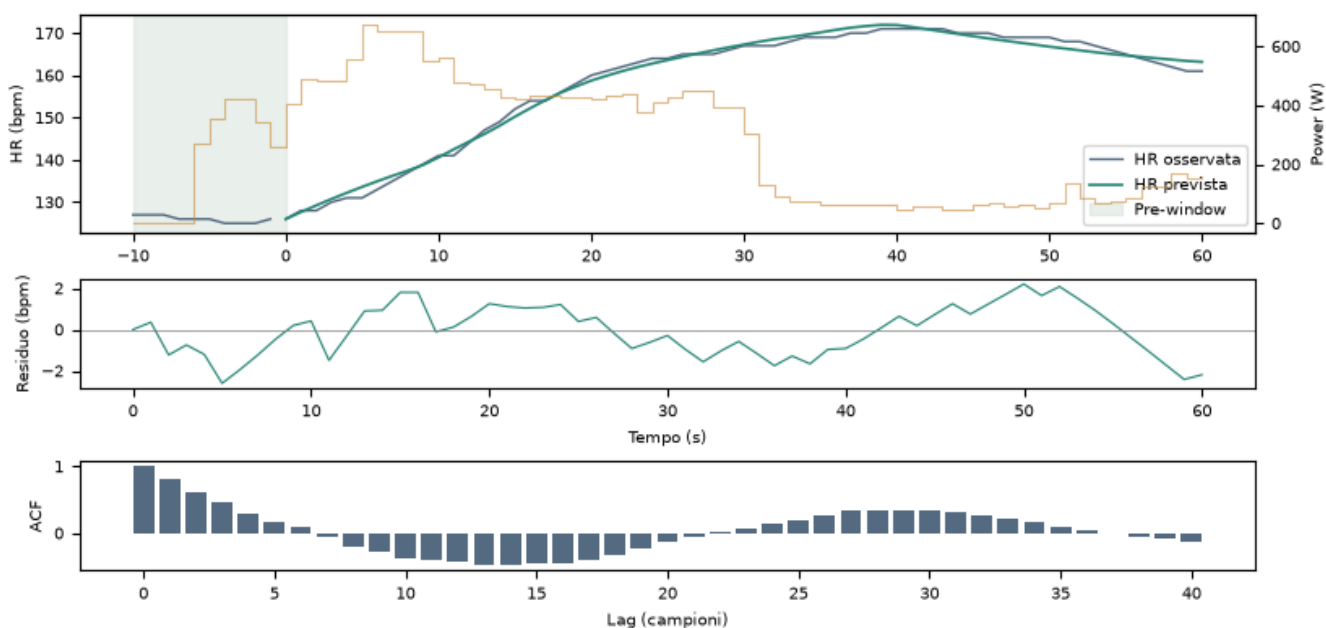
ID: e8702c0b003c461e22e469f6 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=207.8 W; HR0=126 bpm.

HR(0) osservata fissata; B=167.02 bpm stimato; x(0)=-41.018 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
1.2045	1.0245	-0.071779	0.9924	1.0214
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.071779	1.2123	0.80693	0.17027	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.074332	0.0064786	0.061634 / 0.08703	8.7157	no	no
L	8.7582	0.60771	7.5671 / 9.9493	6.9387	no	no
tau	24.391	1.5665	21.321 / 27.461	6.4224	no	no
B	167.02	0.72812	165.59 / 168.45	0.43595	no	no

rho=2.1008; Jacobian rank=4/4; cond=1226.6; accordo multistart=no.

Correlazioni: {"K_L": -0.7035690077352864, "K_tau": 0.8996023643399765, "K_B": 0.4295671228774496, "L_tau": -0.8636410322320284, "L_B": -0.7814923655504001, "tau_B": 0.7530954492778077}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 88.494 / 964.5

Griglia: 42441 coppie L/tau; SSE multistart originale: 88.494; selezione: multistart.

Segnalazioni: autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260424160919__laps_035-036.csv

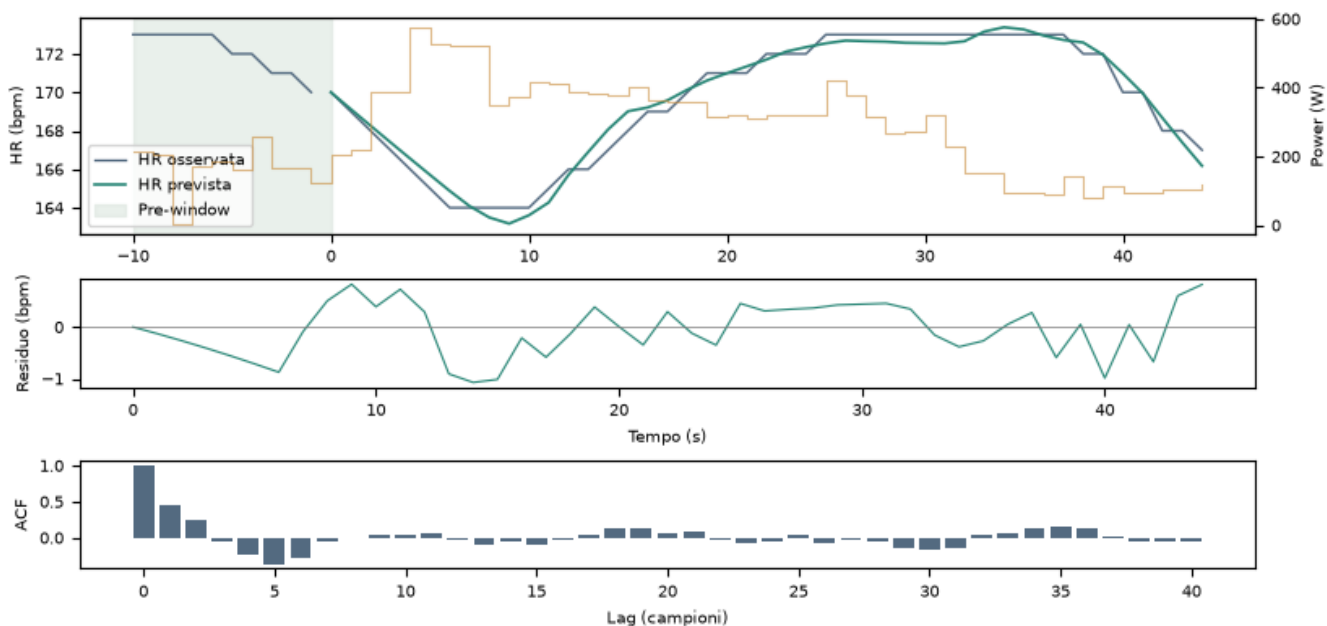
ID: f0184f3a8499a75fb100c8df | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=166.5 W; HR0=170 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=110 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.51228	0.42722	-0.052949	0.97405	1.1343
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.052949	0.51529	0.45661	-0.36138	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.71627	0.61123	-0.48173 / 1.9143	85.335	no	no
L	6.783	0.212	6.3675 / 7.1985	3.1255	no	no
tau	125.7	107.77	-85.525 / 336.92	85.735	no	no
B	60	93.498	-123.26 / 243.26	155.83	si	si

rho=0.29609; Jacobian rank=4/4; cond=51009; accordo multistart=no.

Correlazioni: {"K_L": -0.889404253128163, "K_tau": 0.9995195306358922, "K_B": -0.9999836577427071, "L_tau": -0.886458077395279, "L_B": 0.8907090968359367, "tau_B": -0.999450759240328}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 11.809 / 447.25

Griglia: 42441 coppie L/tau; SSE multistart originale: 11.809; selezione: profile_grid.

Segnalazioni: at_bounds, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250708171459__laps_019-020.csv

ID: f2eef454433d0e73fdda2300 | Stato: fitted

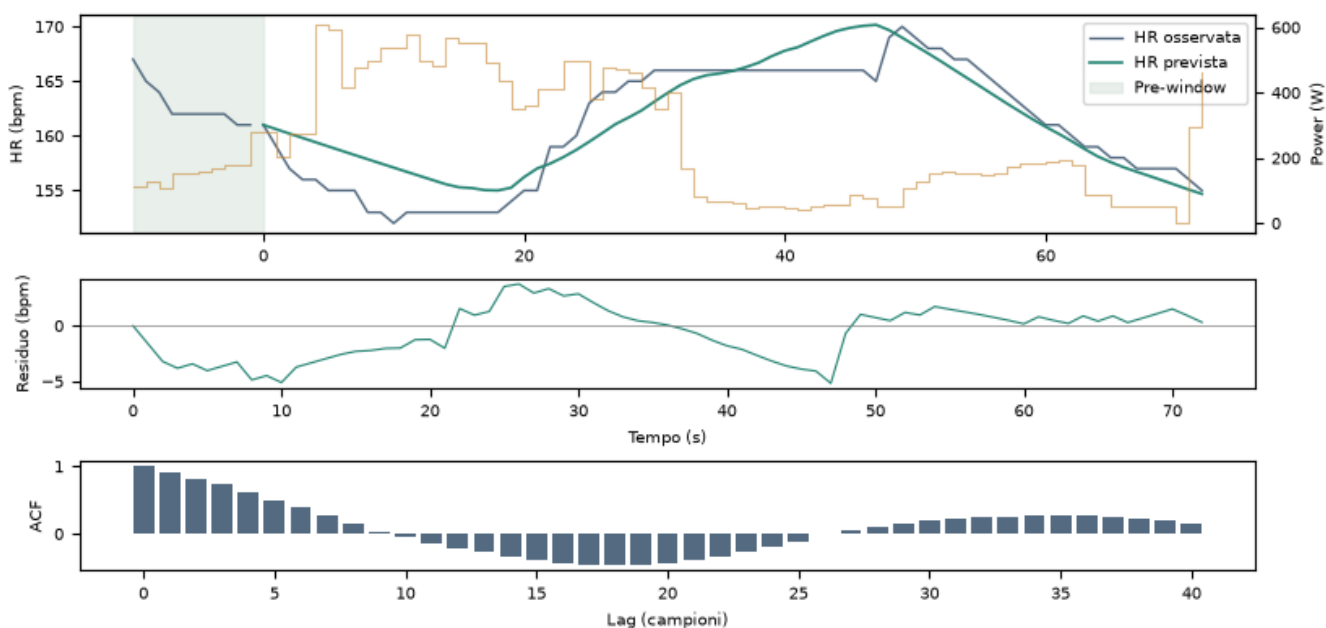
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=162.1 W; HR0=161 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=101 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.3605	1.9174	-0.61238	0.80697	0.85972

residual_mean	residual_sd	ACF lag 1	ACF lag 5
-0.61238	2.2955	0.91273	0.50631



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.77324	2.6901	-4.4994 / 6.0459	347.91	no	no
L	14.724	1.2848	12.205 / 17.242	8.7264	no	no
tau	253.39	885.24	-1481.7 / 1988.5	349.35	no	no
B	60	353.66	-633.18 / 753.18	589.44	si	si

rho=0.22604; Jacobian rank=4/4; cond=72169; accordo multistart=no.

Correlazioni: {"K_L": -0.928167993860725, "K_tau": 0.999849787252099, "K_B": -0.9999841907736114, "L_tau": -0.9287811807337129, "L_B": 0.9285519907769932, "tau_B": -0.9998180370184216}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 406.77 / 2037.9

Griglia: 42441 coppie L/tau; SSE multistart originale: 406.77; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/260427170343__laps_028-029.csv

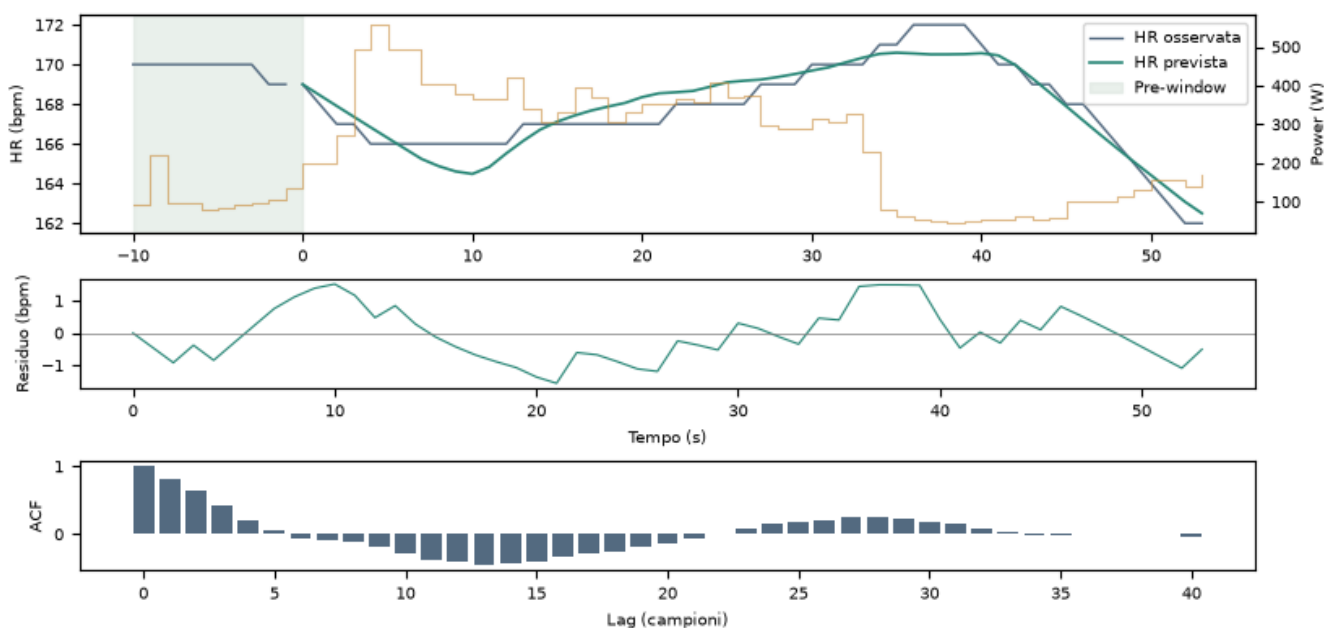
ID: f73906d8288c959166c3ec64 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=110.2 W; HR0=169 bpm.

HR(0) osservata fissata; B=60 bpm stimato; x(0)=109 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
0.80564	0.66836	-0.018922	0.87992	0.80951
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
-0.018922	0.81298	0.82105	0.050035	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.57032	1.8677	-3.0904 / 4.2311	327.49	no	no
L	7.3537	0.50801	6.358 / 8.3494	6.9082	no	no
tau	199.62	658.93	-1091.9 / 1491.1	330.09	no	no
B	60	354.17	-634.18 / 754.18	590.29	si	si

rho=0.22866; Jacobian rank=4/4; cond=1.6643e+05; accordo multistart=no.

Correlazioni: {"K_L": -0.8944828015554681, "K_tau": 0.9998866391807038, "K_B": -0.9999964001610995, "L_tau": -0.8949554526360605, "L_B": 0.8949816195519024, "tau_B": -0.9998963928916054}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 35.049 / 325.44

Griglia: 42441 coppie L/tau; SSE multistart originale: 35.049; selezione: profile_grid.

Segnalazioni: at_bounds, autocorrelated_residuals, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260501094312__laps_002-003.csv

ID: fa162b94f965b809d369523a | Stato: fitted

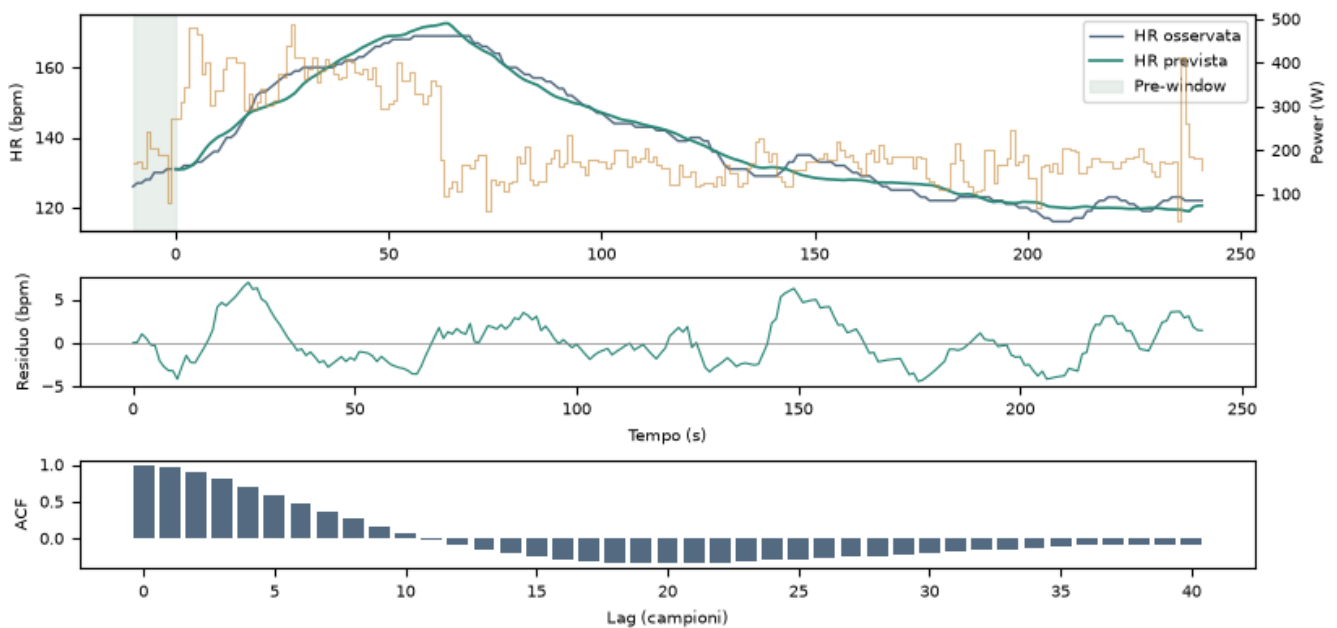
optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=188.7 W; HR0=131 bpm.

HR(0) osservata fissata; B=124.84 bpm stimato; $x(0)$ =6.1586 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.5902	2.1209	0.055166	0.97689	1.0129

residual_mean	residual_sd	ACF lag 1	ACF lag 5
0.055166	2.595	0.95691	0.59232



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.43765	0.011526	0.41506 / 0.46024	2.6336	no	no
L	1.7117	0.37969	0.96746 / 2.4558	22.183	no	no
tau	68.429	2.5907	63.351 / 73.507	3.7859	no	no
B	124.84	0.55346	123.76 / 125.93	0.44333	no	no

rho=3.4969; Jacobian rank=4/4; cond=746.7; accordo multistart=no.

Correlazioni: {"K_L": -0.5210361757320442, "K_tau": 0.9367038435963179, "K_B": -0.9079010389177332, "L_tau": -0.5936506434754203, "L_B": 0.42799293275483935, "tau_B": -0.8866259229283919}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1623.7 / 69875

Griglia: 42441 coppie L/tau; SSE multistart originale: 1623.7; selezione: multistart.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G2/260523080618__laps_004-005.csv

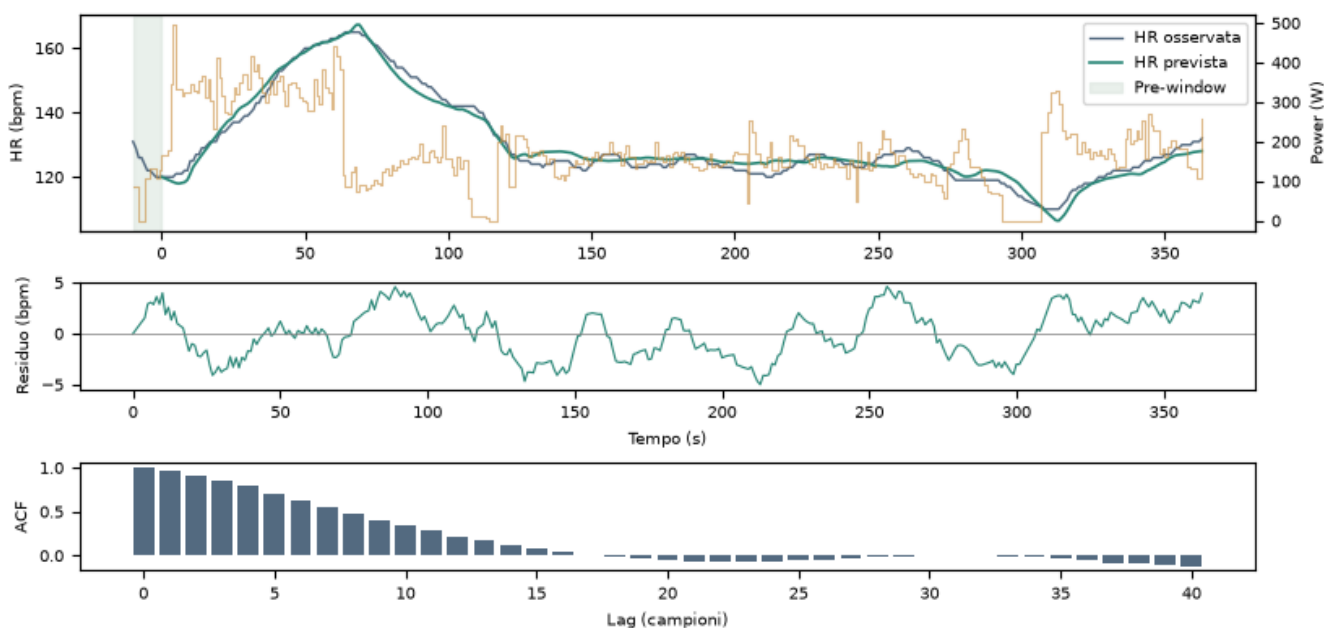
ID: fb76005fcd8e06b8b62c73d5 | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=88.8 W; HR0=120 bpm.

HR(0) osservata fissata; B=101.57 bpm stimato; x(0)=18.426 bpm. B e un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
2.3089	1.9684	0.013122	0.97153	1.1079
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.013122	2.3121	0.95686	0.70798	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.33713	0.004869	0.32758 / 0.34667	1.4442	no	no
L	5.6364	0.22313	5.199 / 6.0737	3.9587	no	no
tau	45.84	1.0276	43.826 / 47.854	2.2416	no	no
B	101.57	0.45504	100.68 / 102.47	0.44798	no	no

rho=7.7959; Jacobian rank=4/4; cond=787.35; accordo multistart=no.

Correlazioni: {"K_L": -0.38991748382404645, "K_tau": 0.8103446953863018, "K_B": -0.9536040754144764, "L_tau": -0.5370695449815887, "L_B": 0.38698317231037066, "tau_B": -0.7921221643667086}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 1919.1 / 66579

Griglia: 42441 coppie L/tau; SSE multistart originale: 1934.4; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, recording_gaps, irregular_sampling, equivalent_predictions_different_parameters, multistart_disagreement

Scheda segmento

dataset_segments_G1_G2_extended_10s.zip::UtD_G1/250708171459__laps_023-024.csv

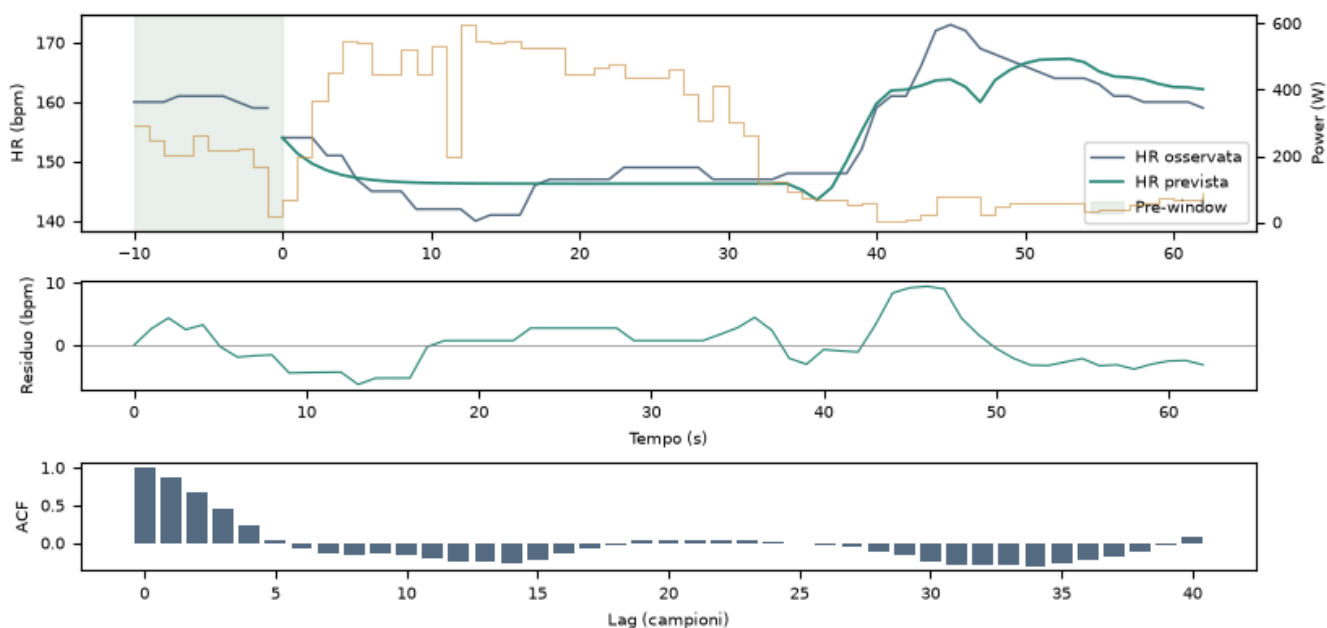
ID: fe33ee2b04d7e7cbec9ae47b | Stato: fitted

optimizer_success: si | identification_valid: si

Pre-window disponibile: si; usata: si; campioni: 10. P0=204.4 W; HR0=154 bpm.

HR(0) osservata fissata; B=146.27 bpm stimato; x(0)=7.731 bpm. B è un livello matematico di equilibrio, senza interpretazione fisiologica.

RMSE	MAE	bias	R2	amplitude_ratio
3.5995	2.8826	0.068691	0.84361	0.71868
residual_mean	residual_sd	ACF lag 1	ACF lag 5	
0.068691	3.6278	0.87136	0.037394	



Parametro	Stima	SE	CI95	RSE %	Bound	Near
K	0.065816	0.0040571	0.057864 / 0.073767	6.1643	no	no
L	34.696	0.29897	34.11 / 35.282	0.8617	no	no
tau	2.4107	0.62578	1.1842 / 3.6372	25.958	no	no
B	146.27	0.63692	145.02 / 147.52	0.43544	no	no

rho=11.326; Jacobian rank=4/4; cond=242.17; accordo multistart=no.

Correlazioni: {"K_L": -0.21211078740264863, "K_tau": 0.4166517547035288, "K_B": -0.6290398503373311, "L_tau": -0.4656312196898386, "L_B": 0.1874553052099934, "tau_B": -0.16117994485604148}

Multistart: 16 tentativi, 16 convergenti; SSE min/max: 816.26 / 4803.7

Griglia: 42441 coppie L/tau; SSE multistart originale: 924.74; selezione: profile_grid.

Segnalazioni: autocorrelated_residuals, equivalent_predictions_different_parameters, multistart_disagreement

Configurazione completa dell'esperimento

Configurazione, split, sorgenti, hash e ambiente della run. I CSV originali restano conservati nel blob store. Manifest integrale e risultati sono inclusi nell'export tabellare ZIP.

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  }
}

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